Graphs and network flows

Challenges of this module



Challenges of this module

* Introduce the notation



Challenges of this module

* Introduce the notation
* How to represent graphs



Challenges of this module

* Introduce the notation
* How to represent graphs
* Create an algorithm for a network design problem



Challenges of this module

* Introduce the notation
* How to represent graphs
* Create an algorithm for a network design problem
* Correctness of the algorithm and optimality conditions



Challenges of this module

* Introduce the notation
* How to represent graphs
* Create an algorithm for a network design problem
* Correctness of the algorithm and optimality conditions
* TSP



Challenges of this module

* Introduce the notation
* How to represent graphs
* Create an algorithm for a network design problem
* Correctness of the algorithm and optimality conditions
* TSP
* ...



Links with previous knowledge



Links with previous knowledge

* Basic algorithmic notation (high level)



Links with previous knowledge

* Basic algorithmic notation (high level)
* Remember the principles of modeling



Links with previous knowledge

* Basic algorithmic notation (high level)
* Remember the principles of modeling
  * decisions are usually associated with the arcs

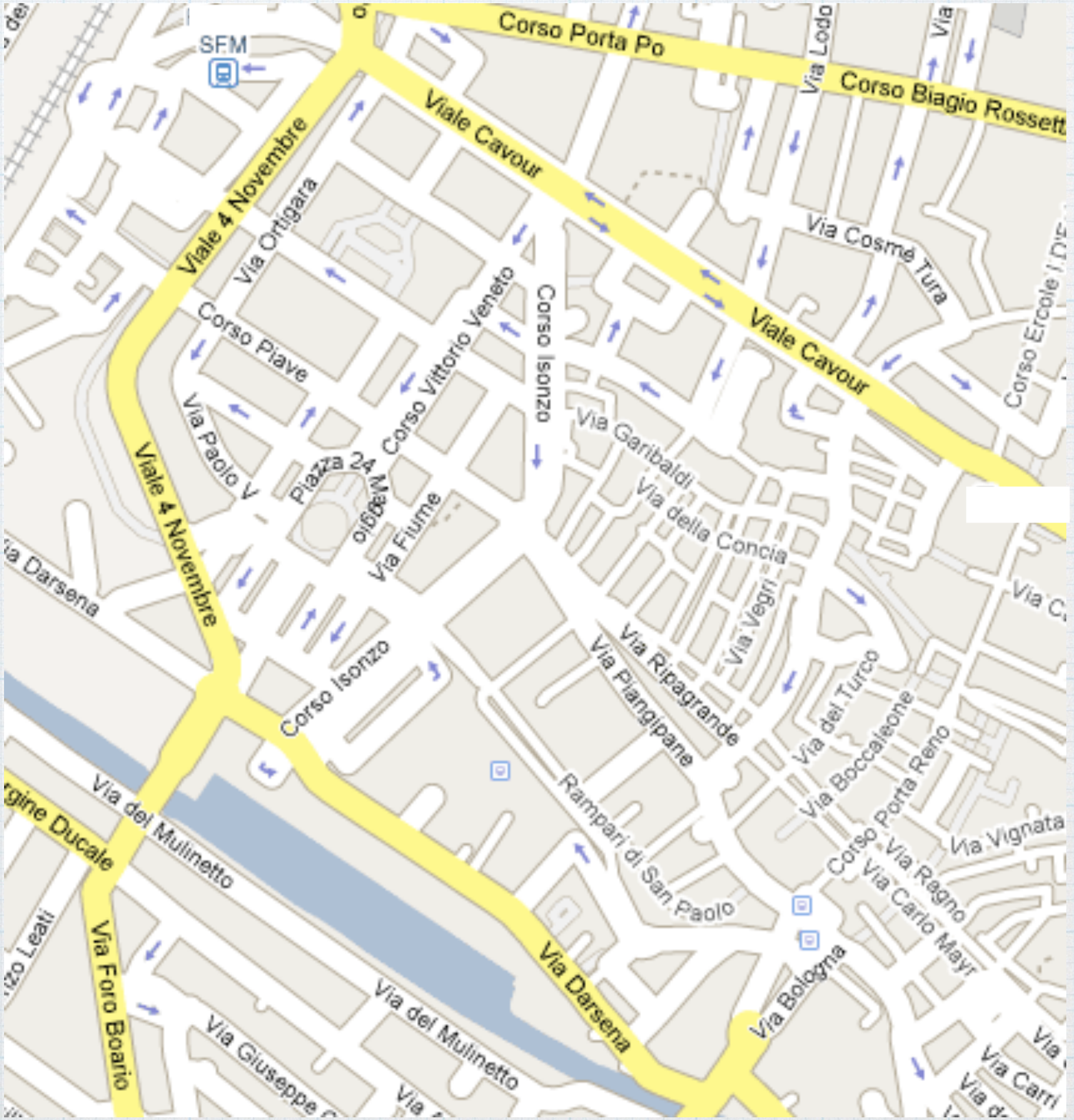


Links with previous knowledge

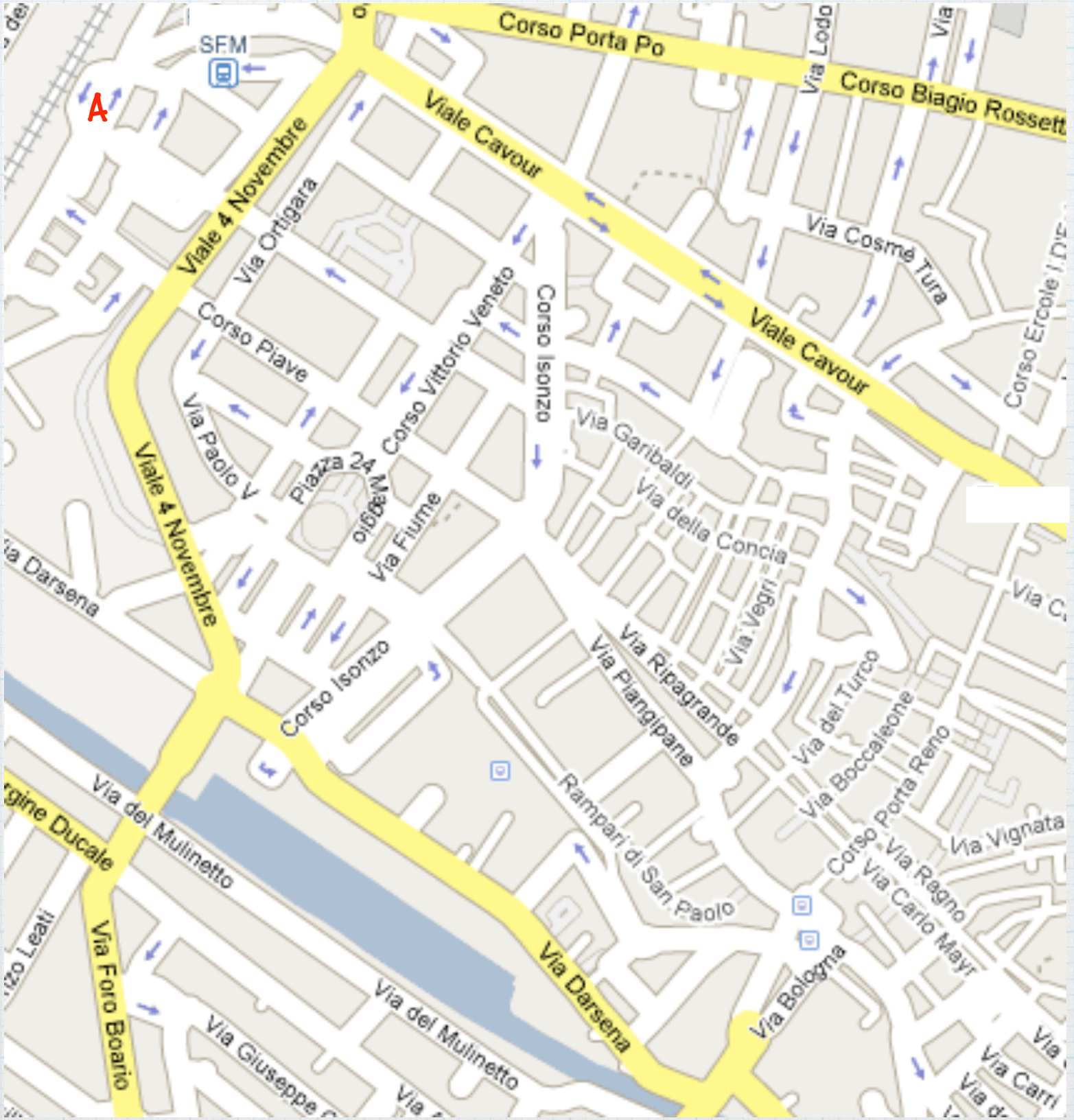
* Basic algorithmic notation (high level)
* Remember the principles of modeling
  * decisions are usually associated with the arcs
* Ask why



Example

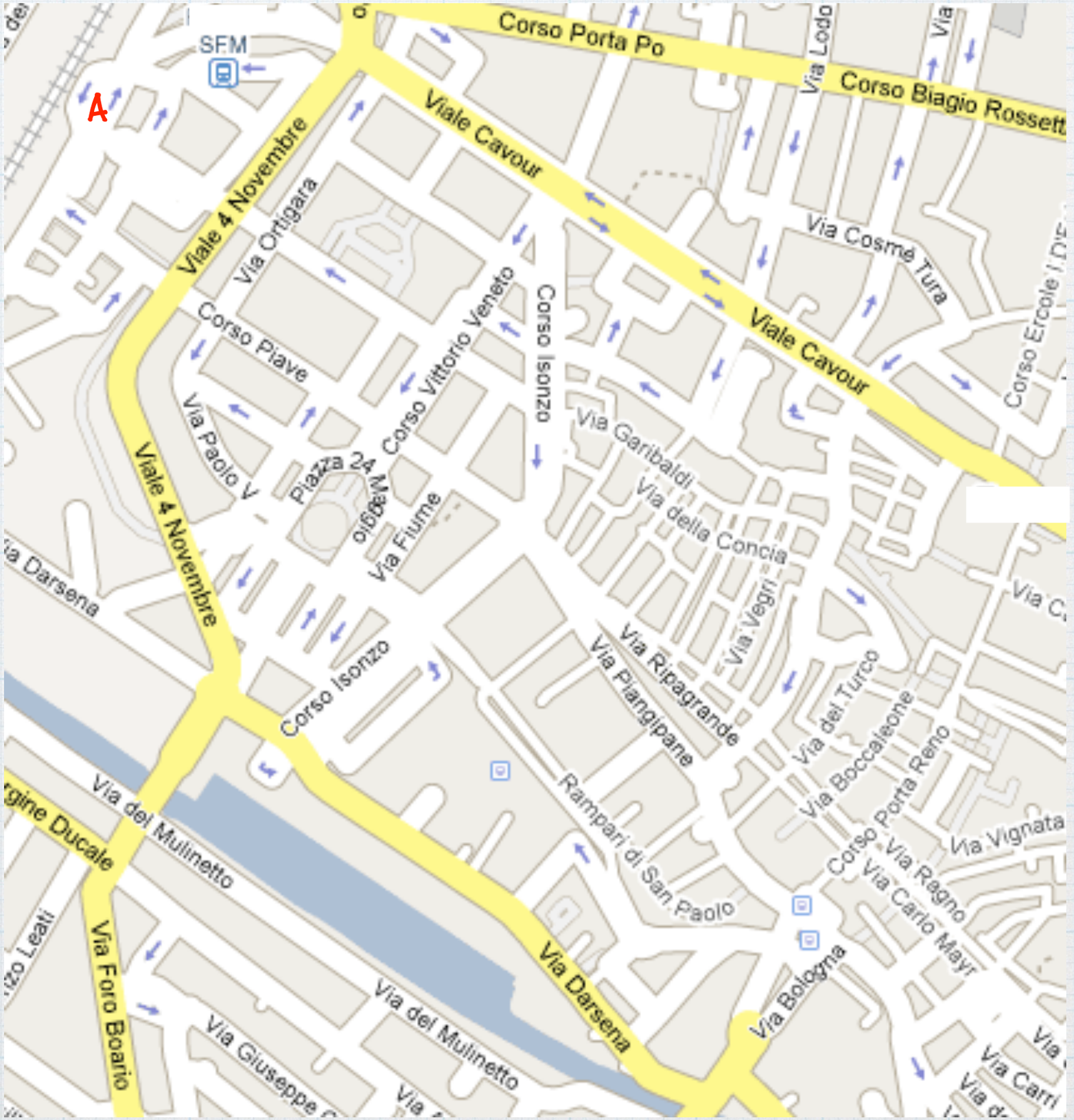

SFM
Corso Porta Po
Corso Biagio Rossett
Viale Cavour
Viale 4 Novembre
Via Ortigara
Via Lodo
Via Cosmè Tura
Corso Ercole I D'Es
Corso Piave
Via Paolo V
Corso Vittorio Veneto
Corso Isonzo
Via Garibaldi
Via della Concia
Piazza 24 Maggio
Via Fiume
Via Darsena
Via Ripagrande
Via Piangipane
Via Vegri
Via del Turco
Via Boccaleone
Corso Porta Reno
Via Vignata
Rampari di San Paolo
Corso Via Regno
Via Carlo Mayr
Via Bologna
Via del Mulinetto
Via Foro Boario
Via Giuseppe
Via Carri

Example


A
SFM
Corso Porta Po
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Viale 4 Novembre
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Corso Piave
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Via Foro Boario
Via Bologna
Via Carri

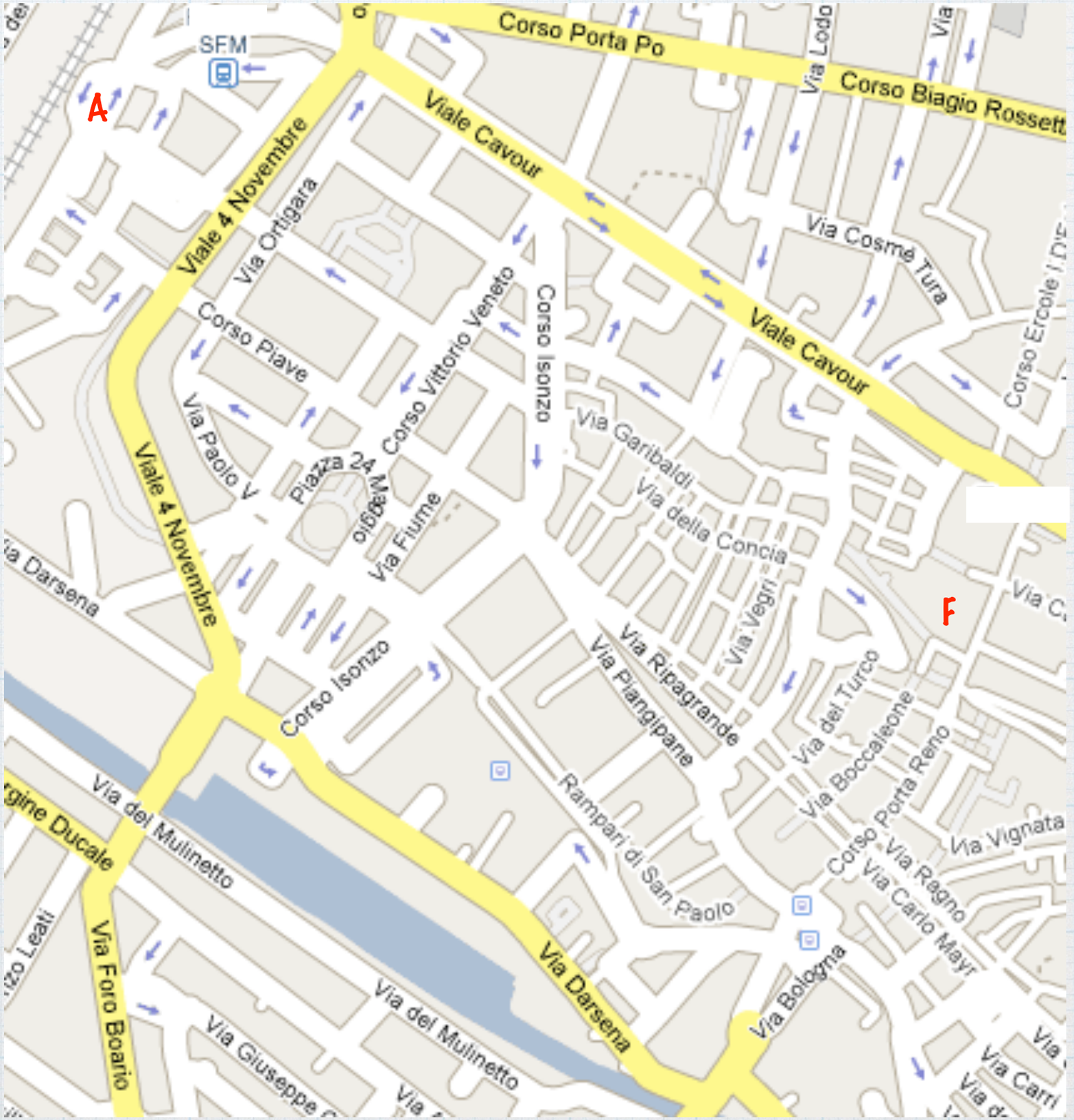
Example

A: starting place



Example

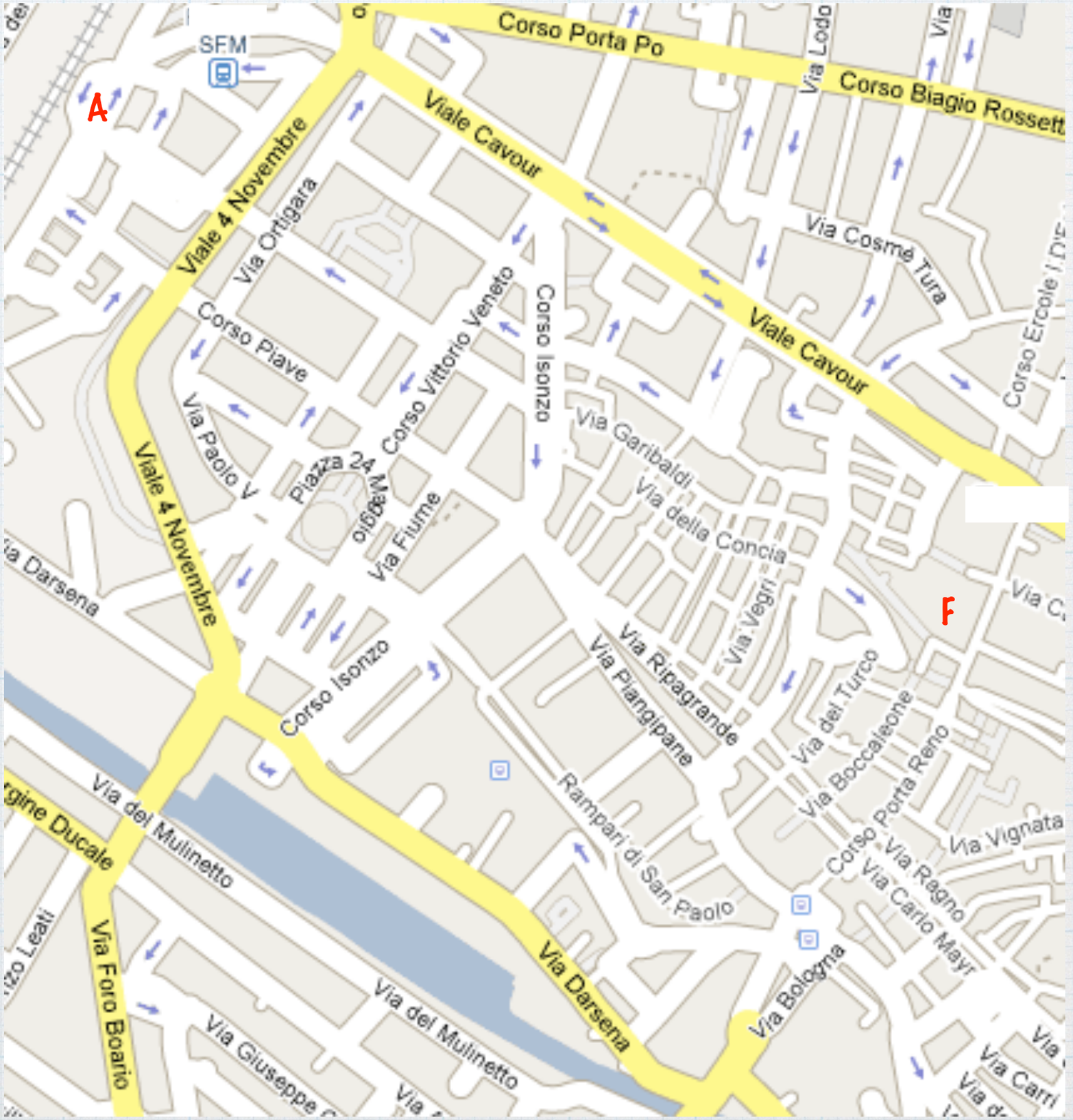
A: starting place



Example

A: starting place

F: destination

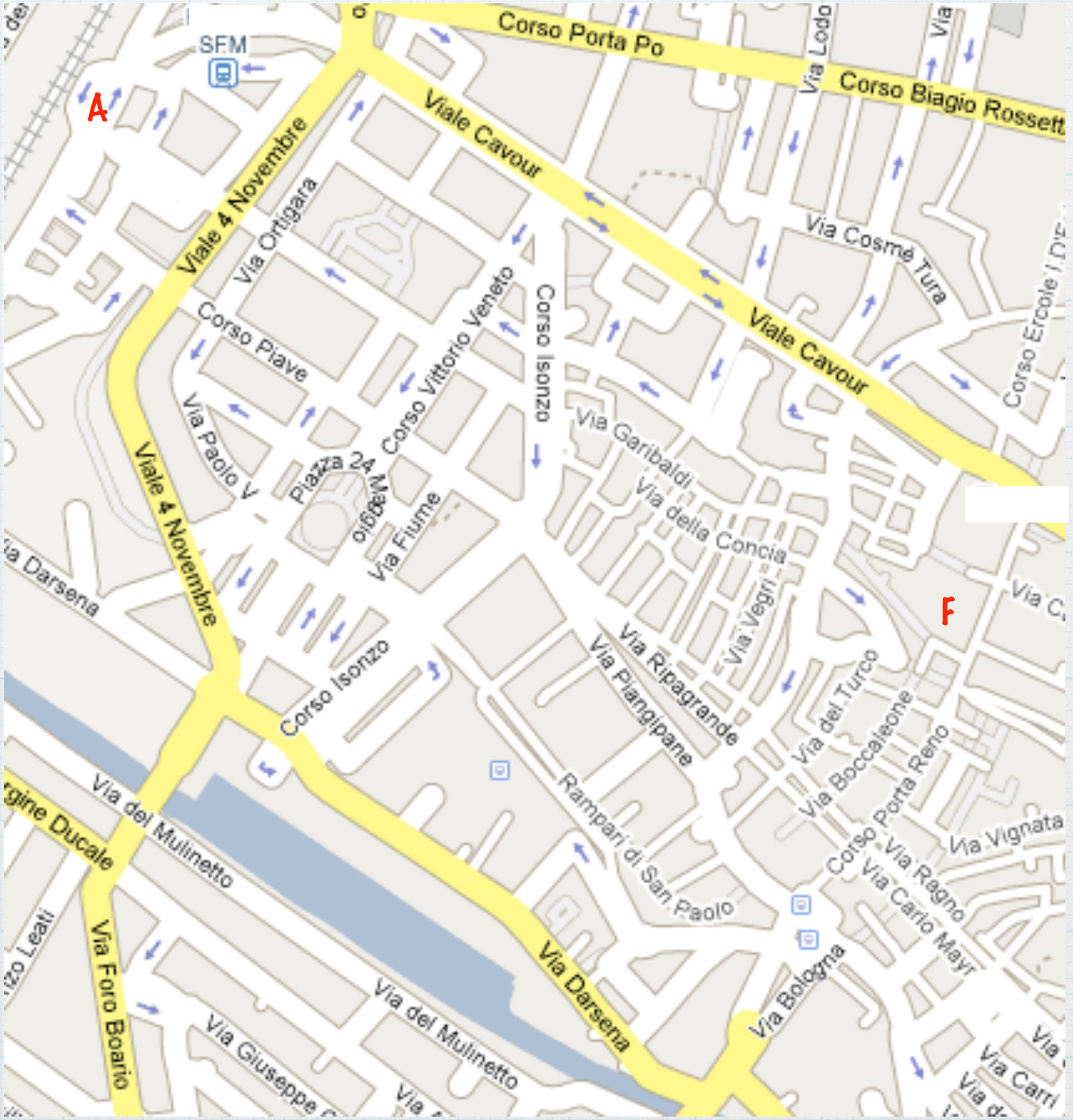


Example

site or intersection

A: starting place

F: destination

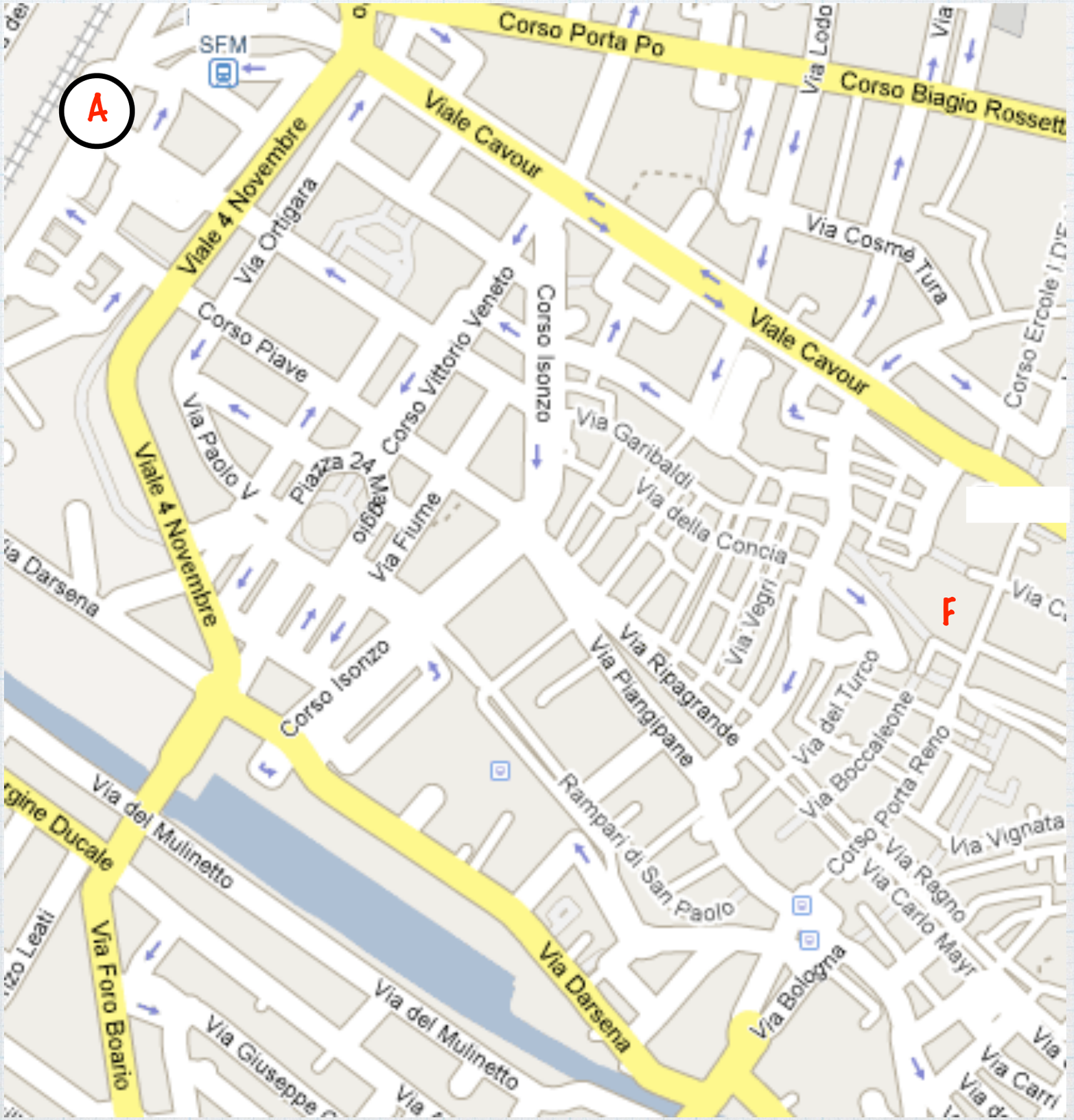


Example

site or intersection

A: starting place

F: destination

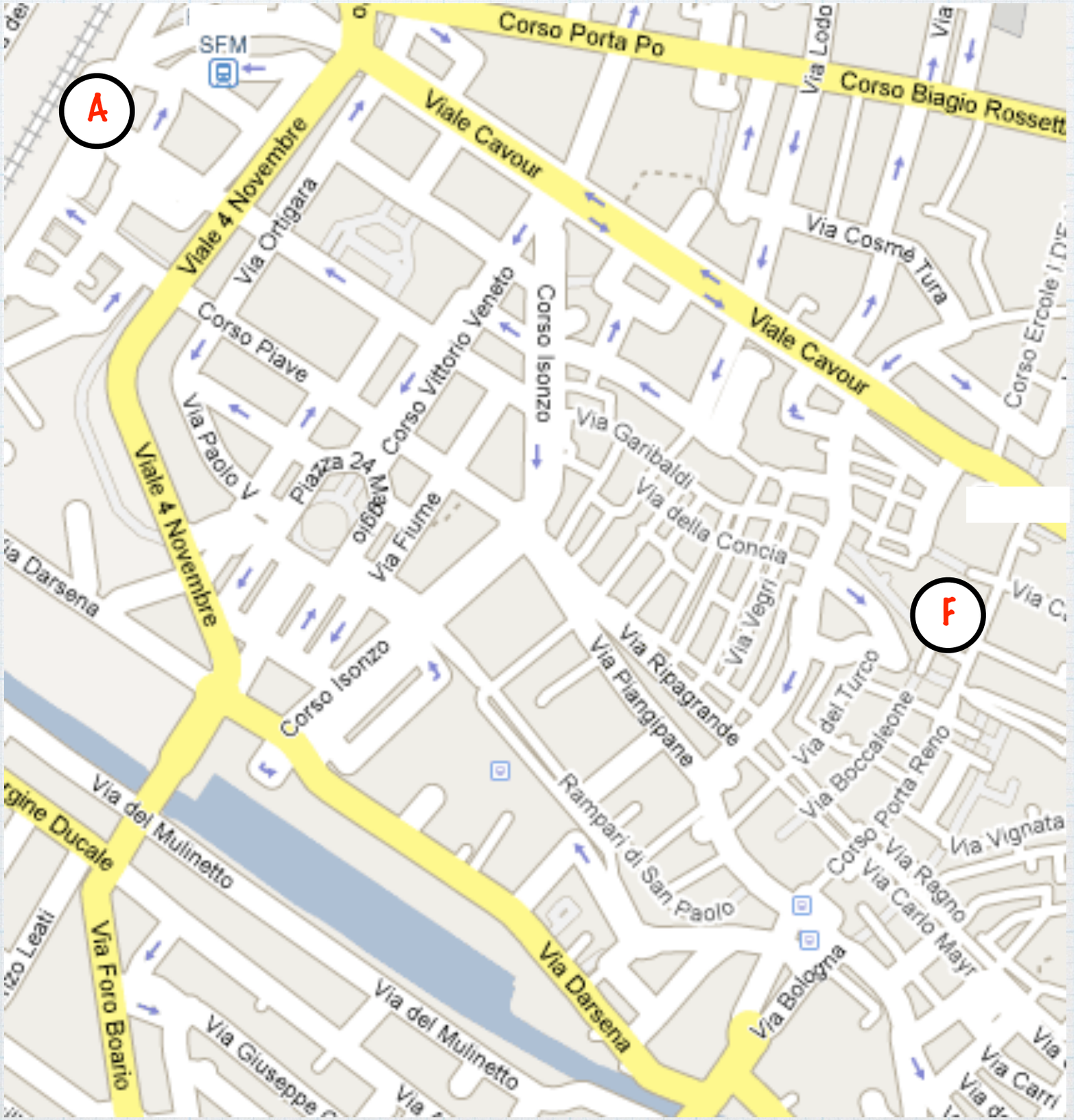


Example

site or intersection

A: starting place

F: destination

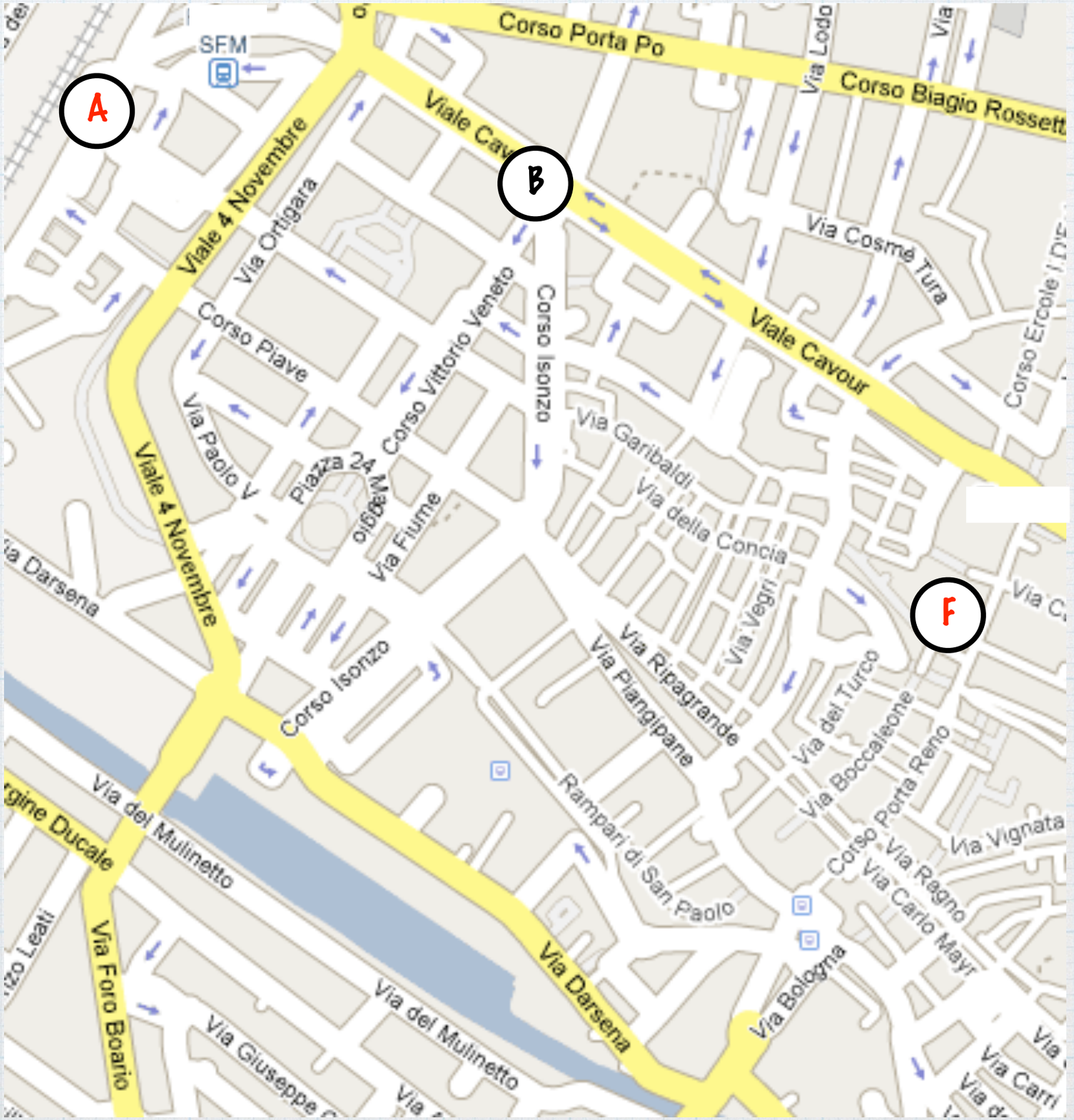


Example

site or intersection

A: starting place

F: destination

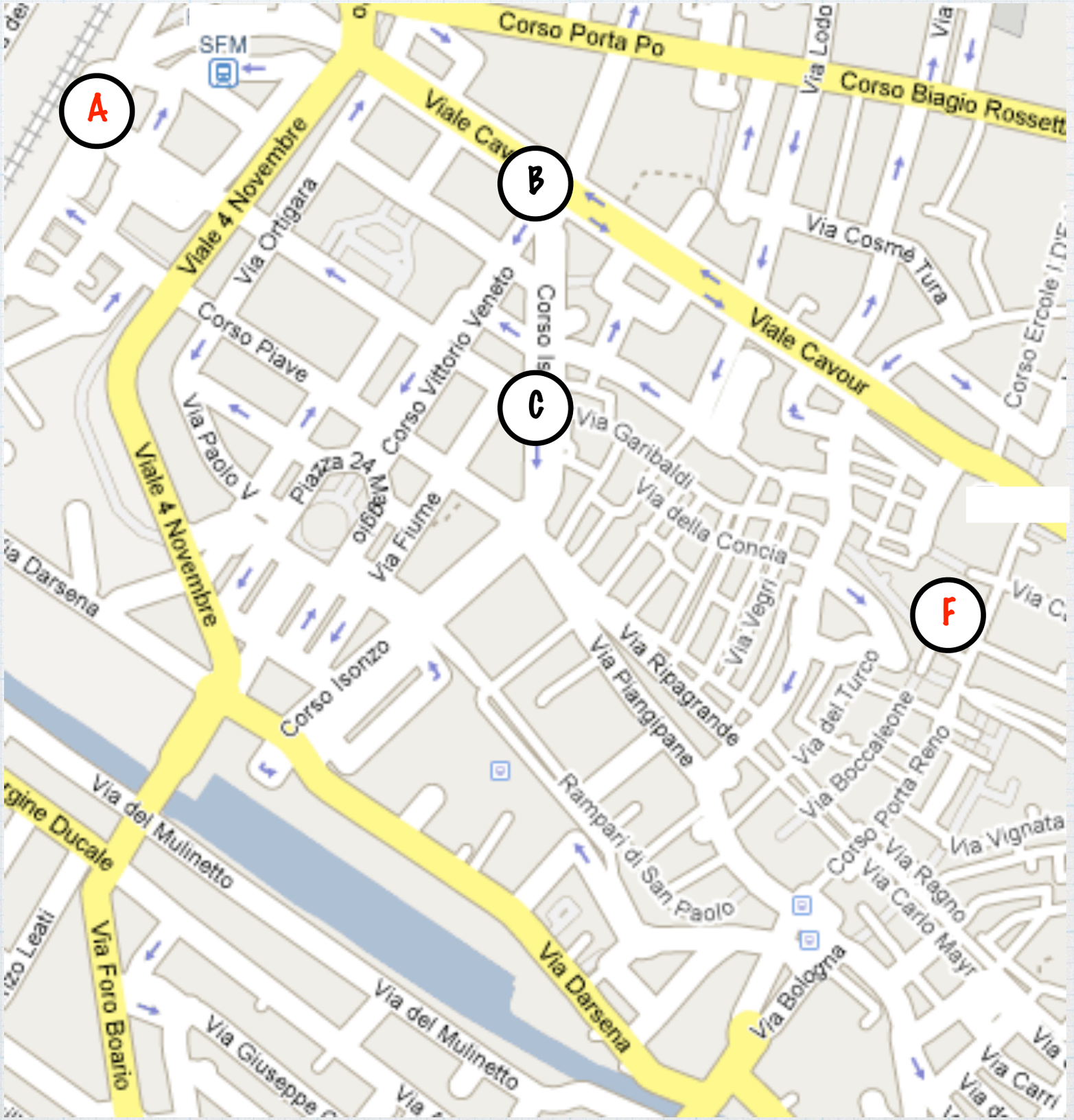


Example

site or intersection

A: starting place

F: destination



Example

site or intersection

A: starting place

F: destination



Example

site or intersection

A: starting place

F: destination



Example

site or intersection

street

A: starting place

F: destination



Example

site or intersection

street

A: starting place

F: destination



Example

site or intersection

street

5 walking distance (min.)

A: starting place

F: destination



Example

site or intersection

street

5 walking distance (min.)

A: starting place

F: destination



Example

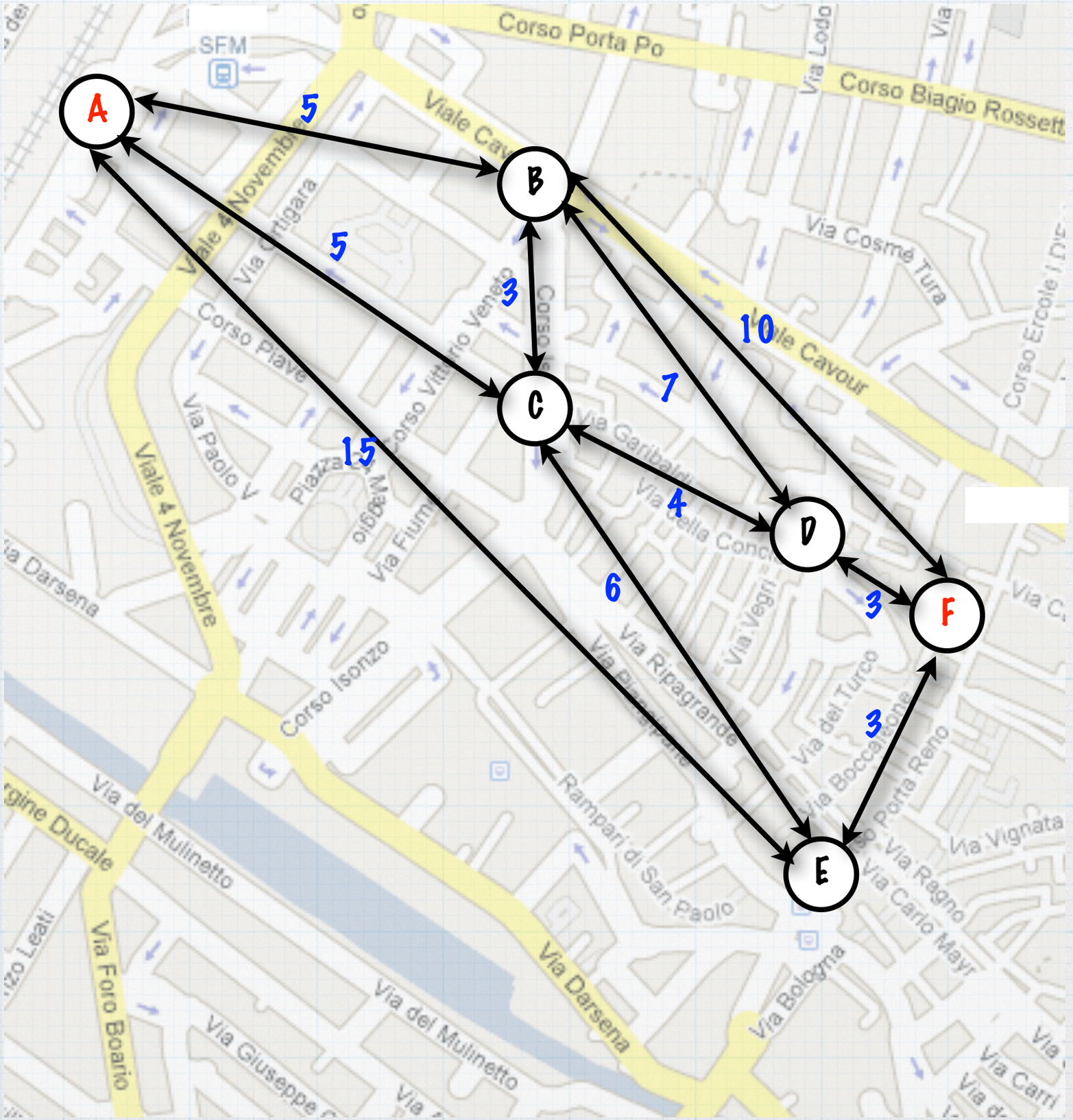
site or intersection

street

5 walking distance (min.)

A: starting place

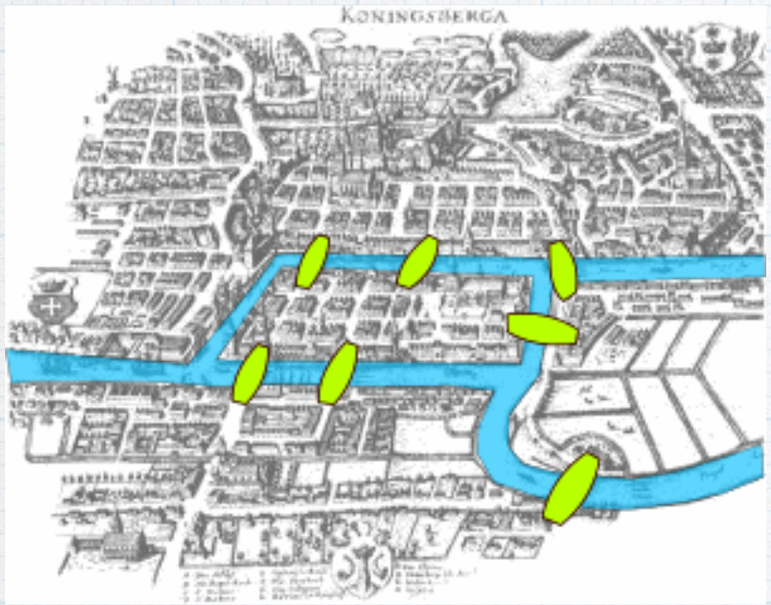
F: destination



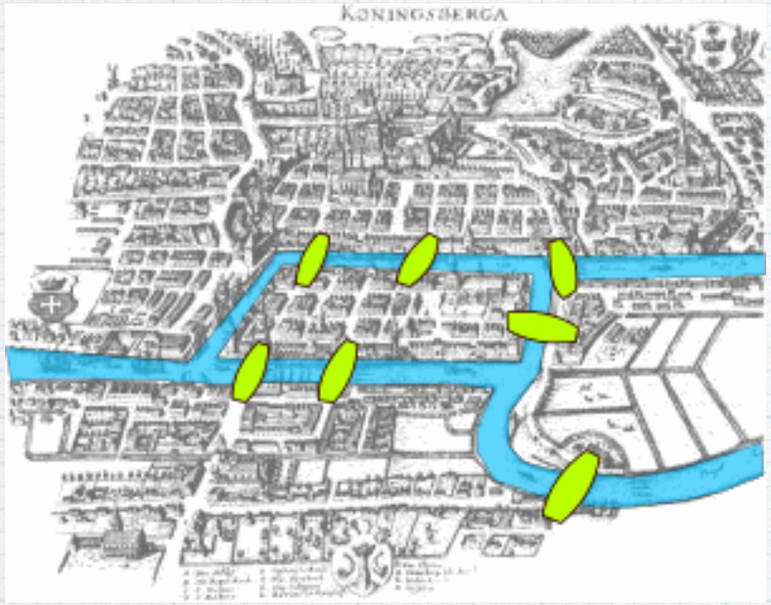
What can we do with the help of graphs?

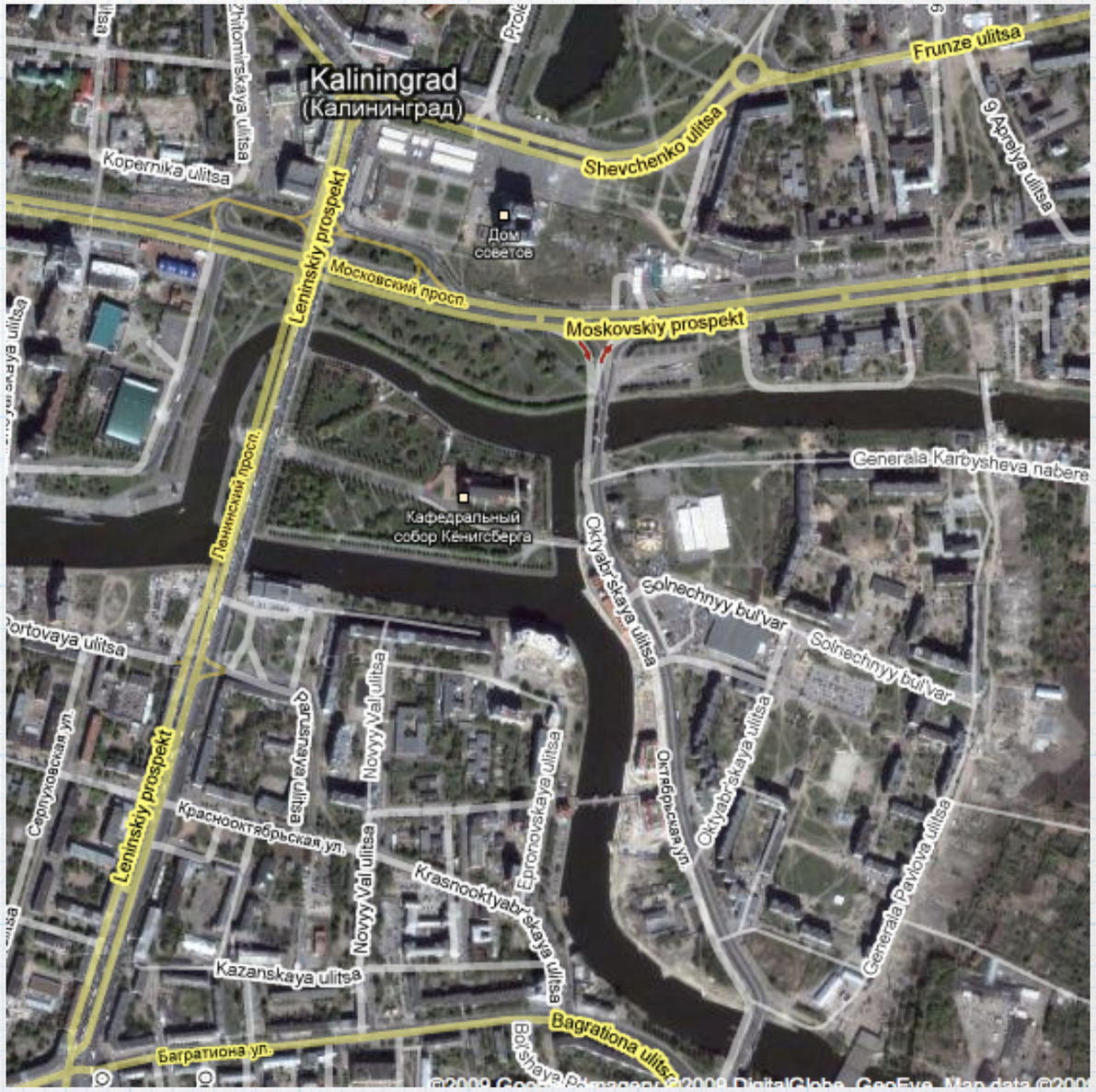


Curiosity: Königsberg bridges

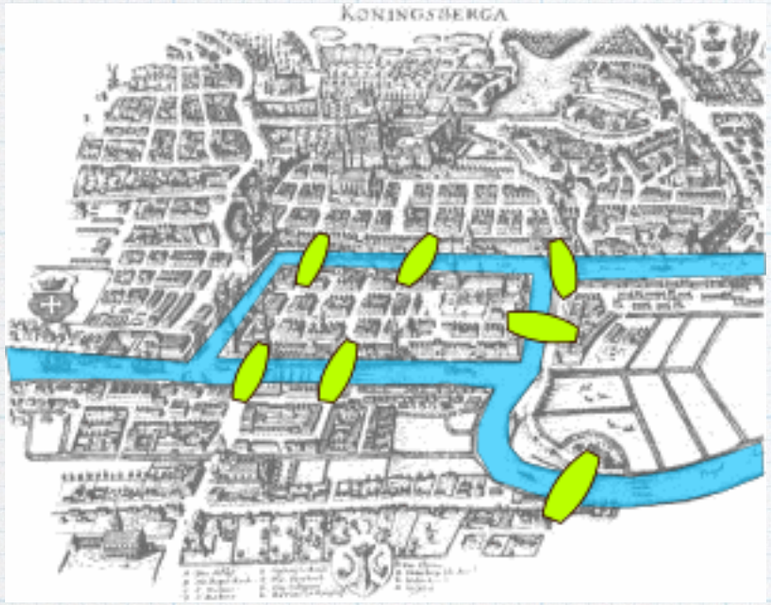


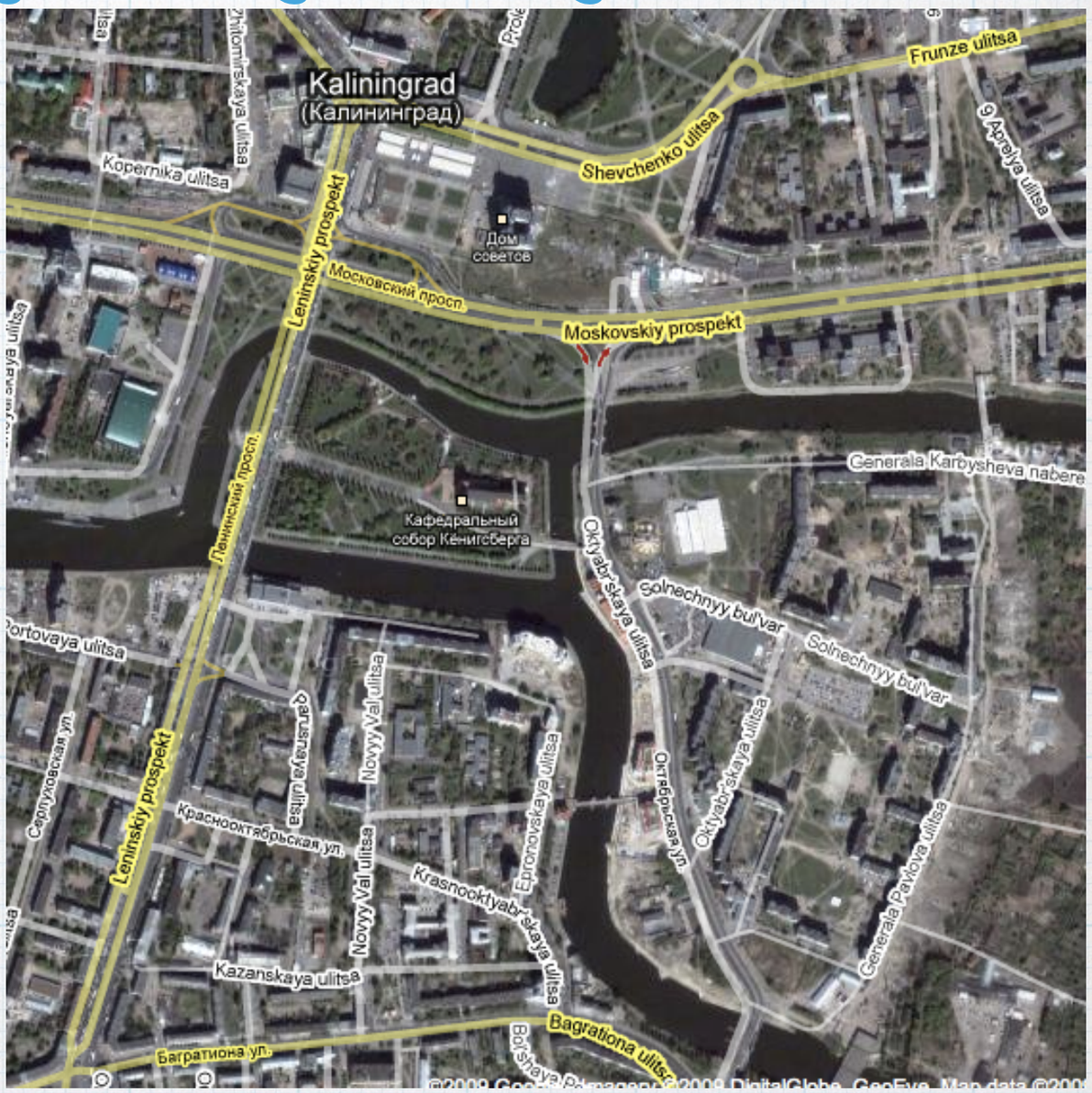
Curiosity: Königsberg bridges





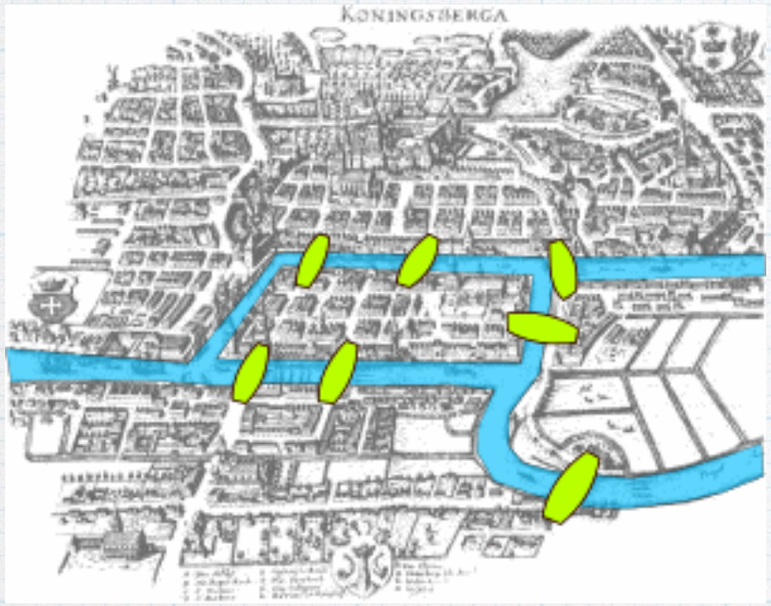
Curiosity: Königsberg bridges





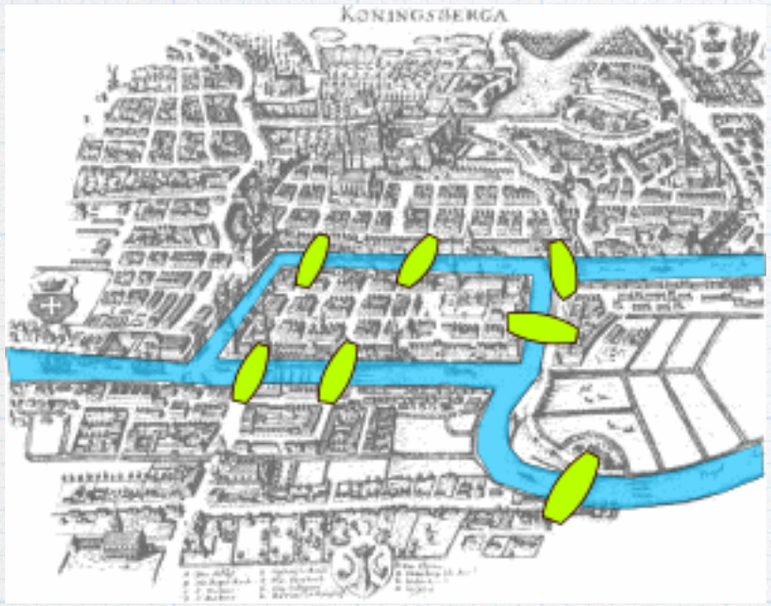
Is it possible to walk over the seven bridges and end up in the starting place?

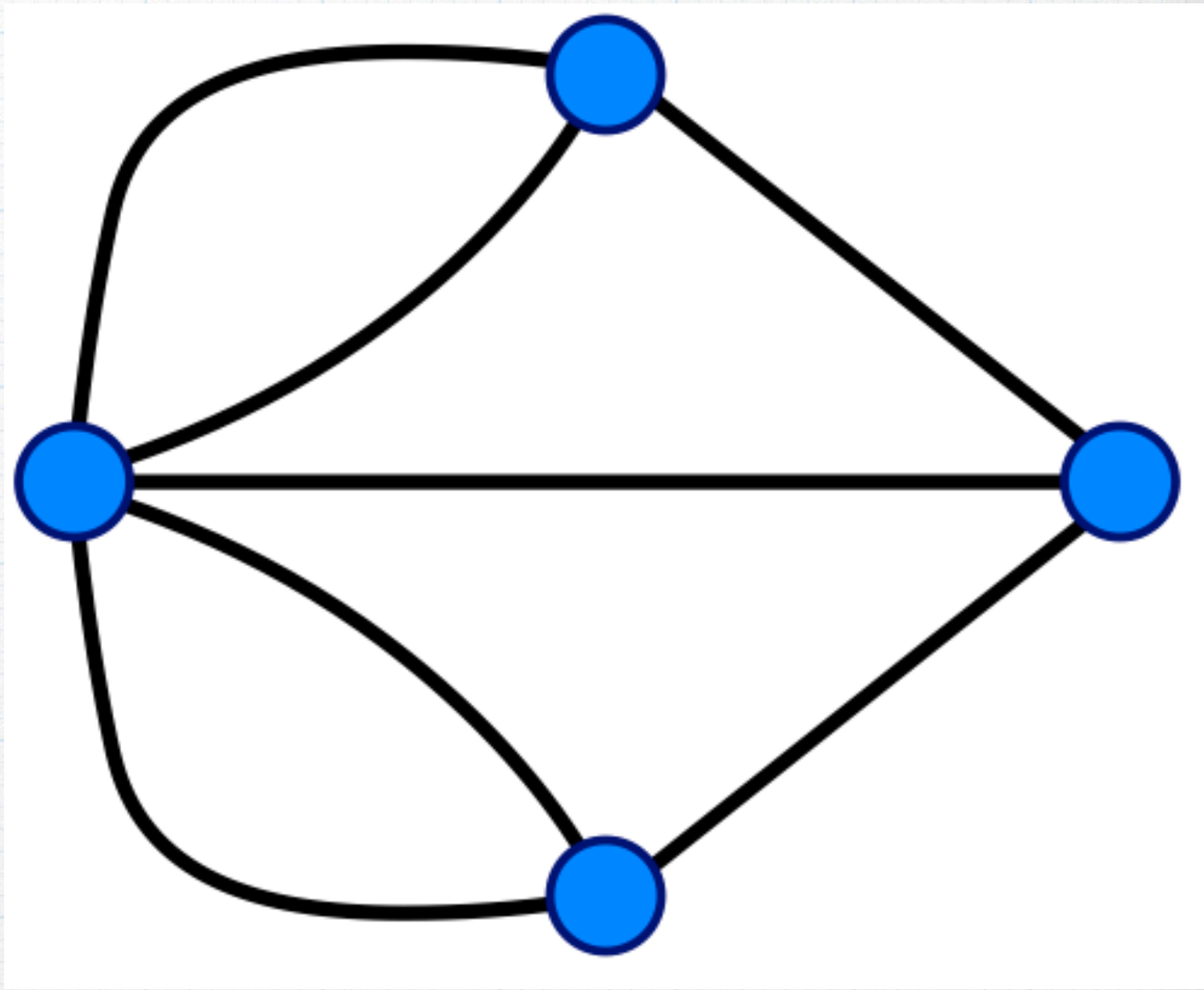
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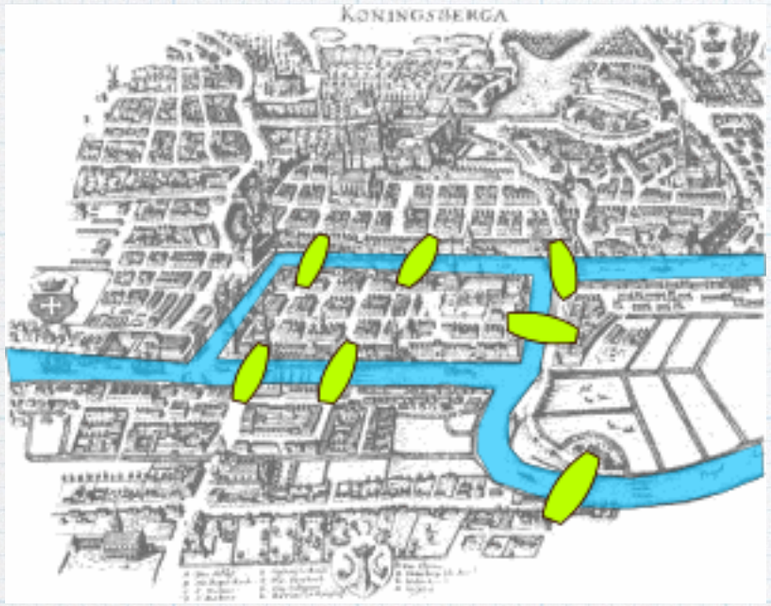
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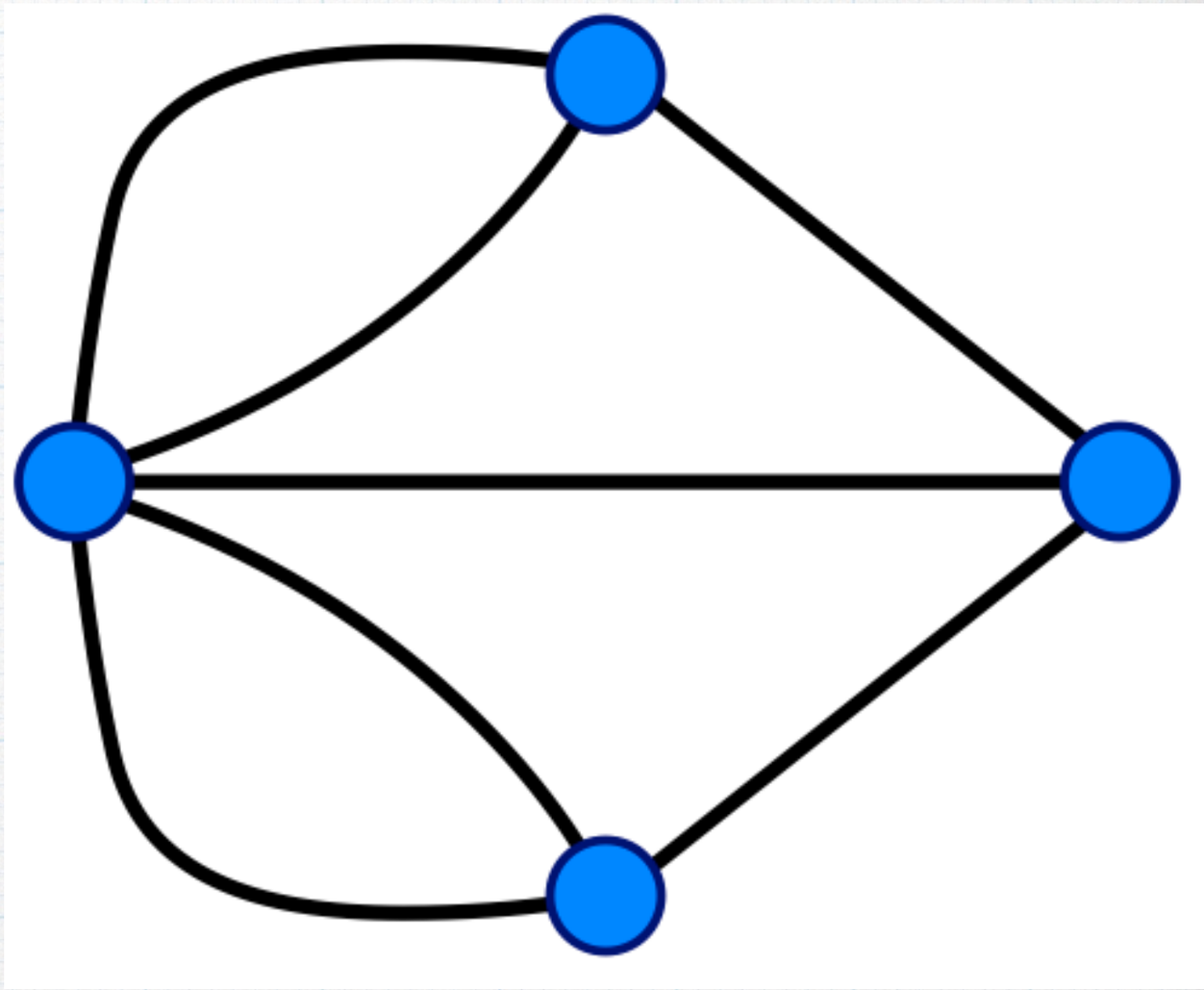




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Curiosity: Königsberg bridges

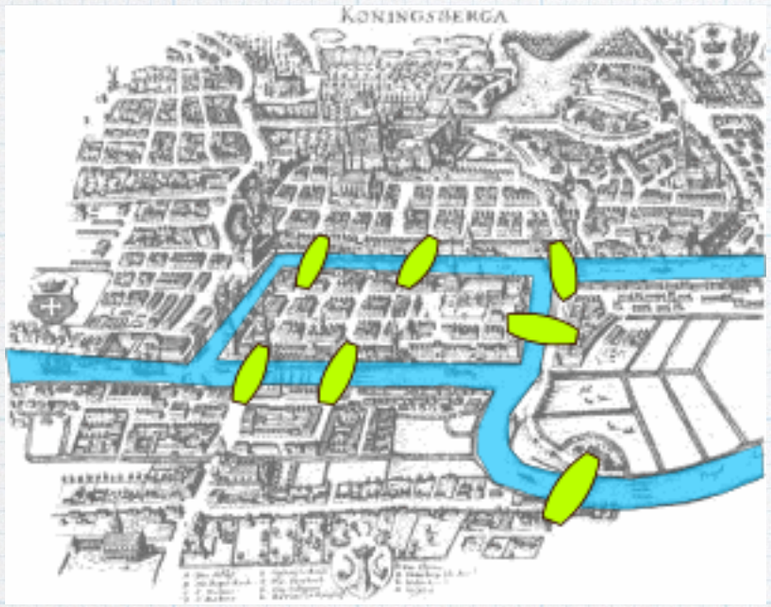


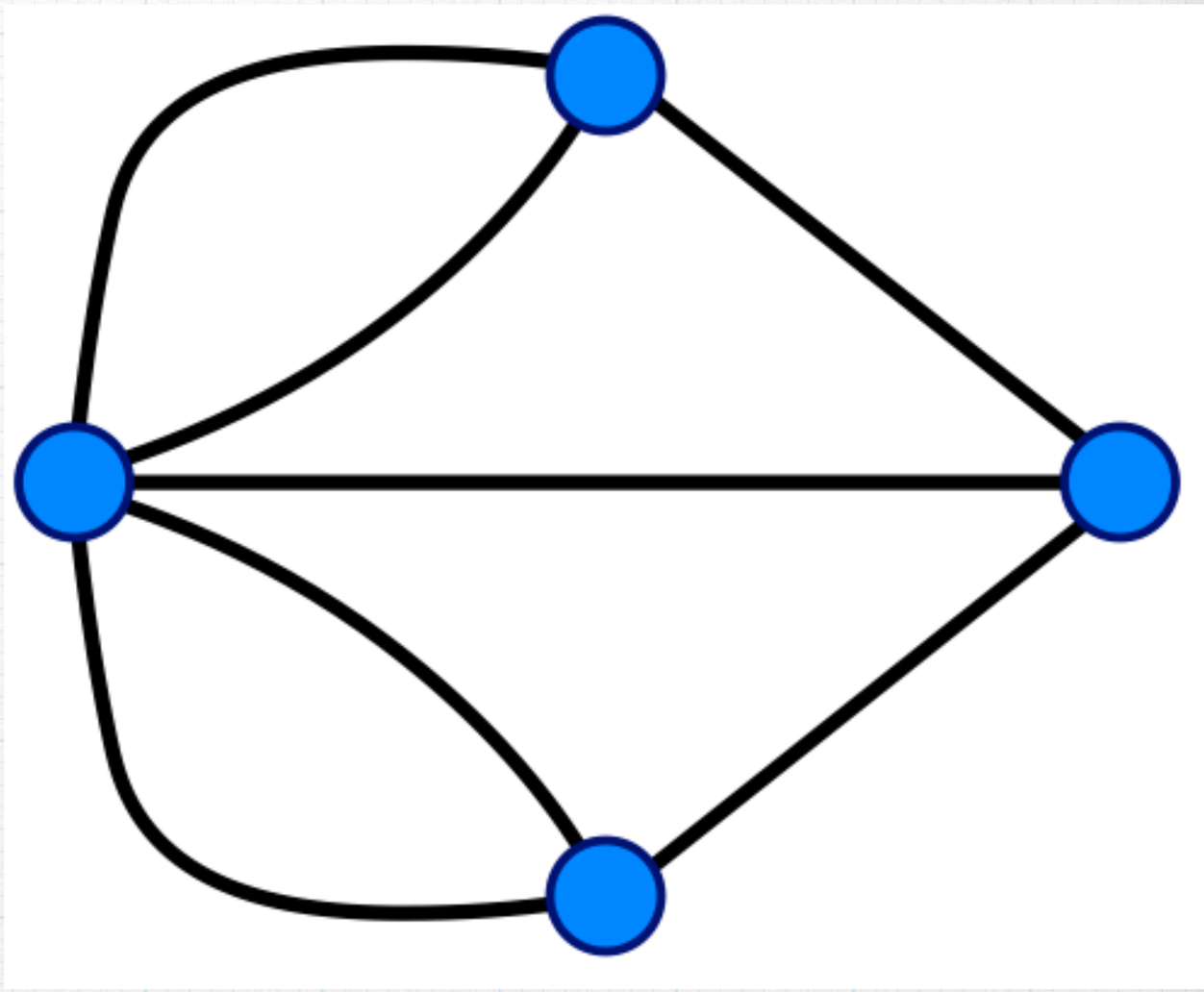


Is it possible to walk over the seven bridges and end up in the starting place?

Is the graph **Eulerian**?

Curiosity: Königsberg bridges



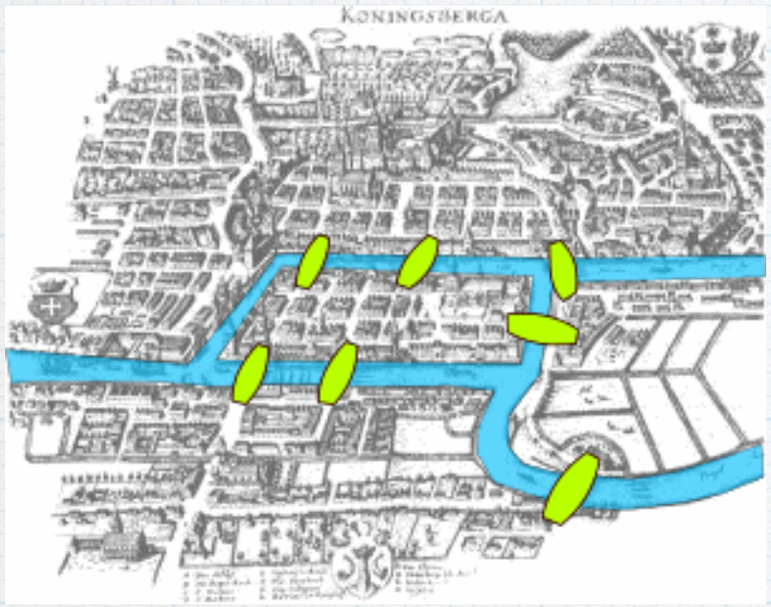


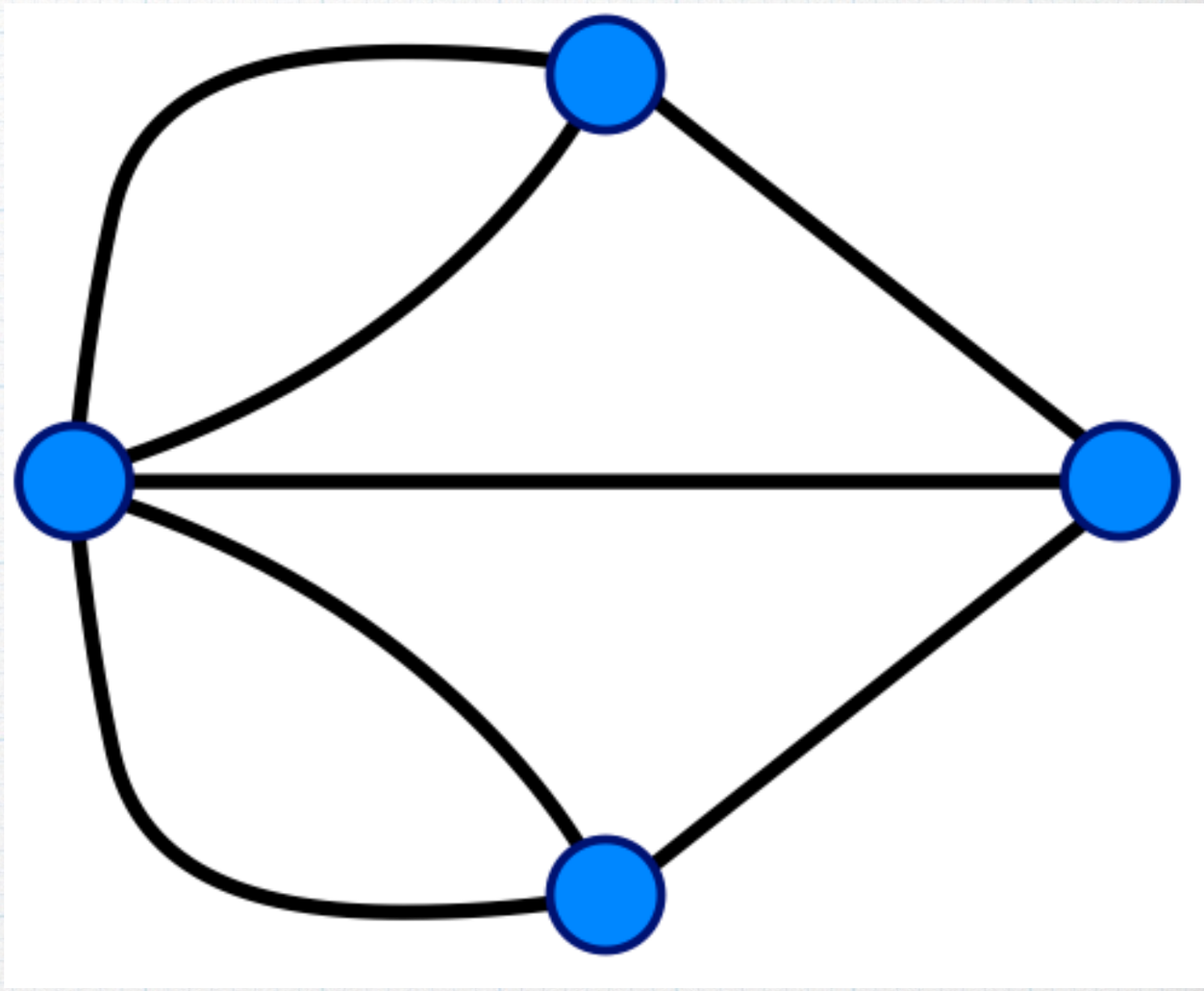
Is it possible to walk over the seven bridges and end up in the starting place?

Is the graph **Eulerian**?

Necessary and sufficient condition on **node degrees** (number of incident arcs)

Curiosity: Königsberg bridges





Is it possible to walk over the seven bridges and end up in the starting place?

Is the graph **Eulerian**?

Necessary and sufficient condition on **node degrees** (number of incident arcs)

Starting milestone of graph theory

Short movie on graph modeling



https://www.youtube.com/watch?v=qmZkf1MssGI&list=PLq8YHwrfUwKdXuAWYM2tQiK2aFAzUx47s&index=4

Short movie on graph modeling

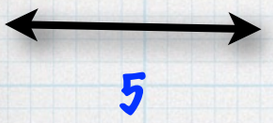


https://www.youtube.com/watch?v=qmZkf1MssGI&list=PLq8YHwrfUwKdXuAWYM2tQiK2aFAzUx47s&index=4

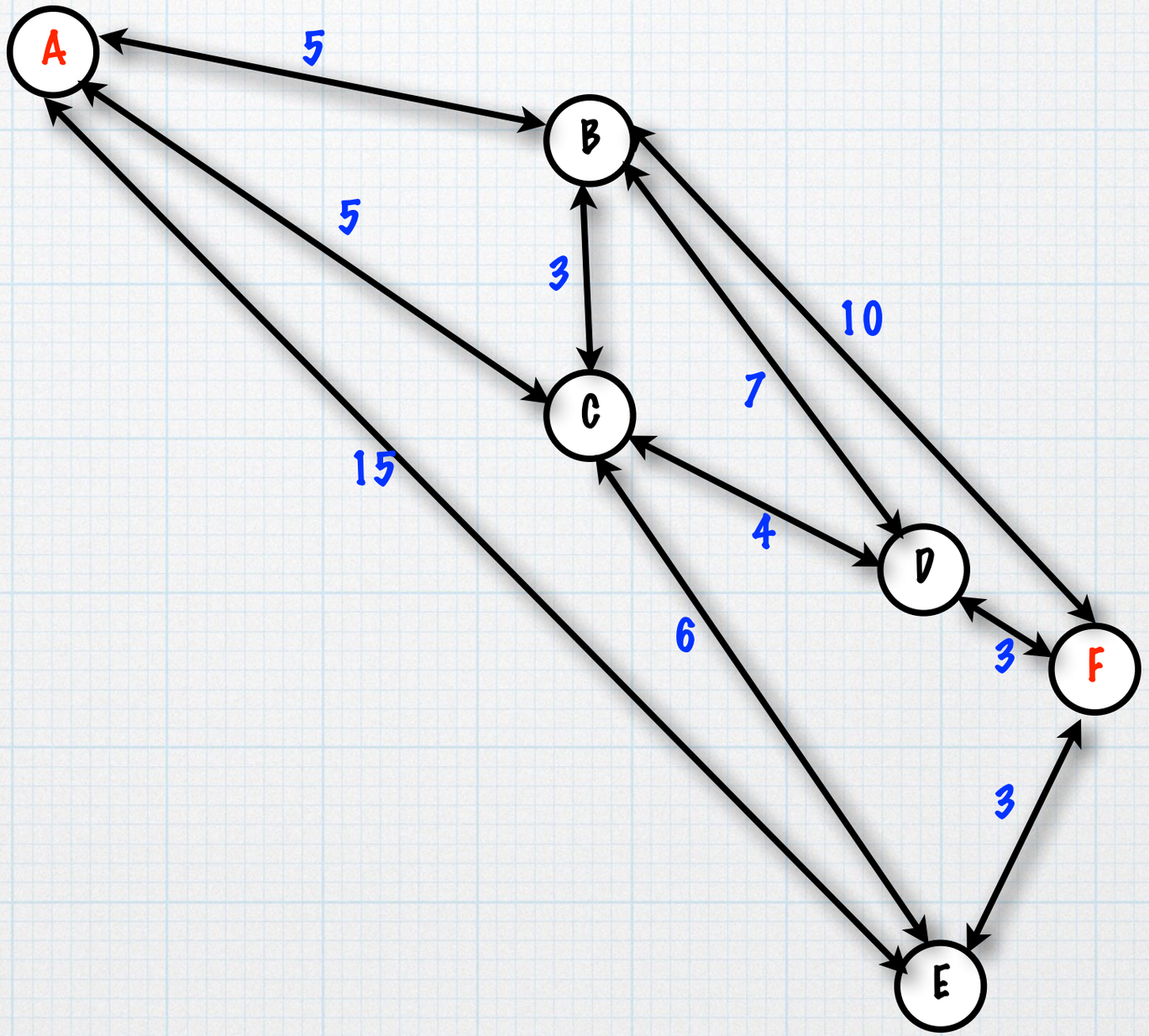
Graph



node / vertex



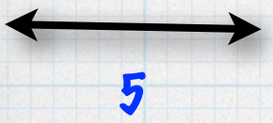
arc / edge
length



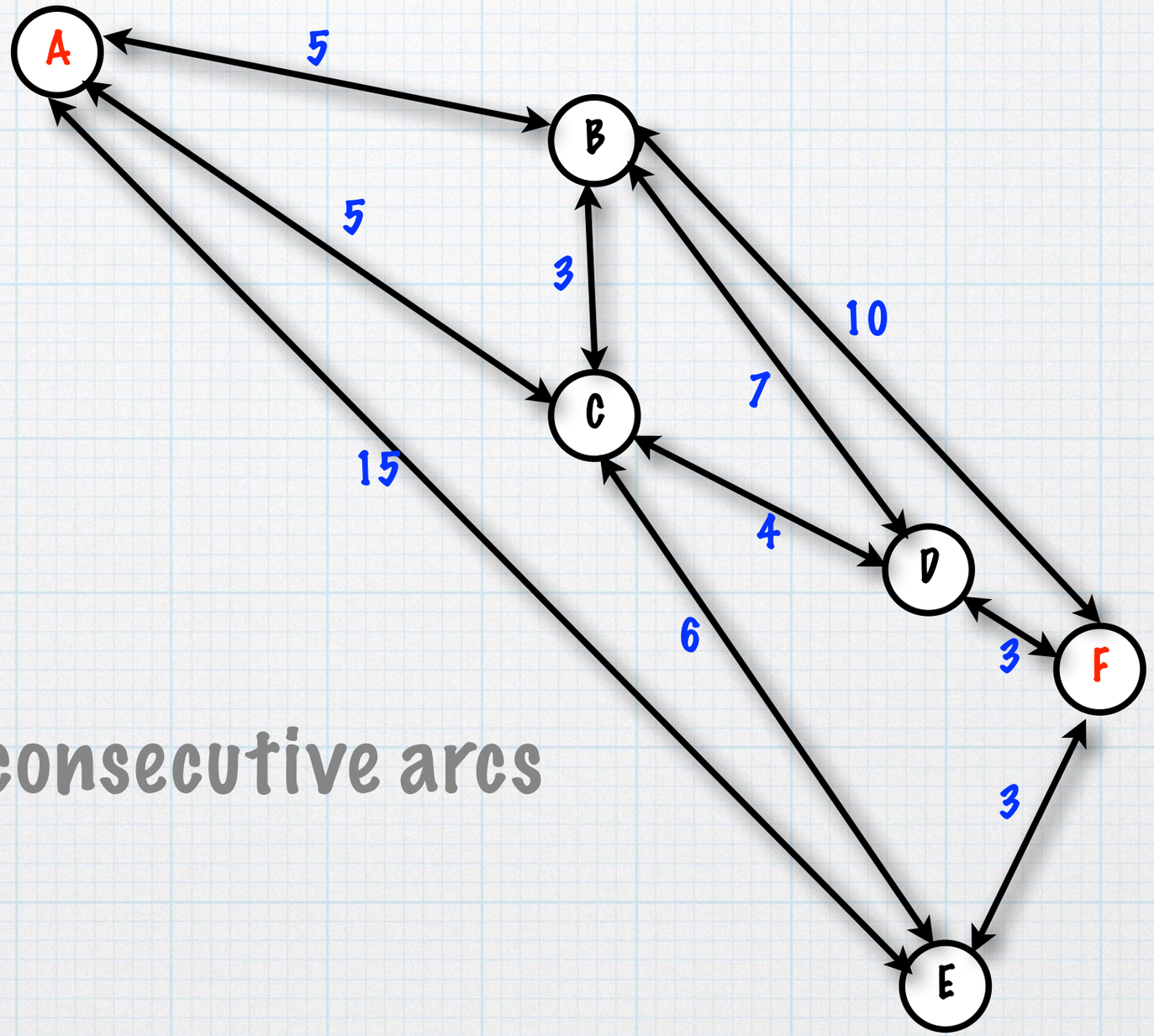
Graph



node / vertex

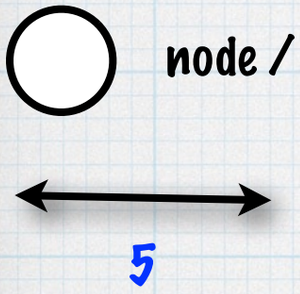


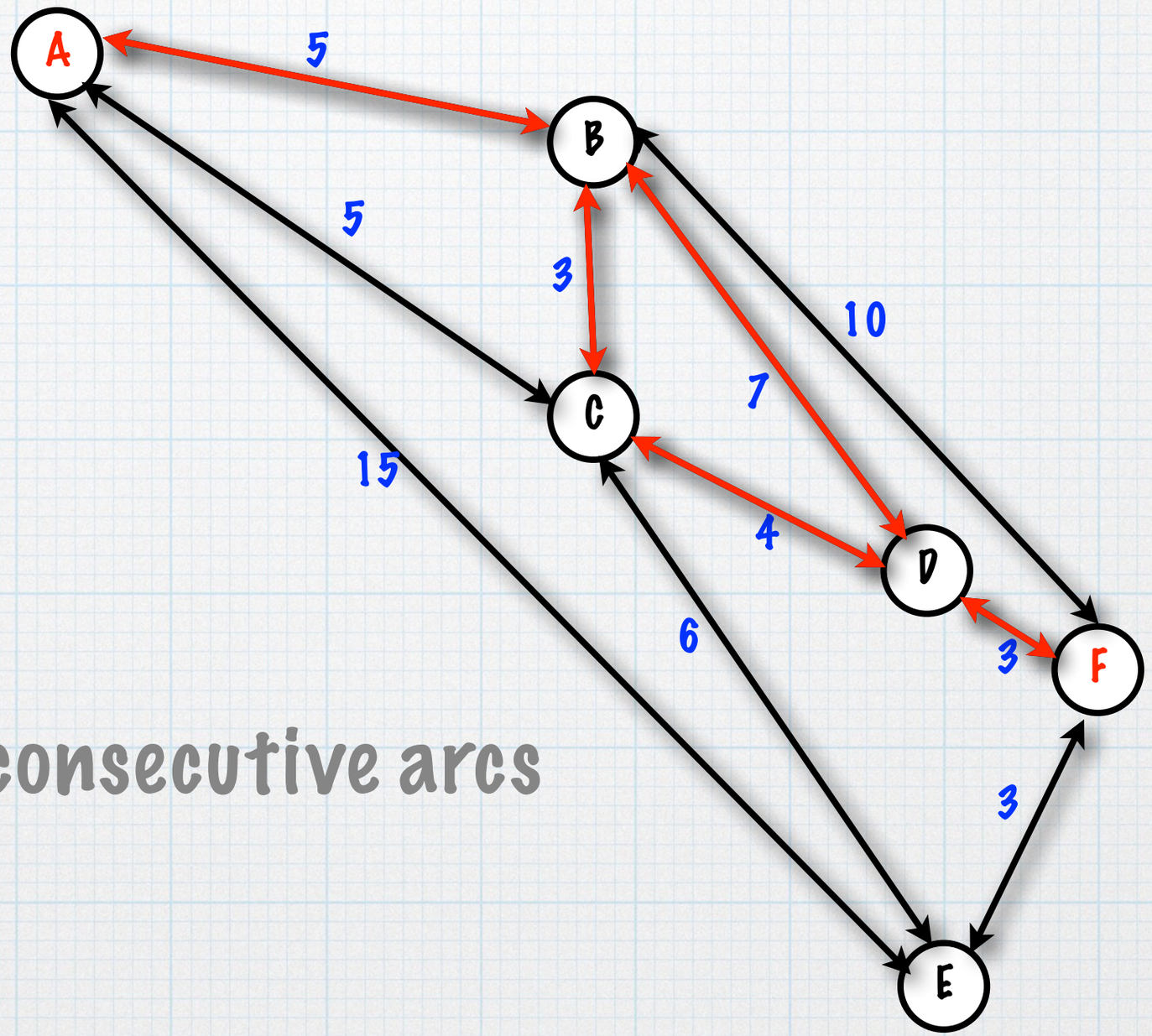
arc / edge
length



path: sequence of consecutive arcs

Graph





path: sequence of consecutive arcs

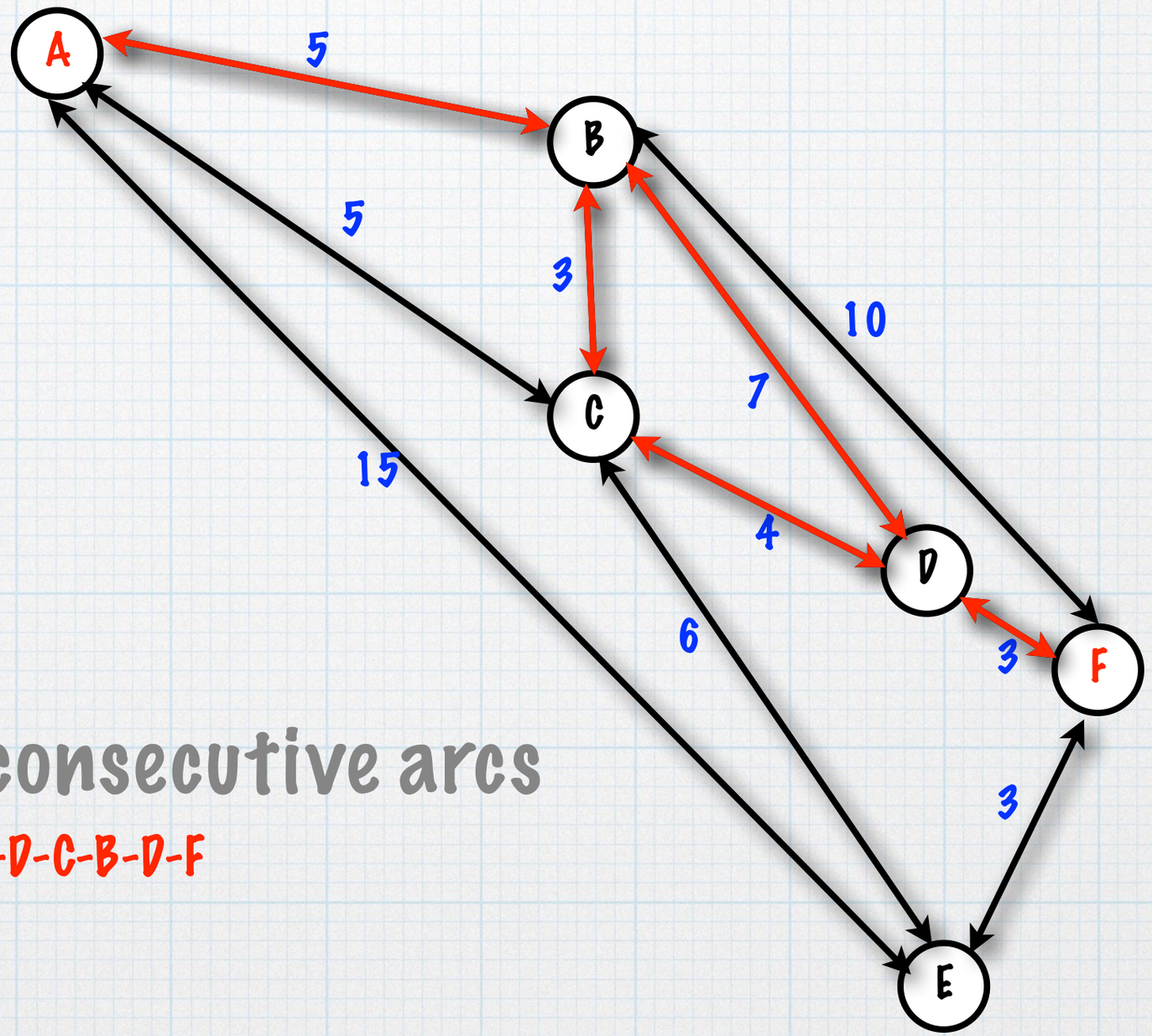
Graph

node / vertex

5

arc / edge

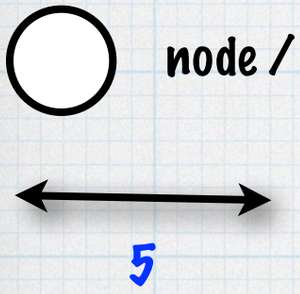
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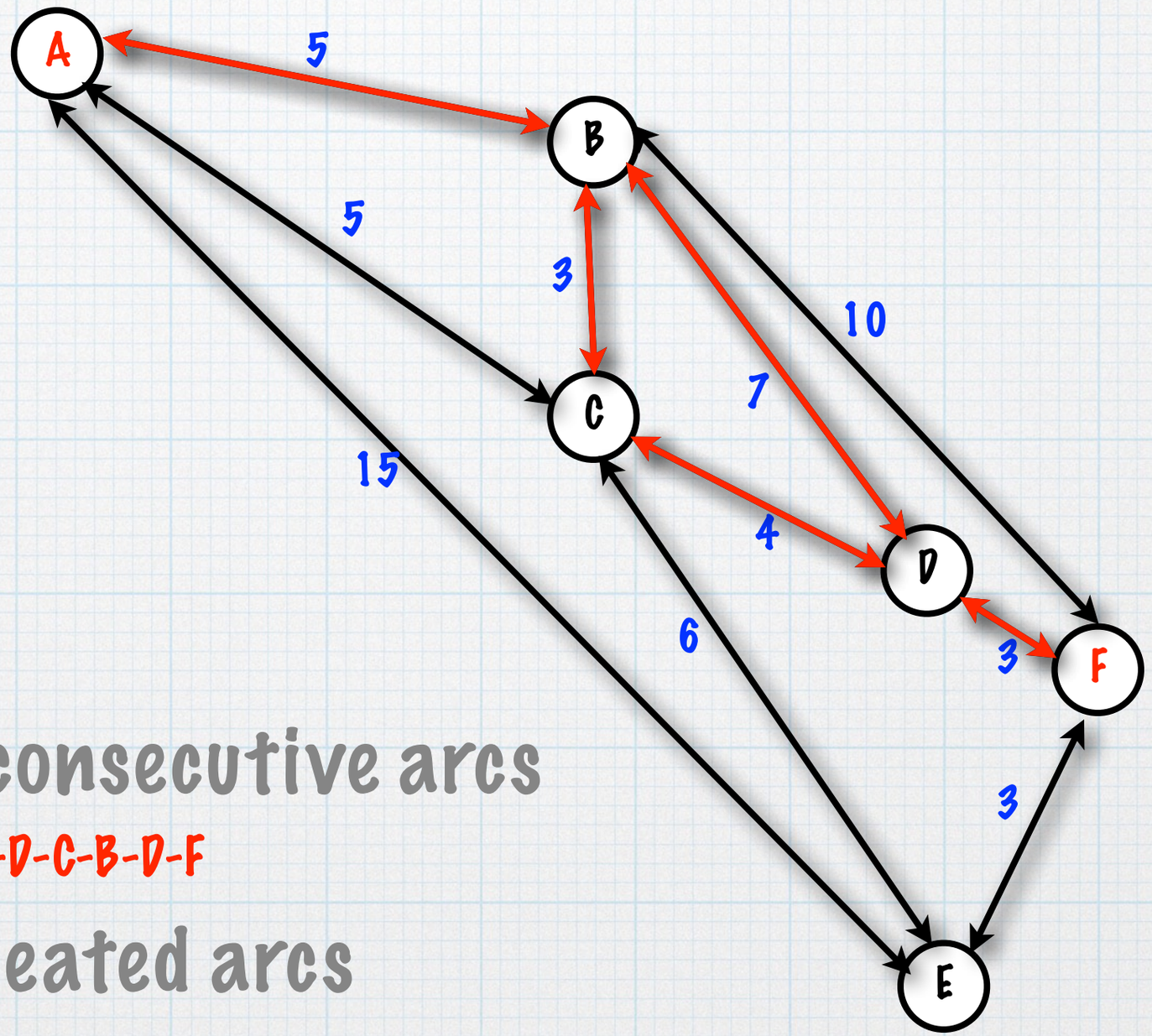


path: sequence of consecutive arcs

A-B-D-C-B-D-F

Graph





path: sequence of consecutive arcs

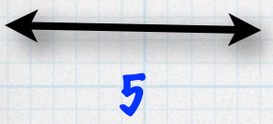
A-B-D-C-B-D-F

simple path: no repeated arcs

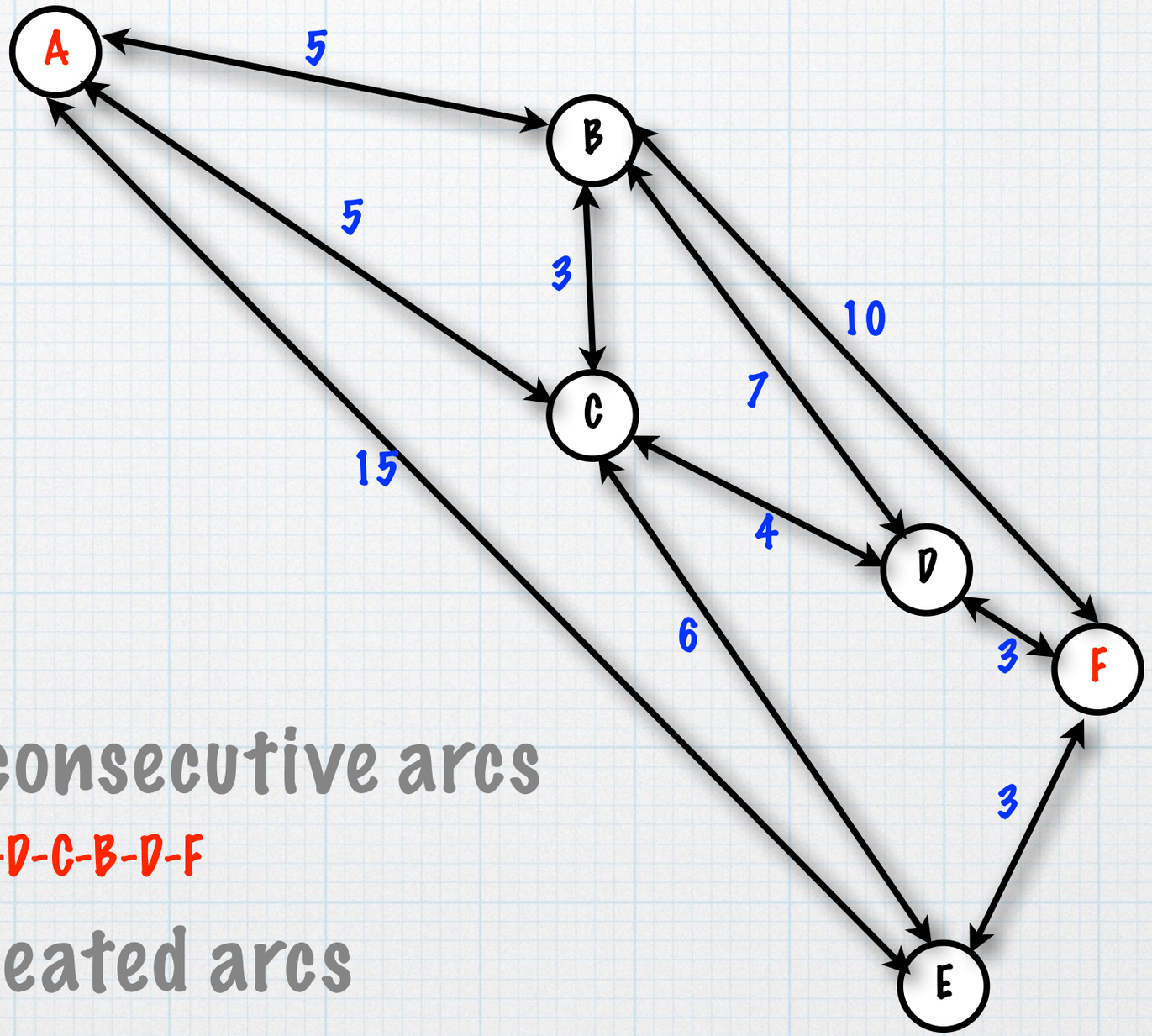
Graph



node / vertex



arc / edge
length



path: sequence of consecutive arcs

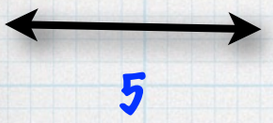
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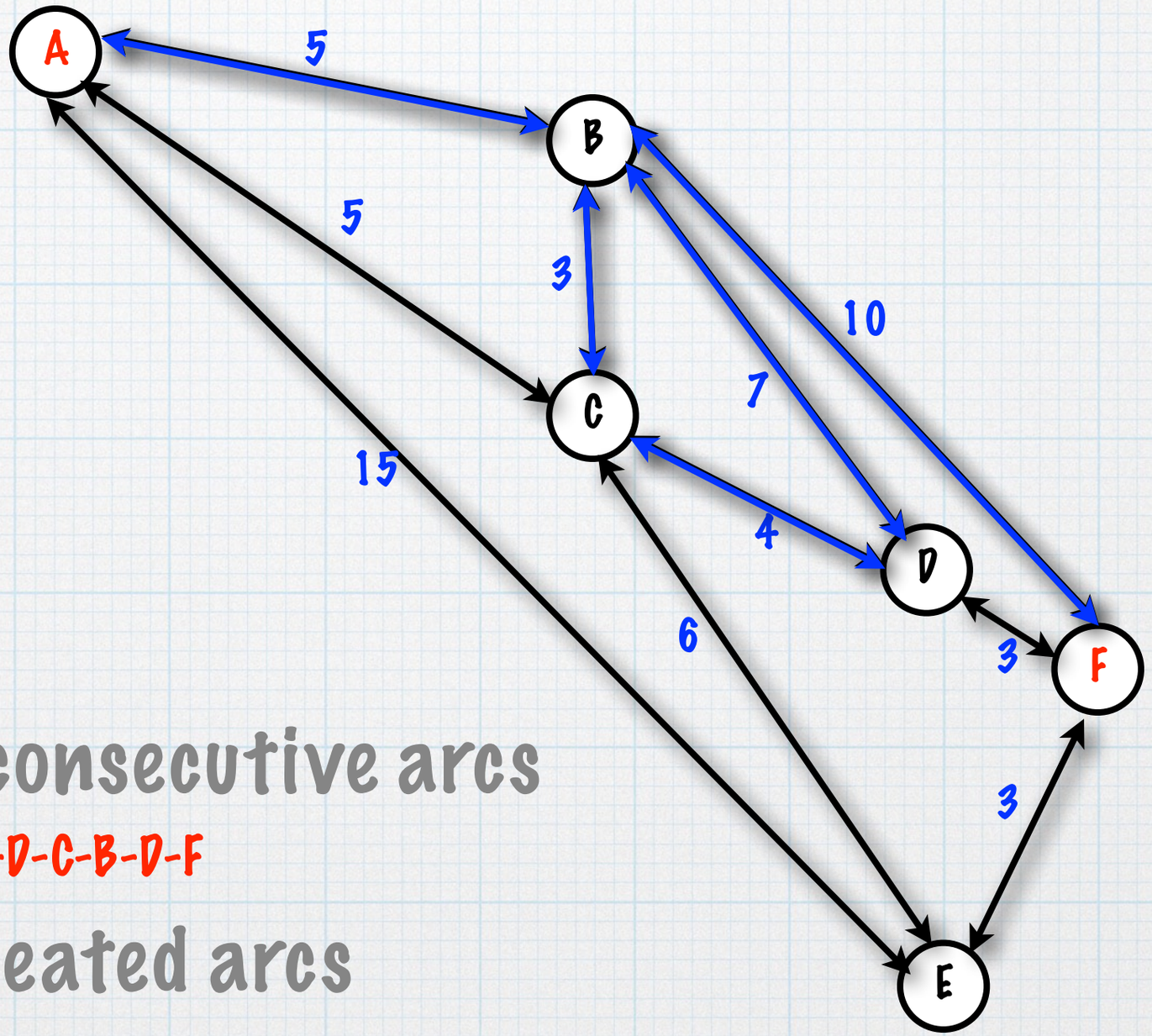
Graph



node / vertex



arc / edge
length



path: sequence of consecutive arcs

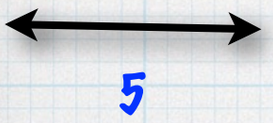
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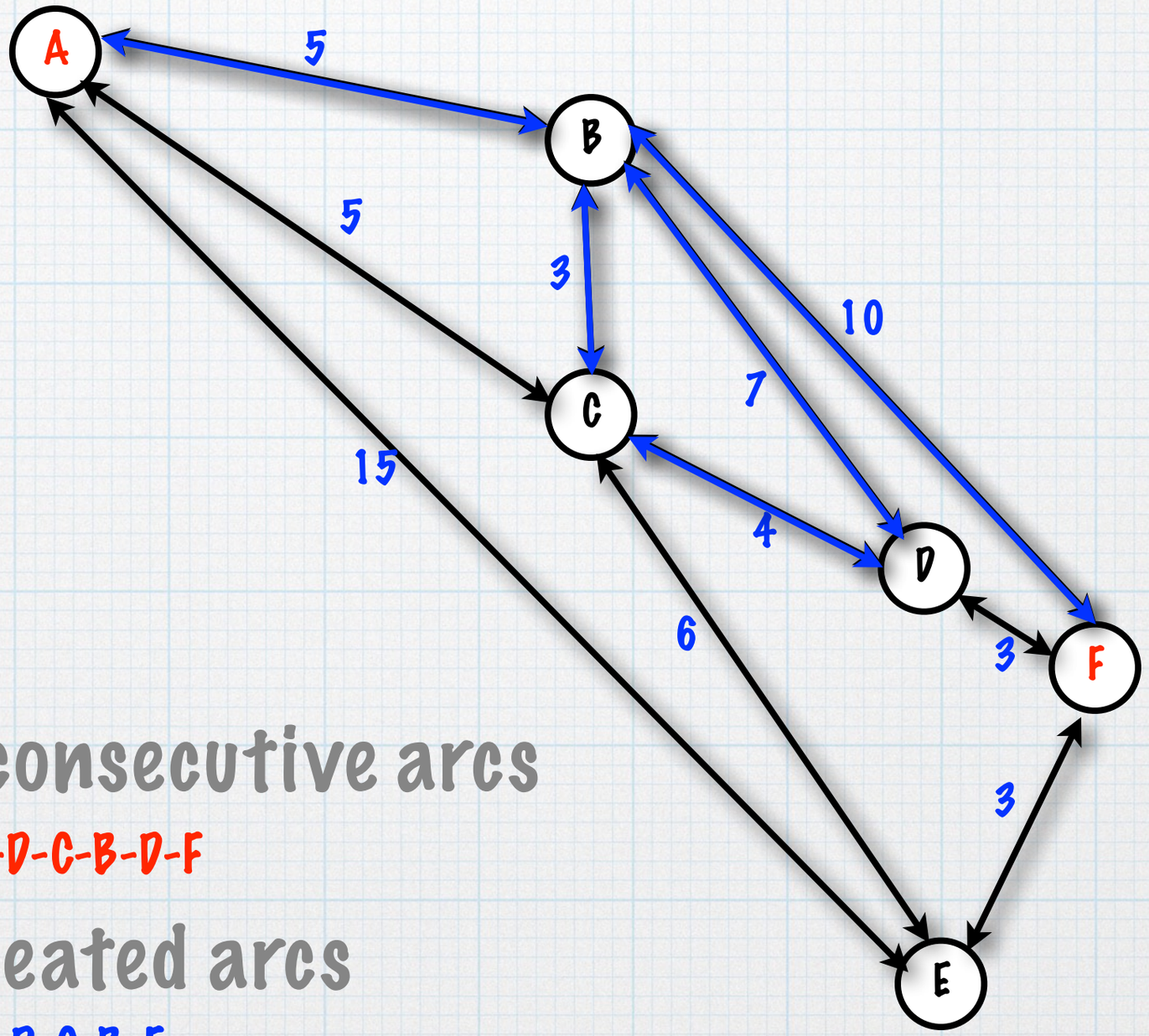
Graph



node / vertex



arc / edge
length



path: sequence of consecutive arcs

A-B-D-C-B-D-F

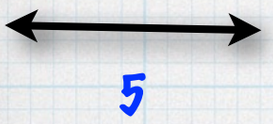
simple path: no repeated arcs

A-B-D-C-B-F

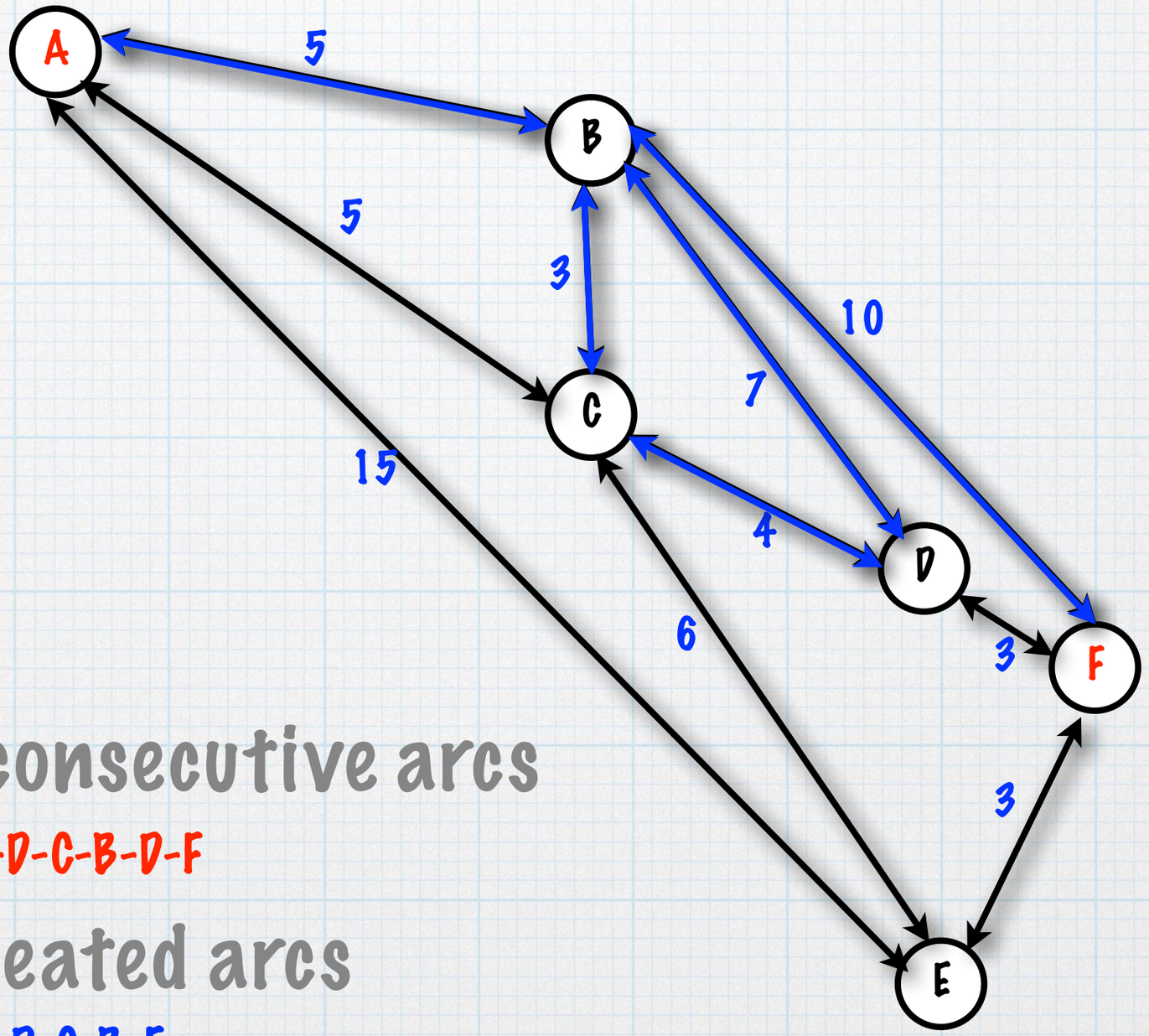
Graph



node / vertex



arc / edge
length



path: sequence of consecutive arcs

A-B-D-C-B-D-F

simple path: no repeated arcs

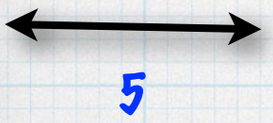
A-B-D-C-B-F

elementary path: no repeated nodes

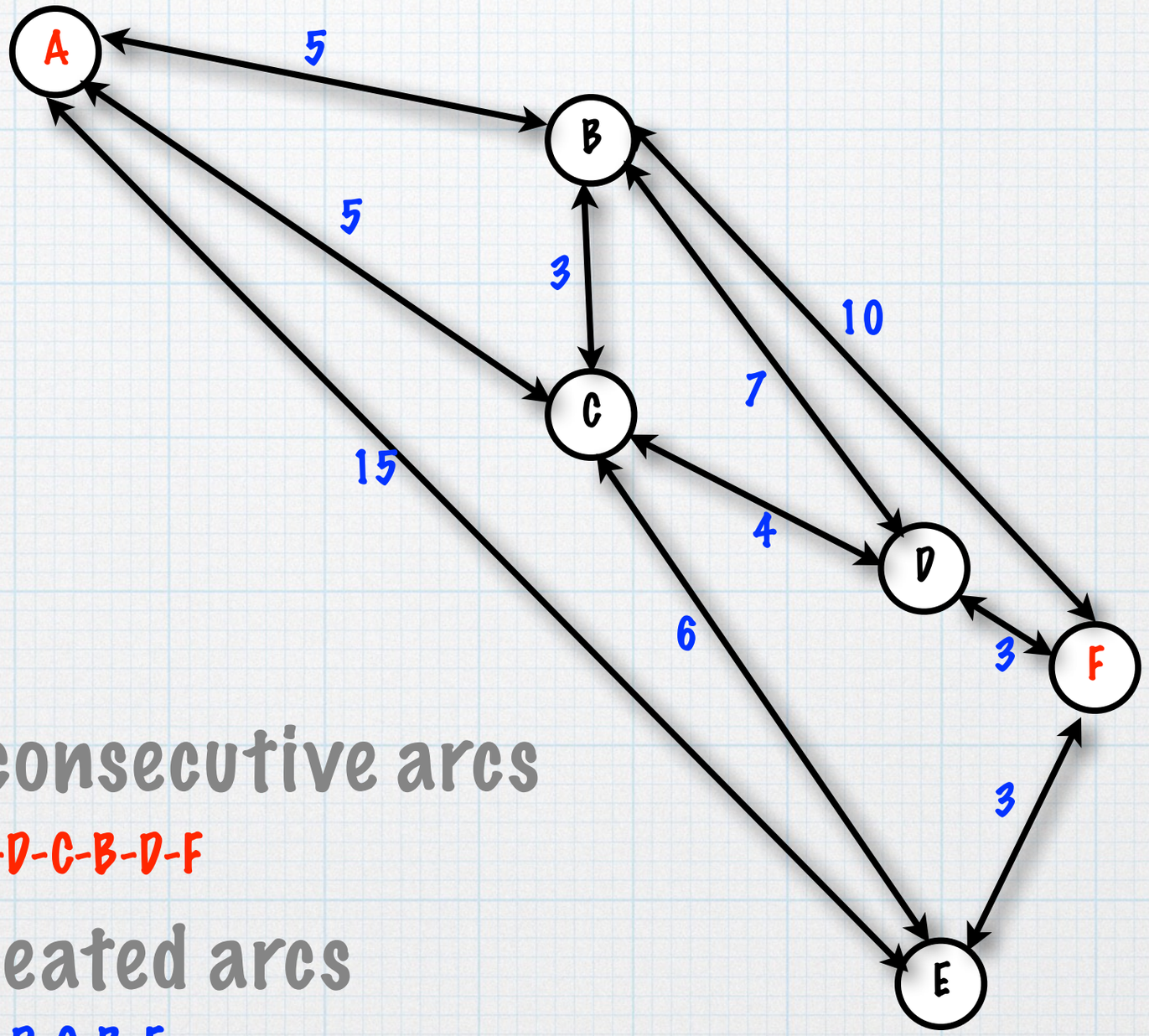
Graph



node / vertex



arc / edge
length



path: sequence of consecutive arcs

A-B-D-C-B-D-F

simple path: no repeated arcs

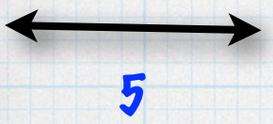
A-B-D-C-B-F

elementary path: no repeated nodes

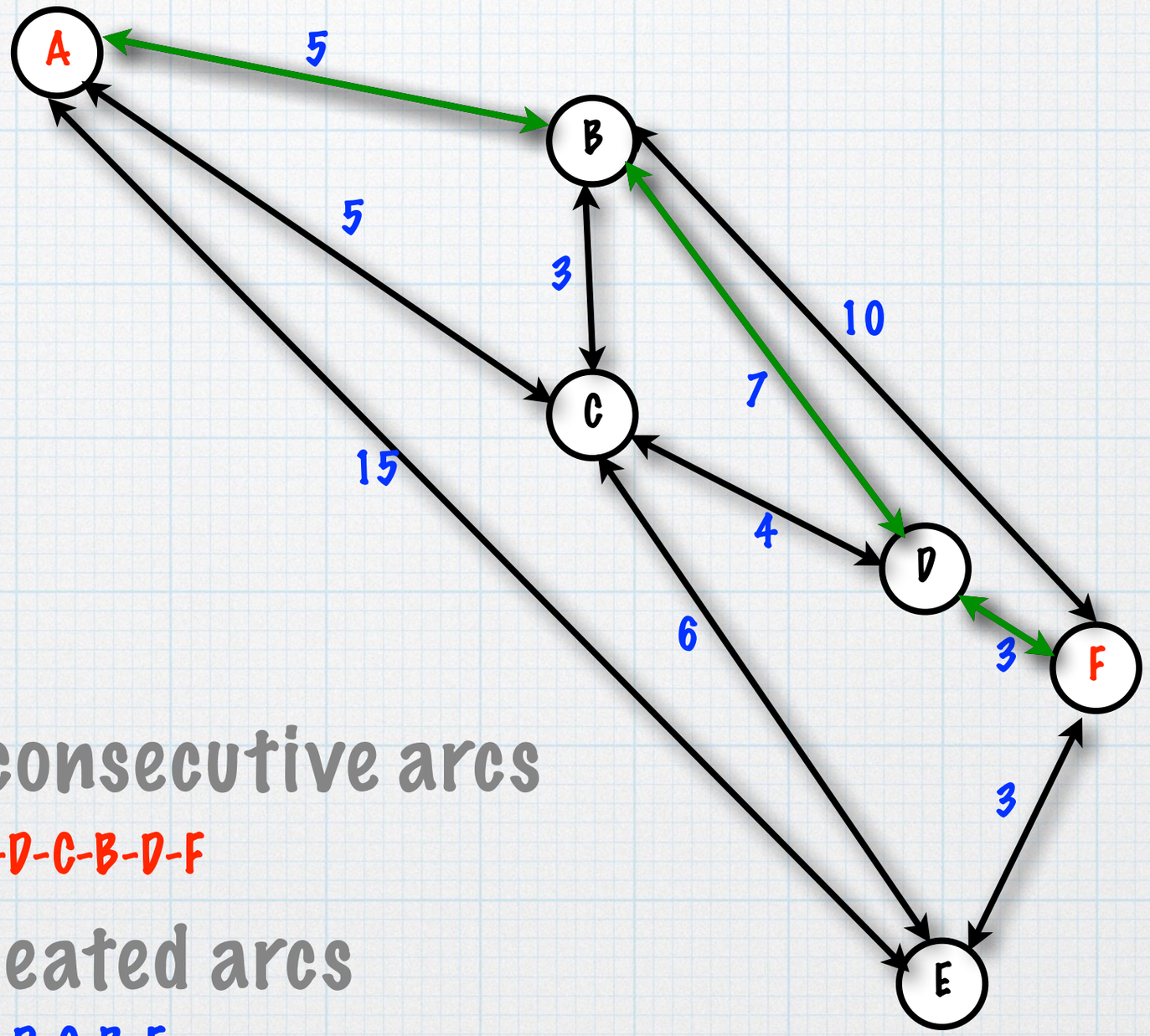
Graph



node / vertex



arc / edge
length



path: sequence of consecutive arcs

A-B-D-C-B-D-F

simple path: no repeated arcs

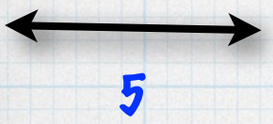
A-B-D-C-B-F

elementary path: no repeated nodes

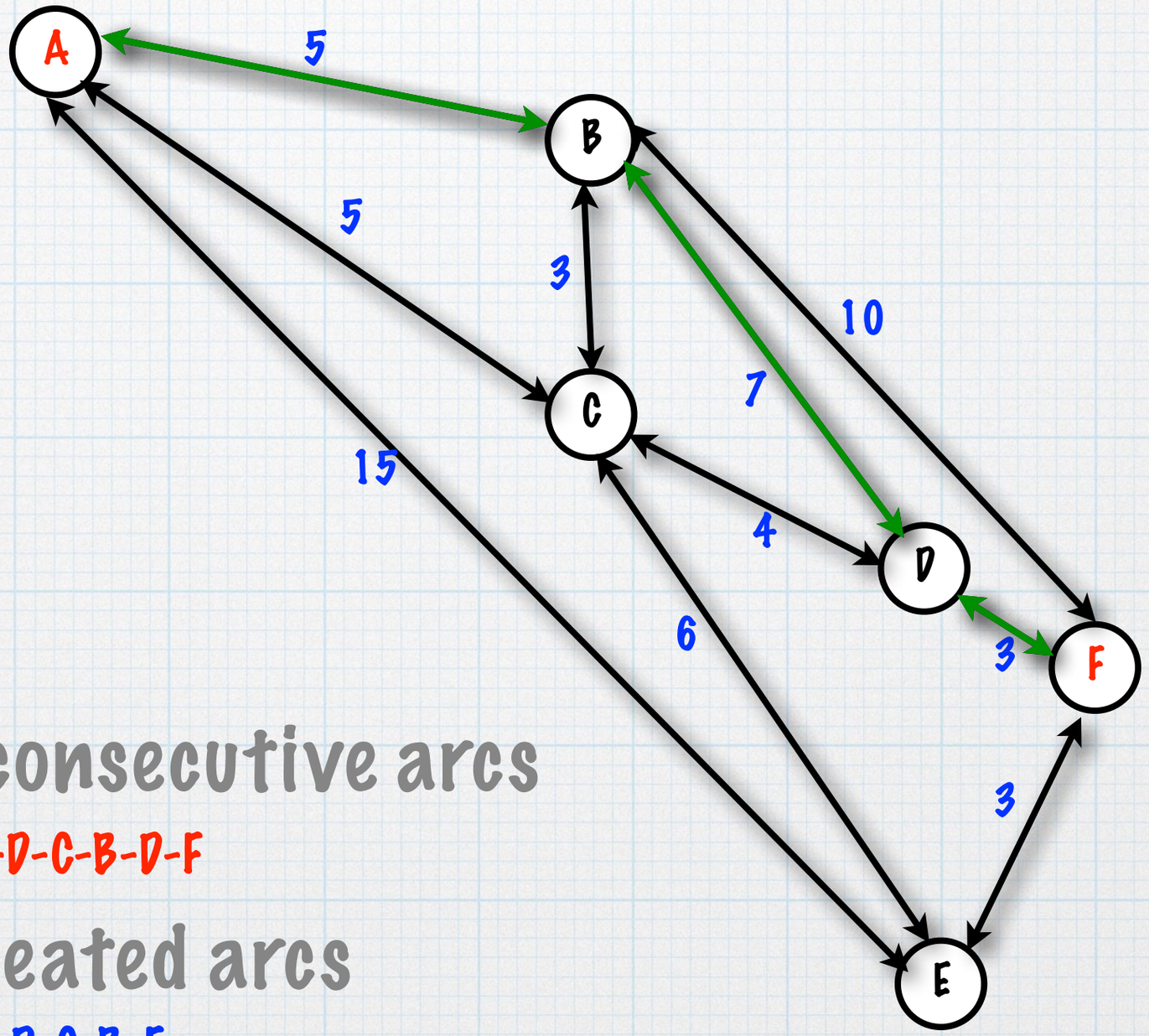
Graph



node / vertex



arc / edge
length



path: sequence of consecutive arcs

A-B-D-C-B-D-F

simple path: no repeated arcs

A-B-D-C-B-F

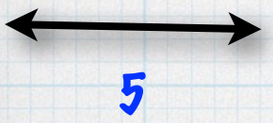
elementary path: no repeated nodes

A-B-D-F

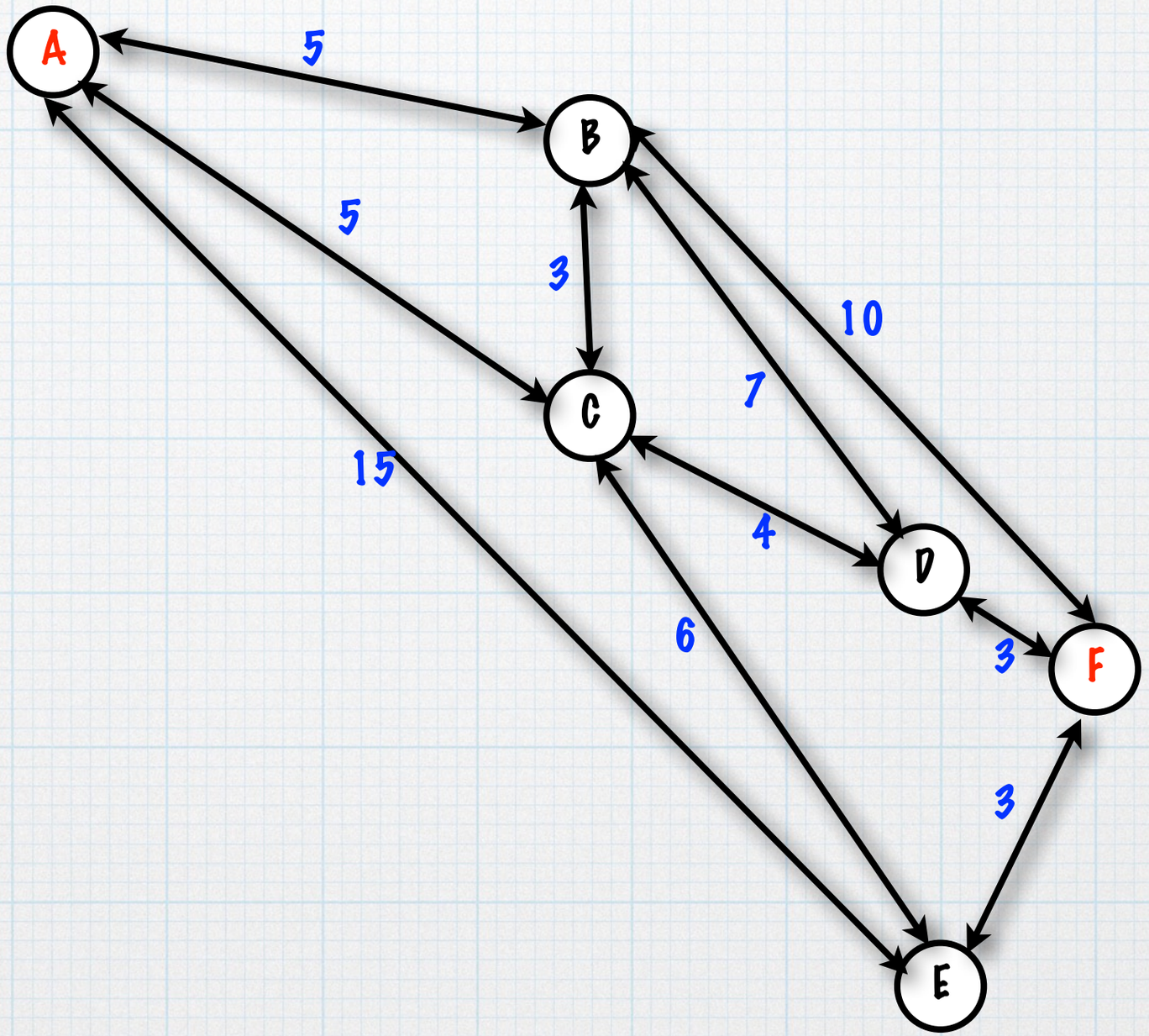
Graph



node / vertex



arc / edge
length



Graph

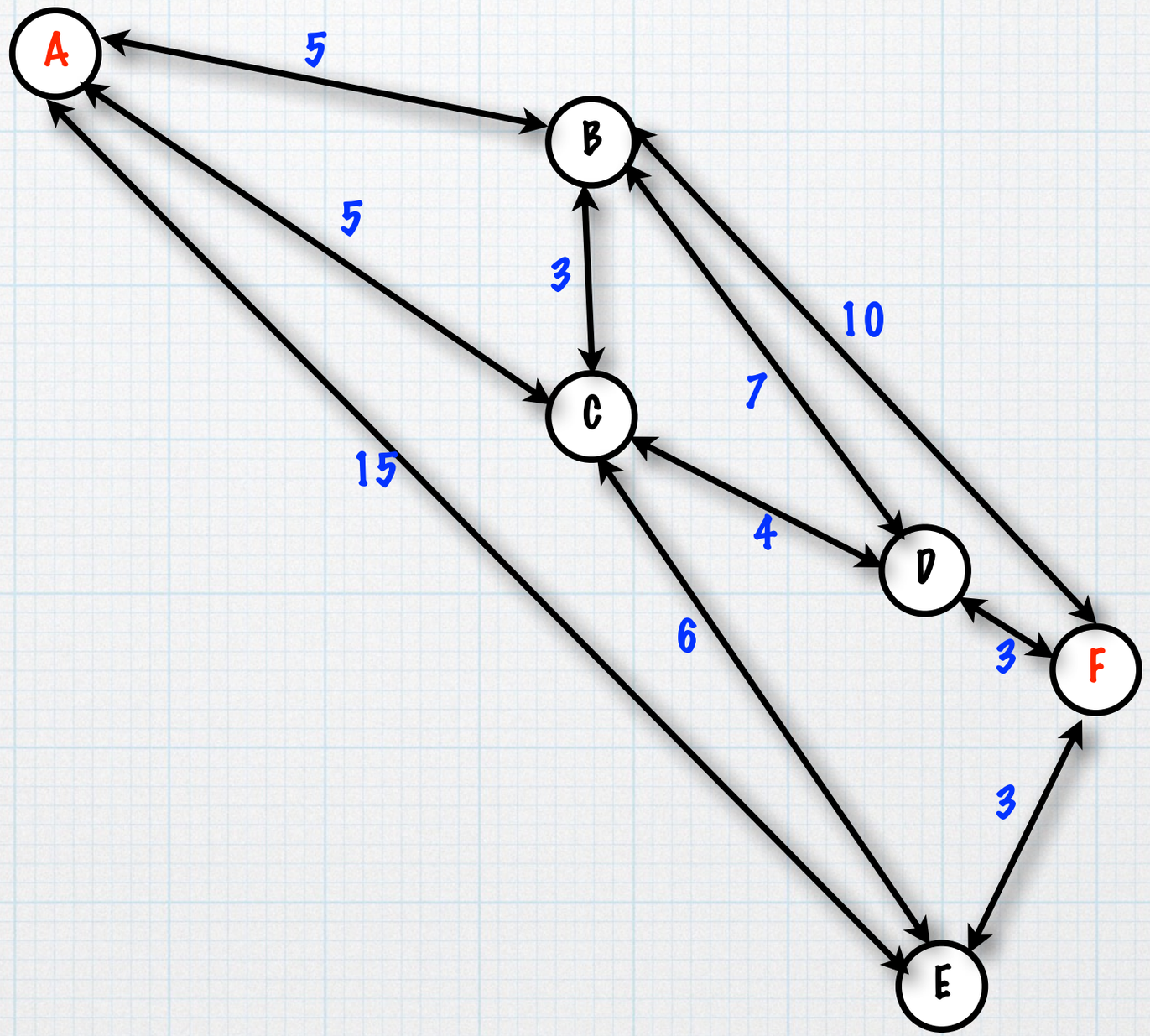
node / vertex

5

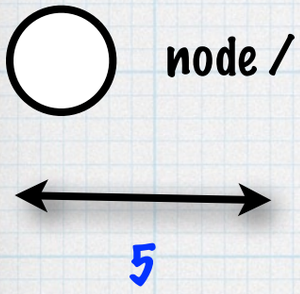
arc / edge

length

cycle: closed path



Graph

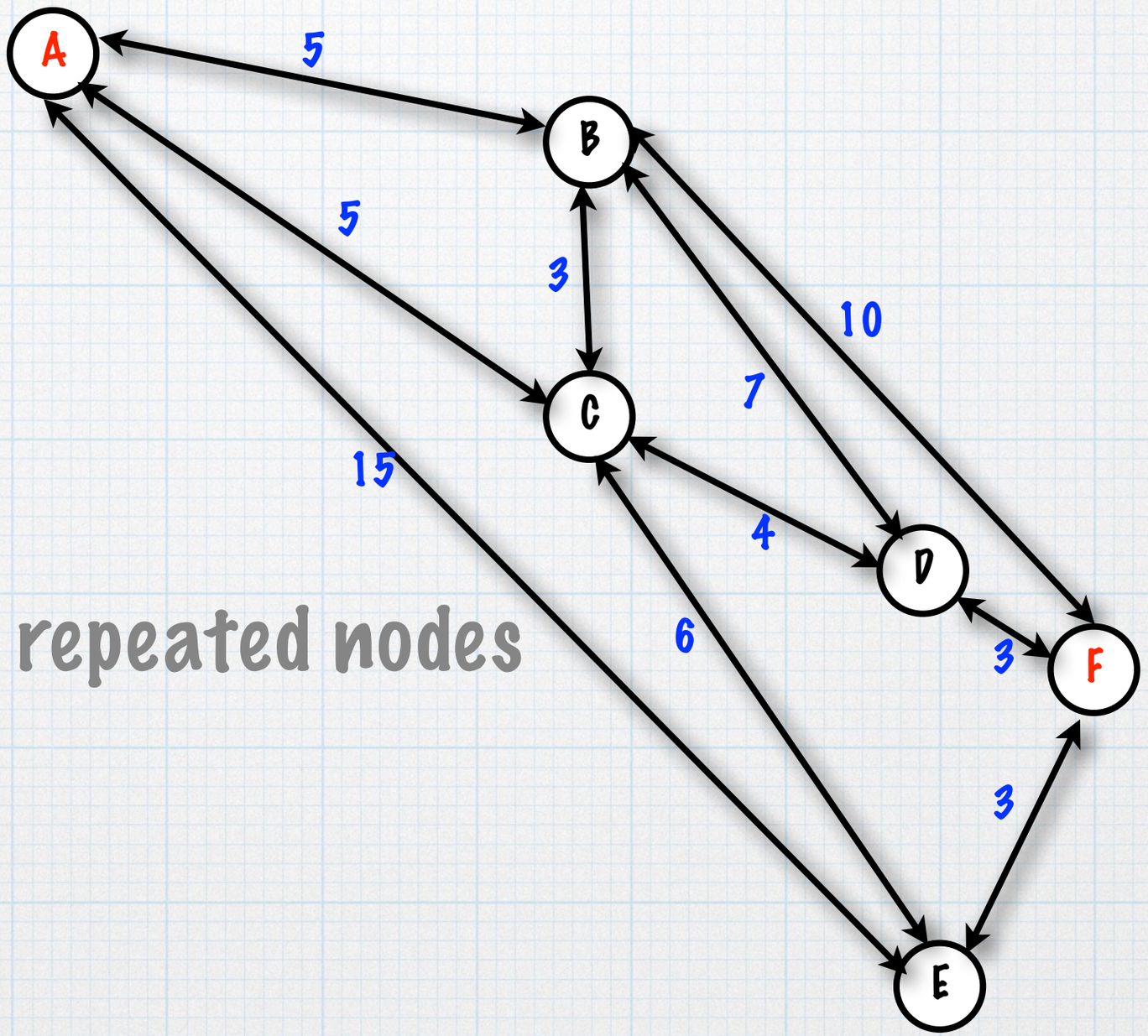


node / vertex

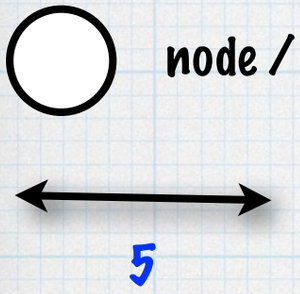
arc / edge
length

cycle: closed path

elementary cycle: no repeated nodes



Graph



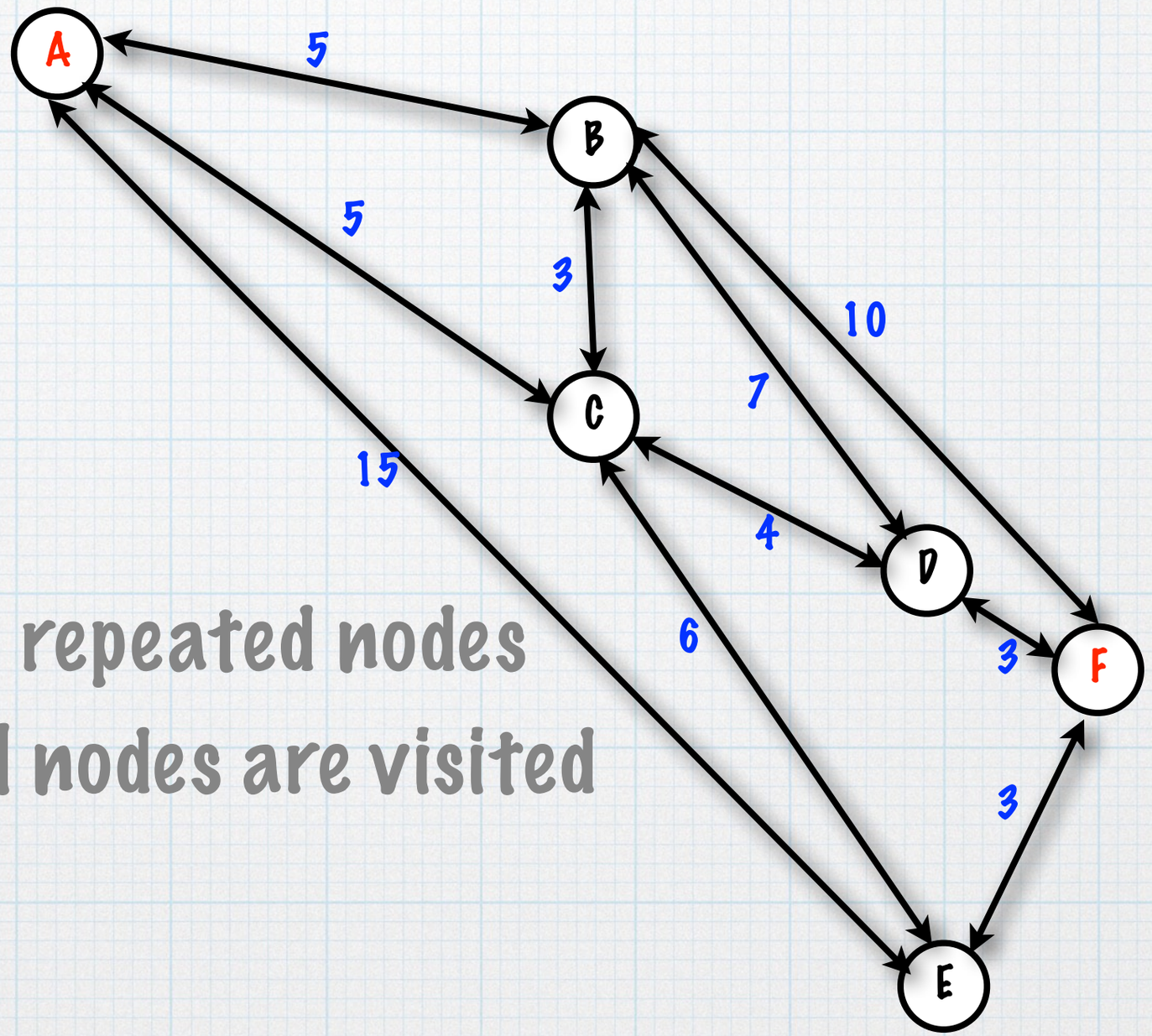
node / vertex

arc / edge
length

cycle: closed path

elementary cycle: no repeated nodes

Hamiltonian cycle: all nodes are visited



Graph

node / vertex

arc / edge

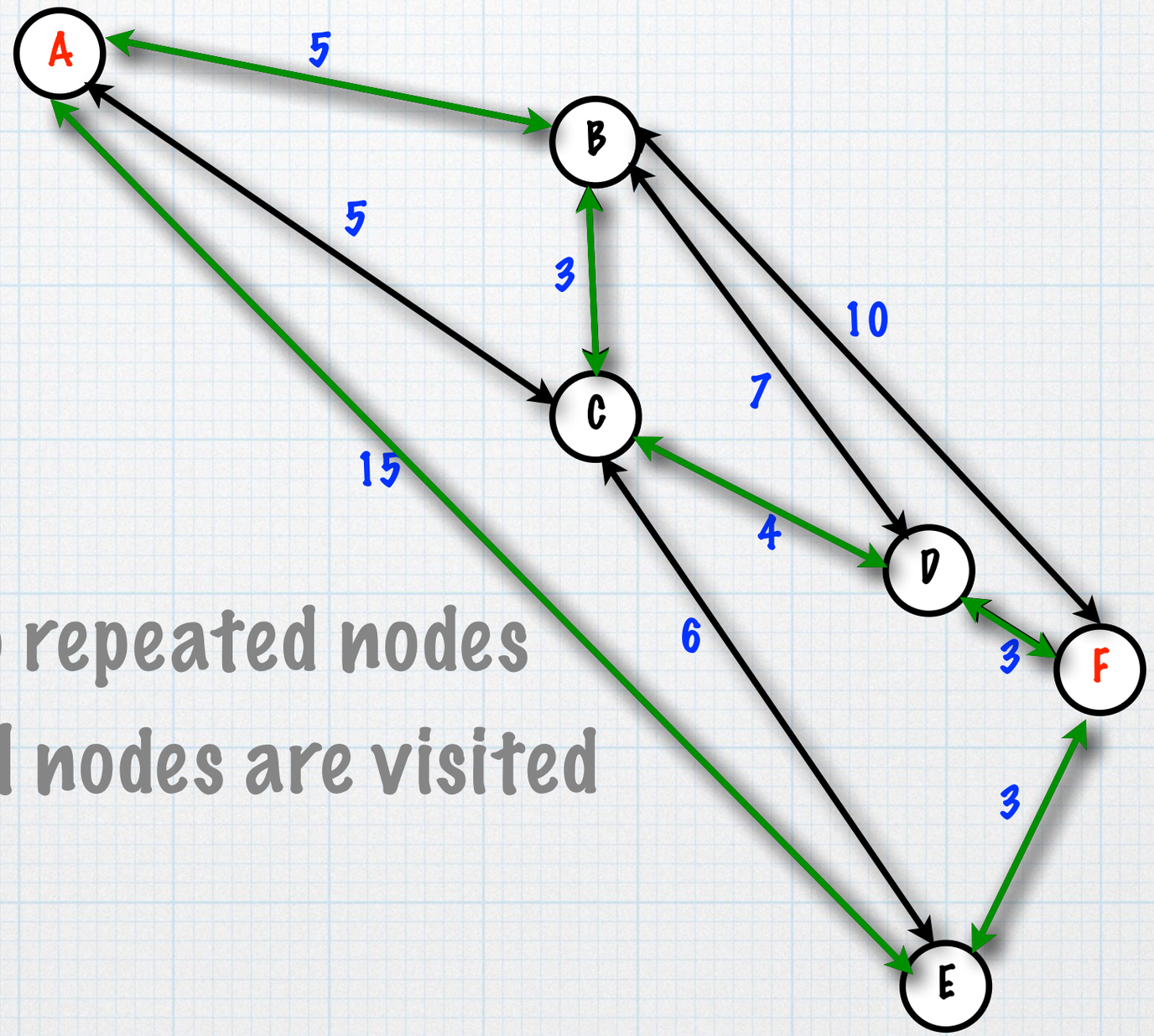
5

length

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Graph

node / vertex

arc / edge

5

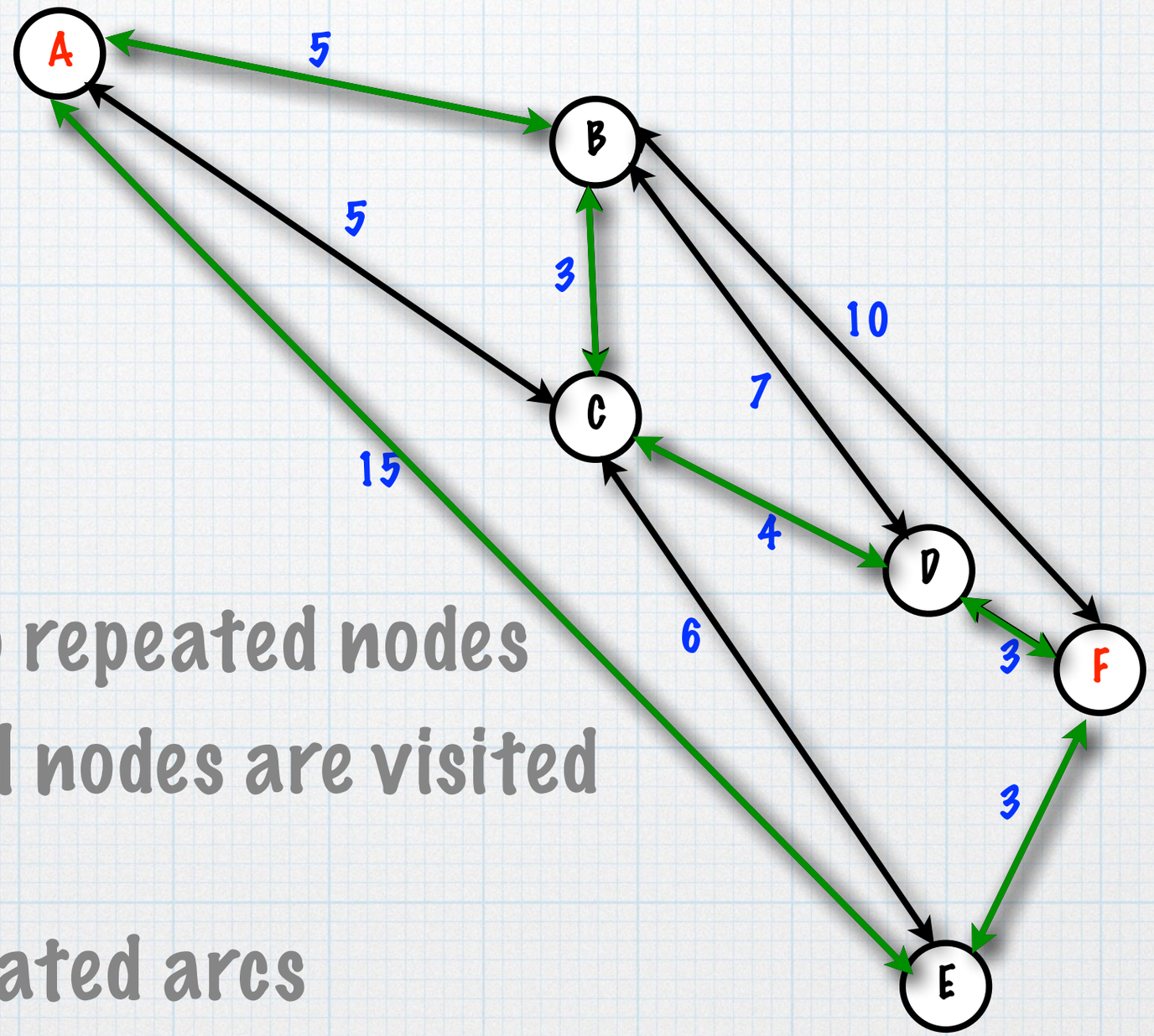
length

cycle: closed path

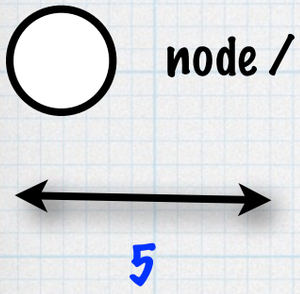
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simple cycle: no repeated arcs



Graph



node / vertex

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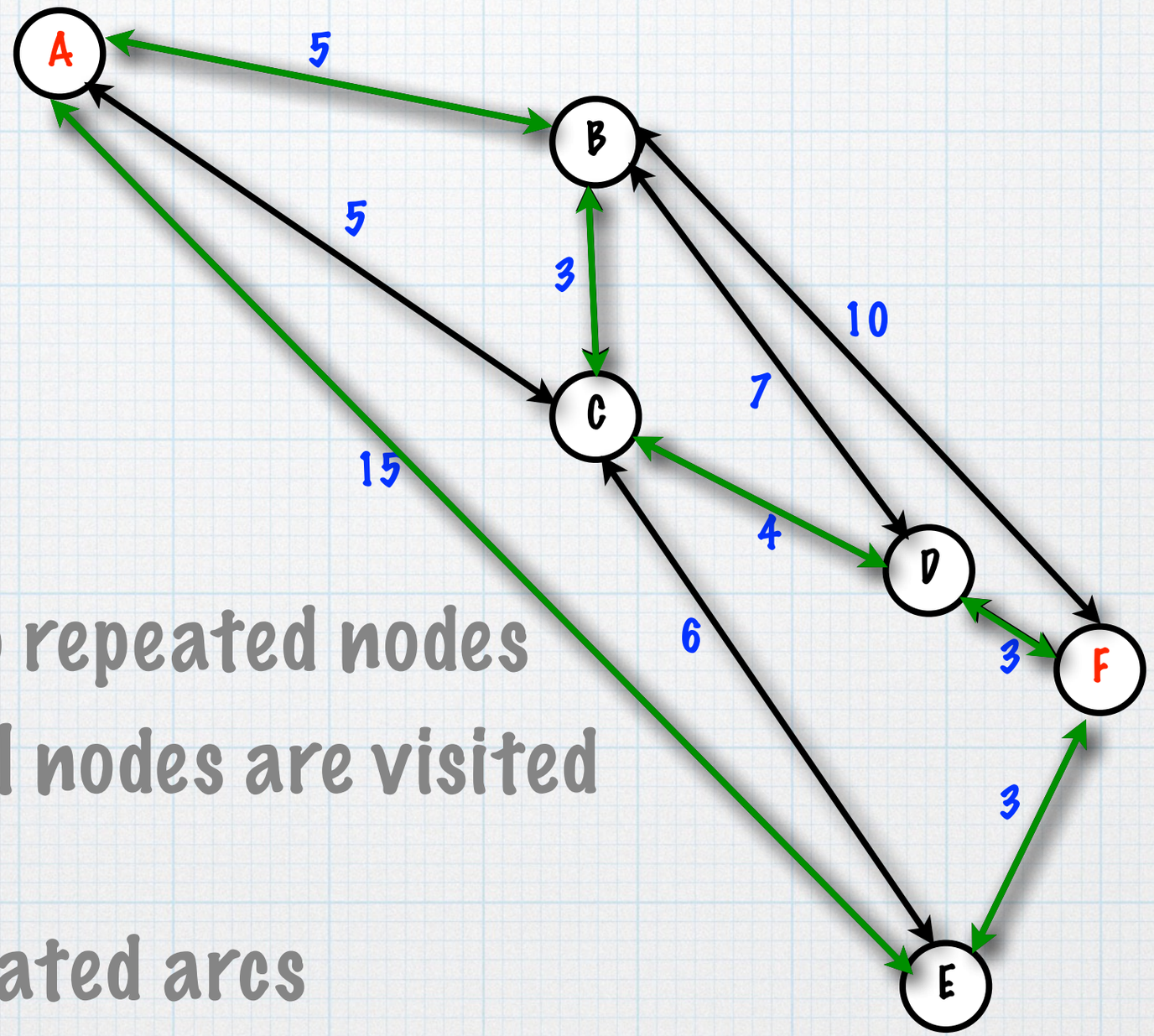
cycle: closed path

elementary cycle: no repeated nodes

Hamiltonian cycle: all nodes are visited

simple cycle: no repeated arcs

Eulerian tour: all arcs are visited once



Undirected vs. directed graphs

Undirected vs. directed graphs

$$G = (N, A)$$

N Set of nodes

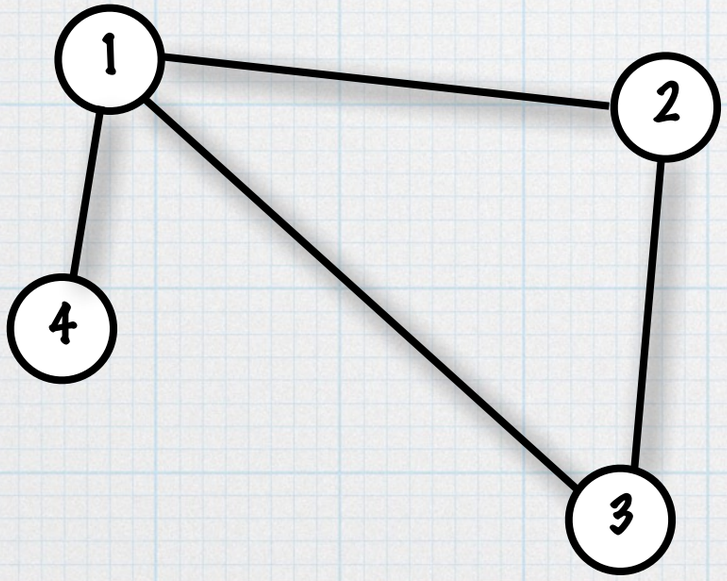
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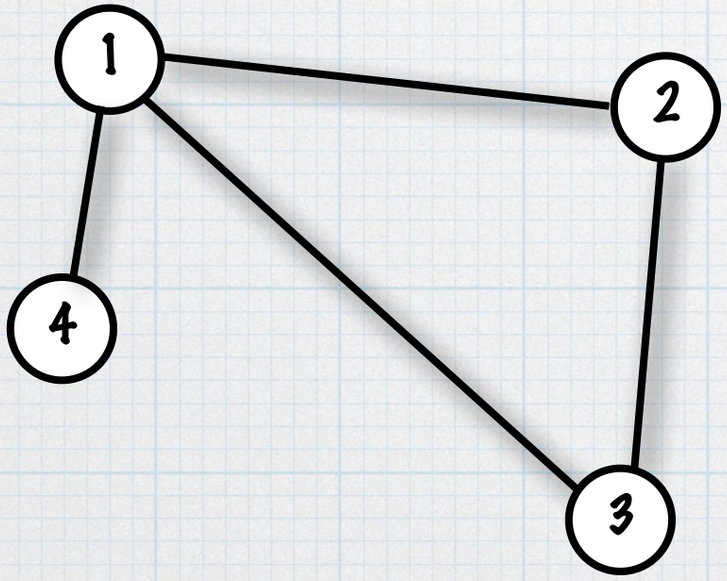
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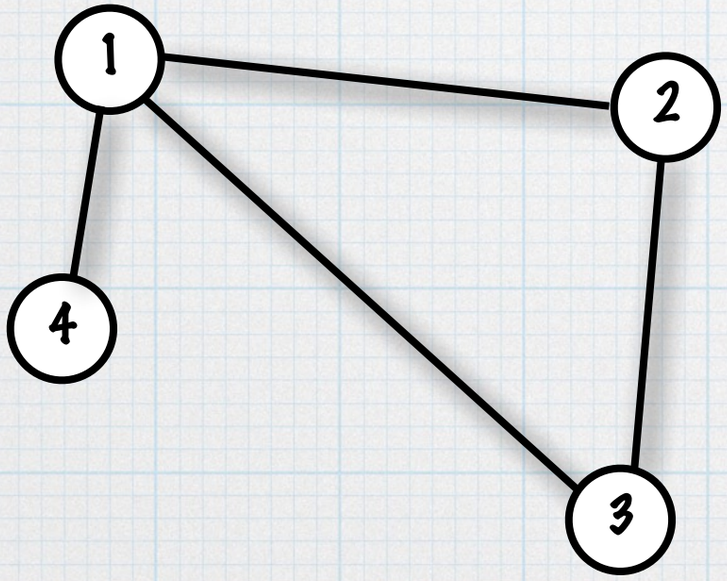
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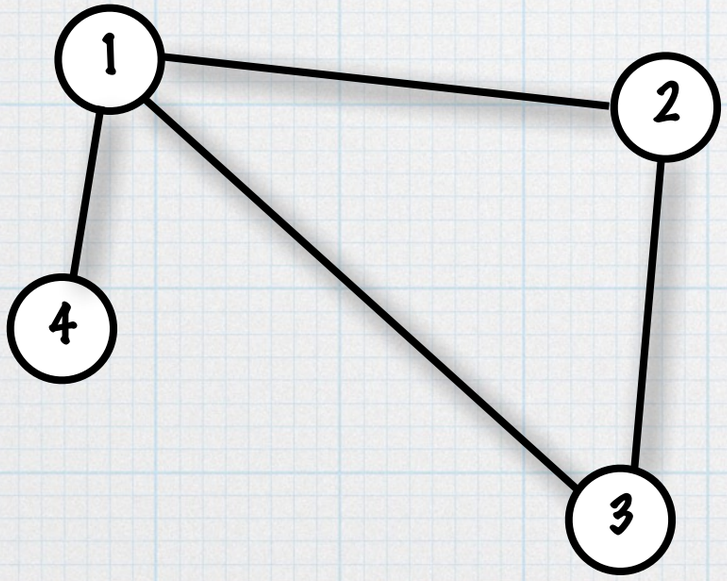
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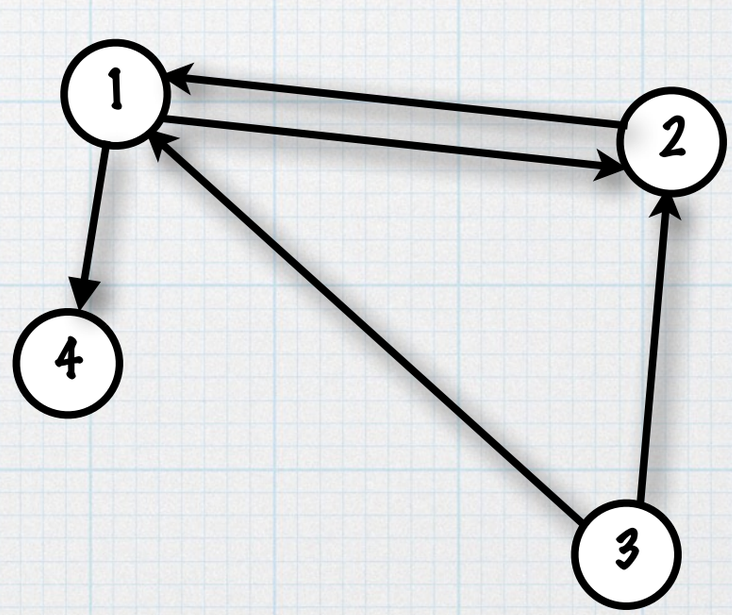
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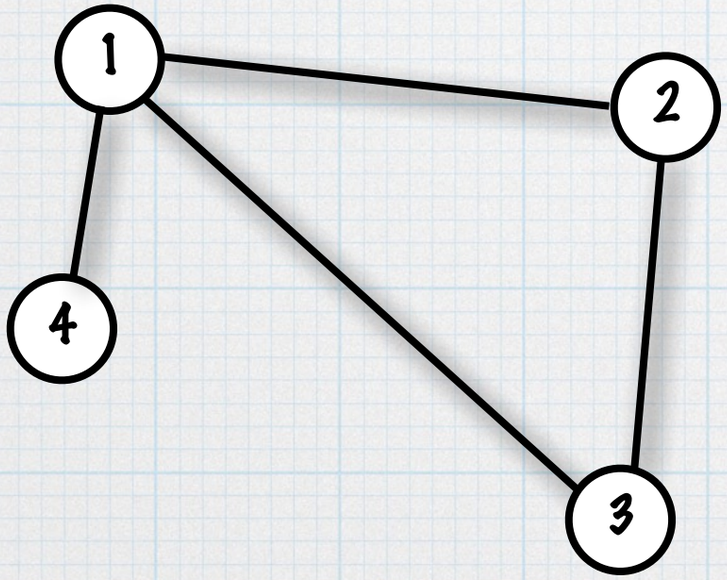
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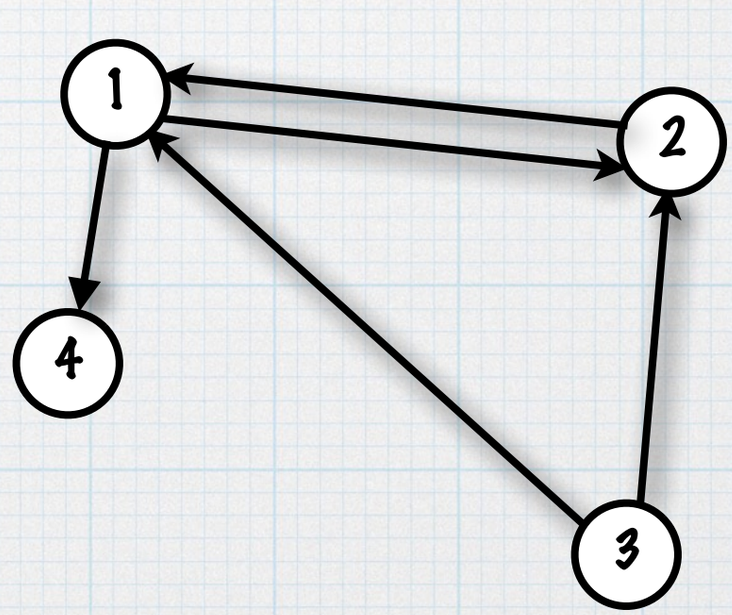
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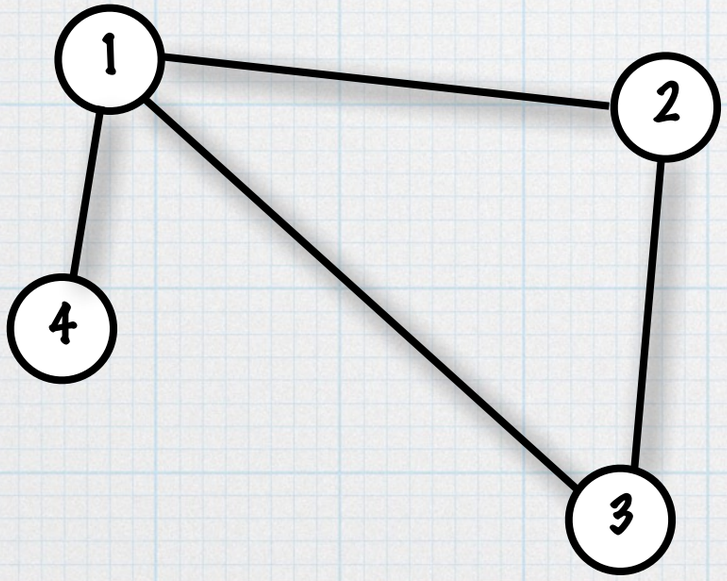
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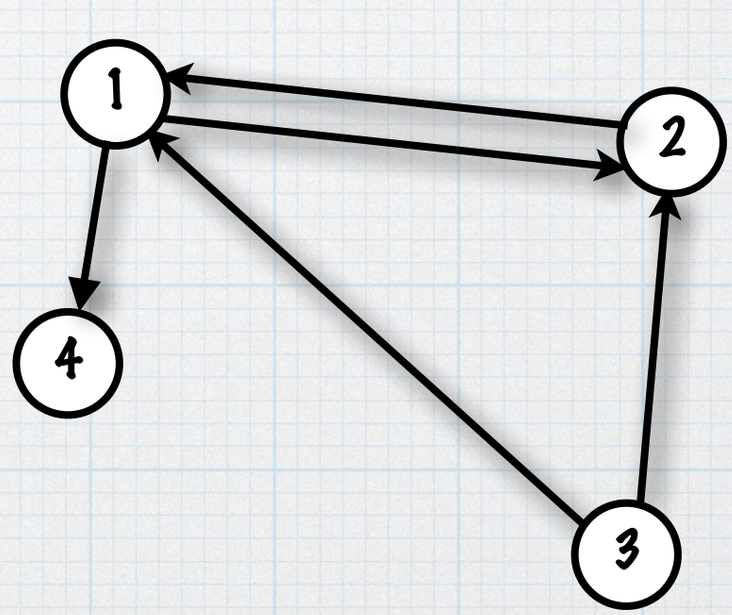
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$$\{i, j\} = \{j, i\}$$

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$$(i, j) \neq (j, i)$$



Notation

$$G = (N, A)$$

N Set of nodes

A Set of arcs (pairs of nodes)

For the remaining notation read

Section 2

Formative assessment activity next time

Rebuild the bridges at minimum cost

Every island must be reachable from any other one



Try to find the optimal solution

Groups of 3-4 persons

Elaborate a strategy

5 minutes

Example: TLC network design

We want to design the Intranet network of Politecnico

Example: TLC network design

We want to design the Intranet network of Politecnico

Politecnico's centers

LC

CO

MiB

CR

MN

MiL

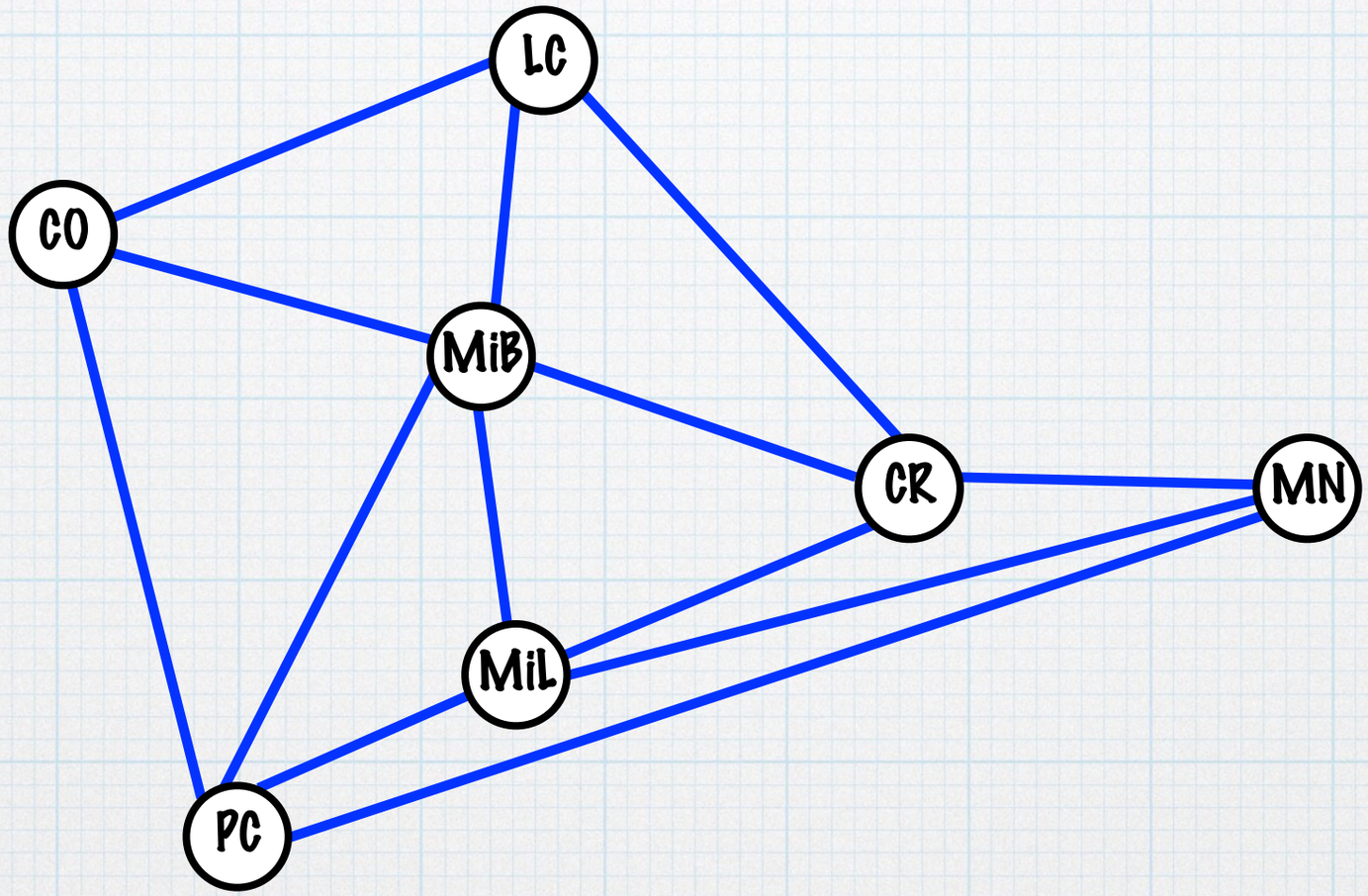
PC

Example: TLC network design

We want to design the Intranet network of Politecnico

Politecnico's centers

Potential links



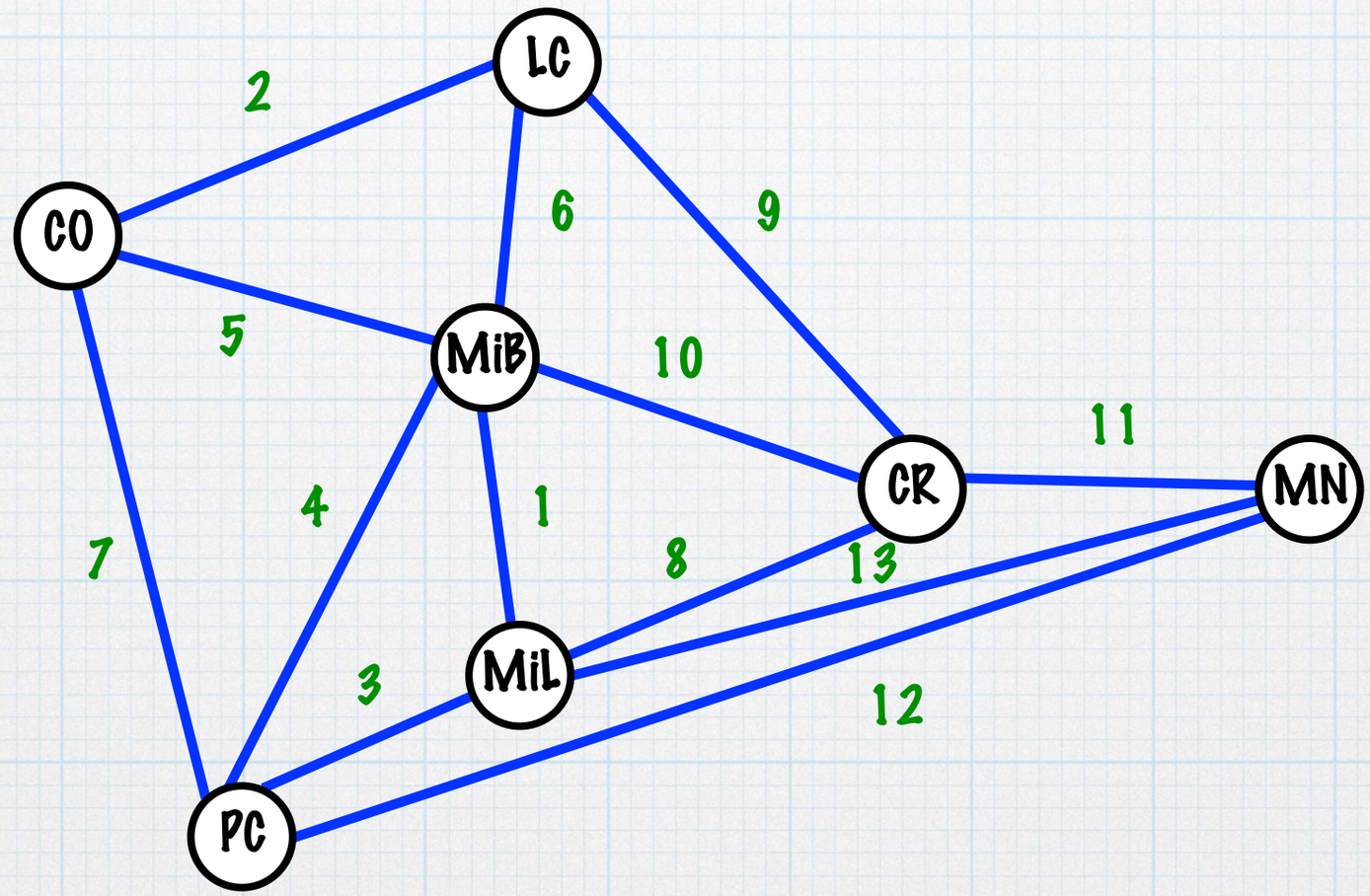
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Installation costs

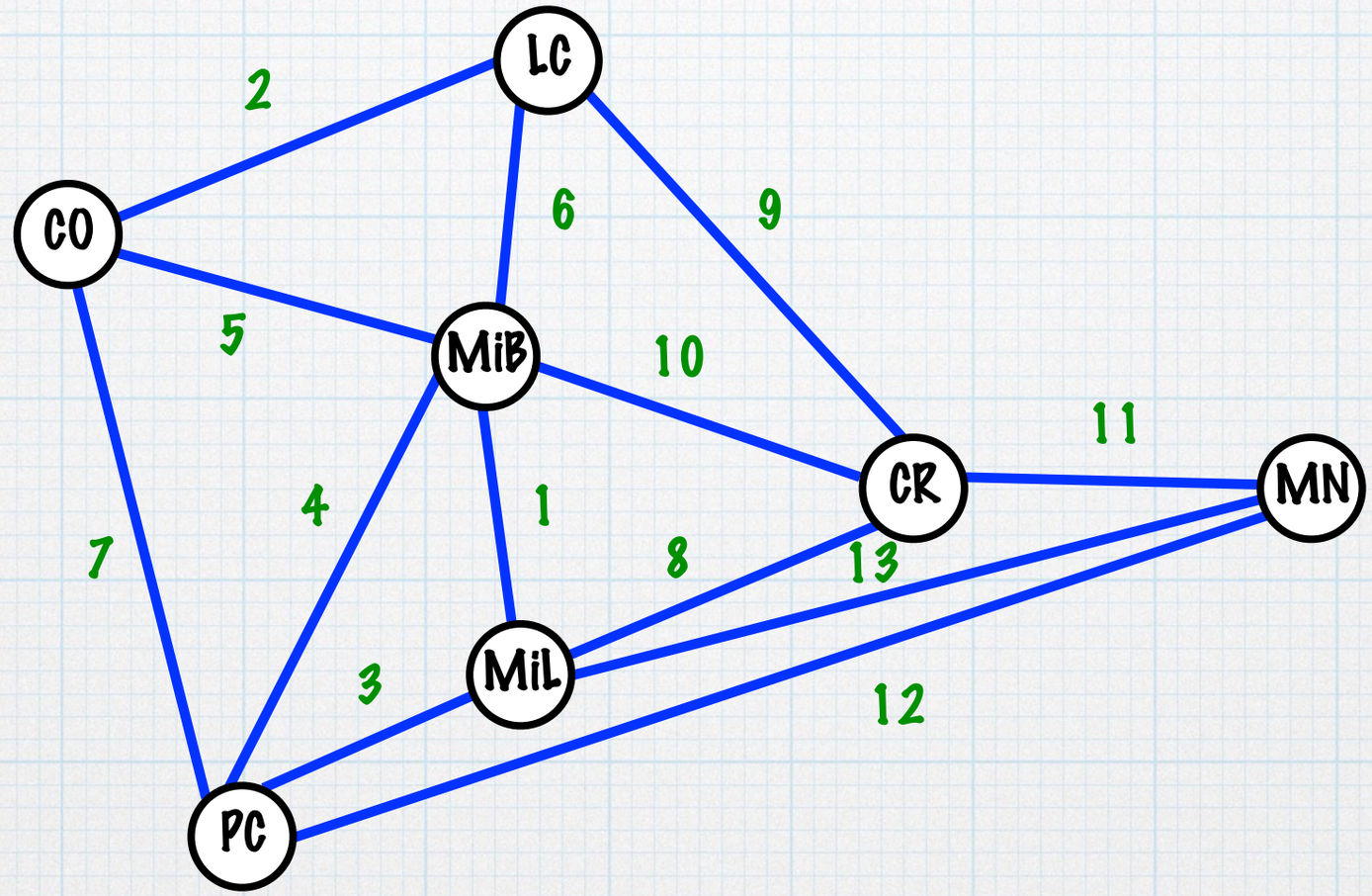


Example: TLC network design

We want to design the Intranet network of Politecnico

Politecnico's centers

Potential links



Installation costs

Problem: select a subset of arcs so that each node is **connected** to any other node by at least one path, minimizing the installation cost

Abstraction of the problem

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Given an undirected graph $G = (N, A)$

Abstraction of the problem

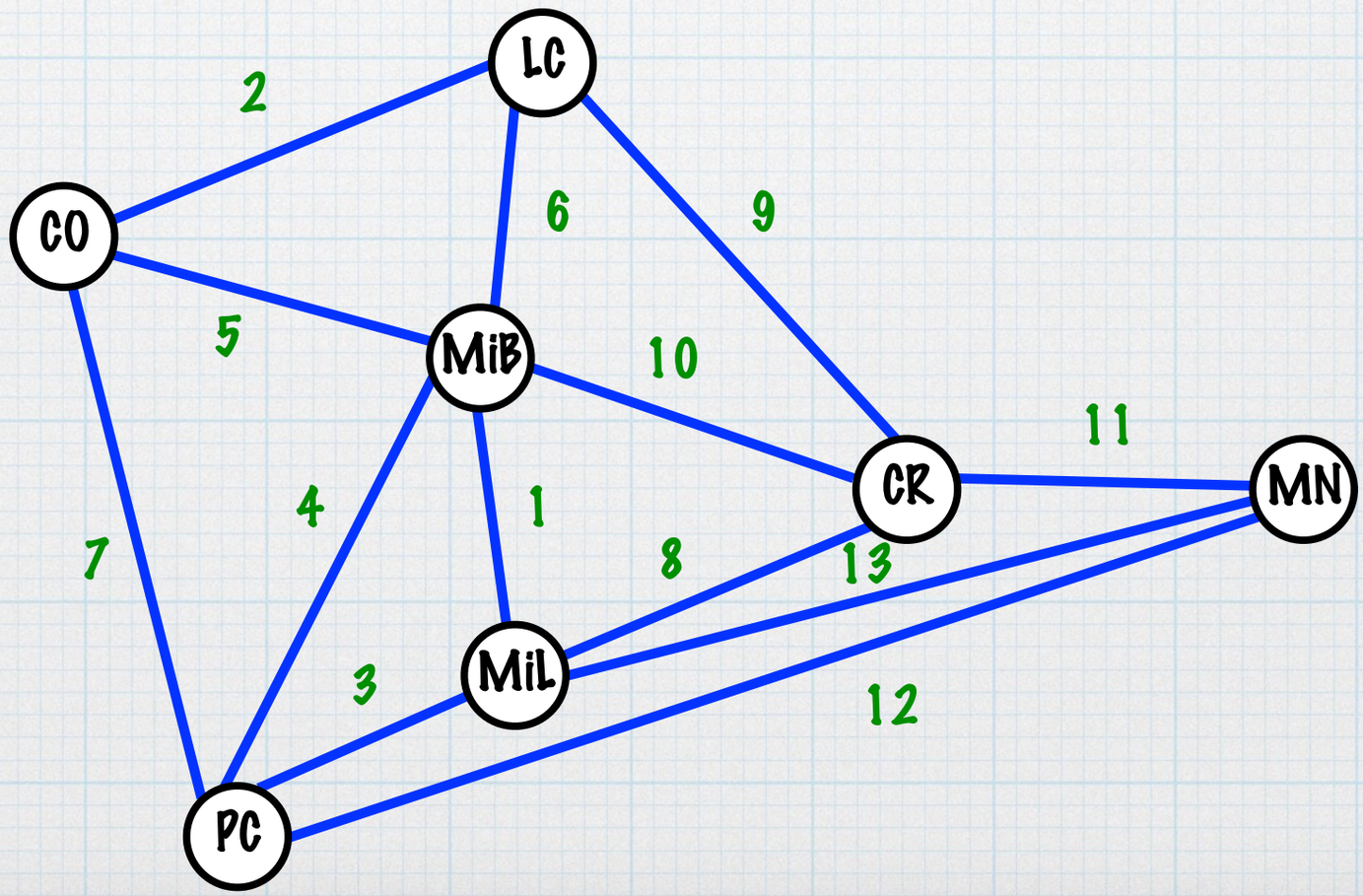
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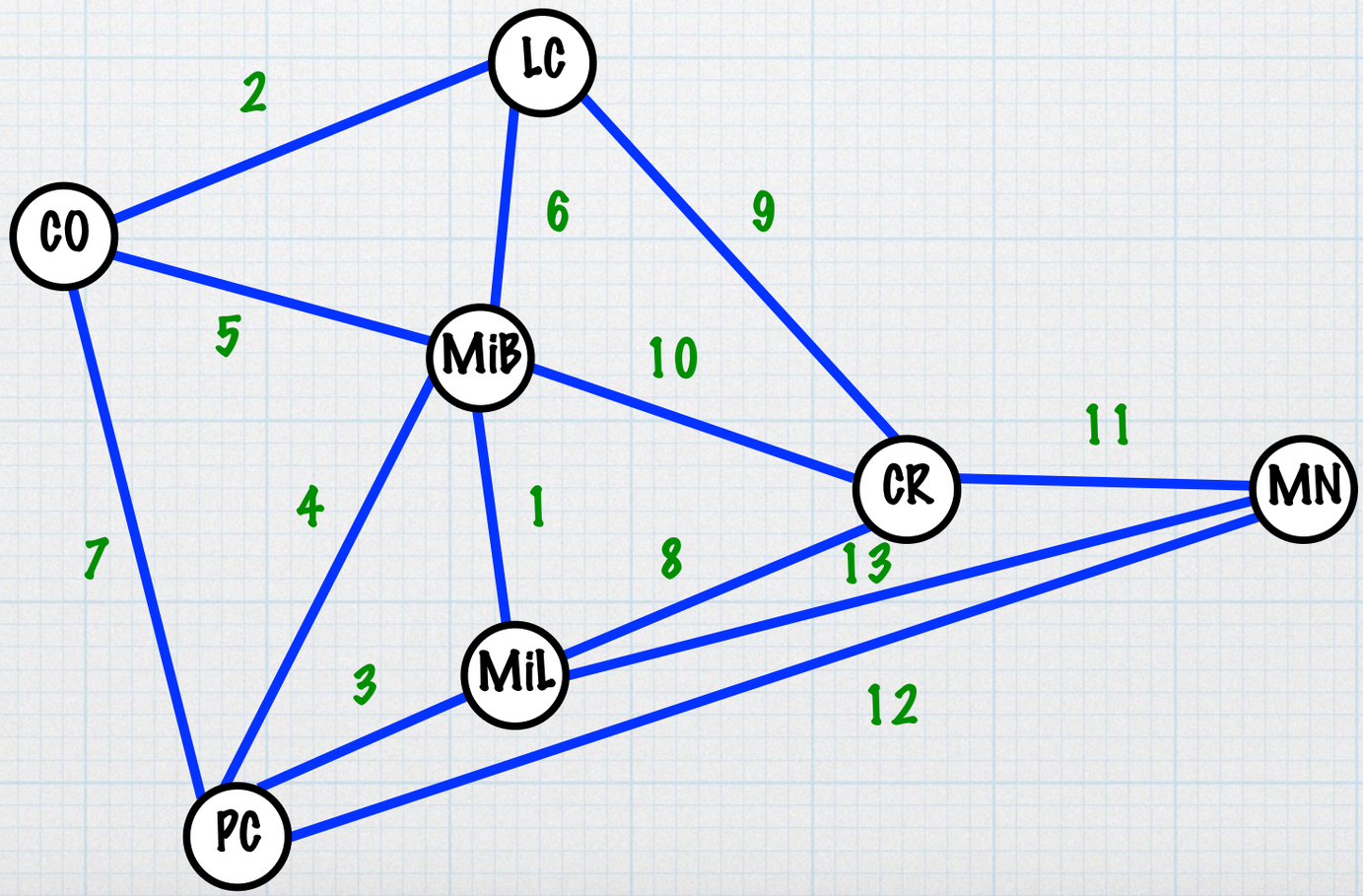


Abstraction of the problem

Given an undirected graph $G = (N, A)$

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find a connected subgraph $G' = (N, A'), A' \subseteq A$

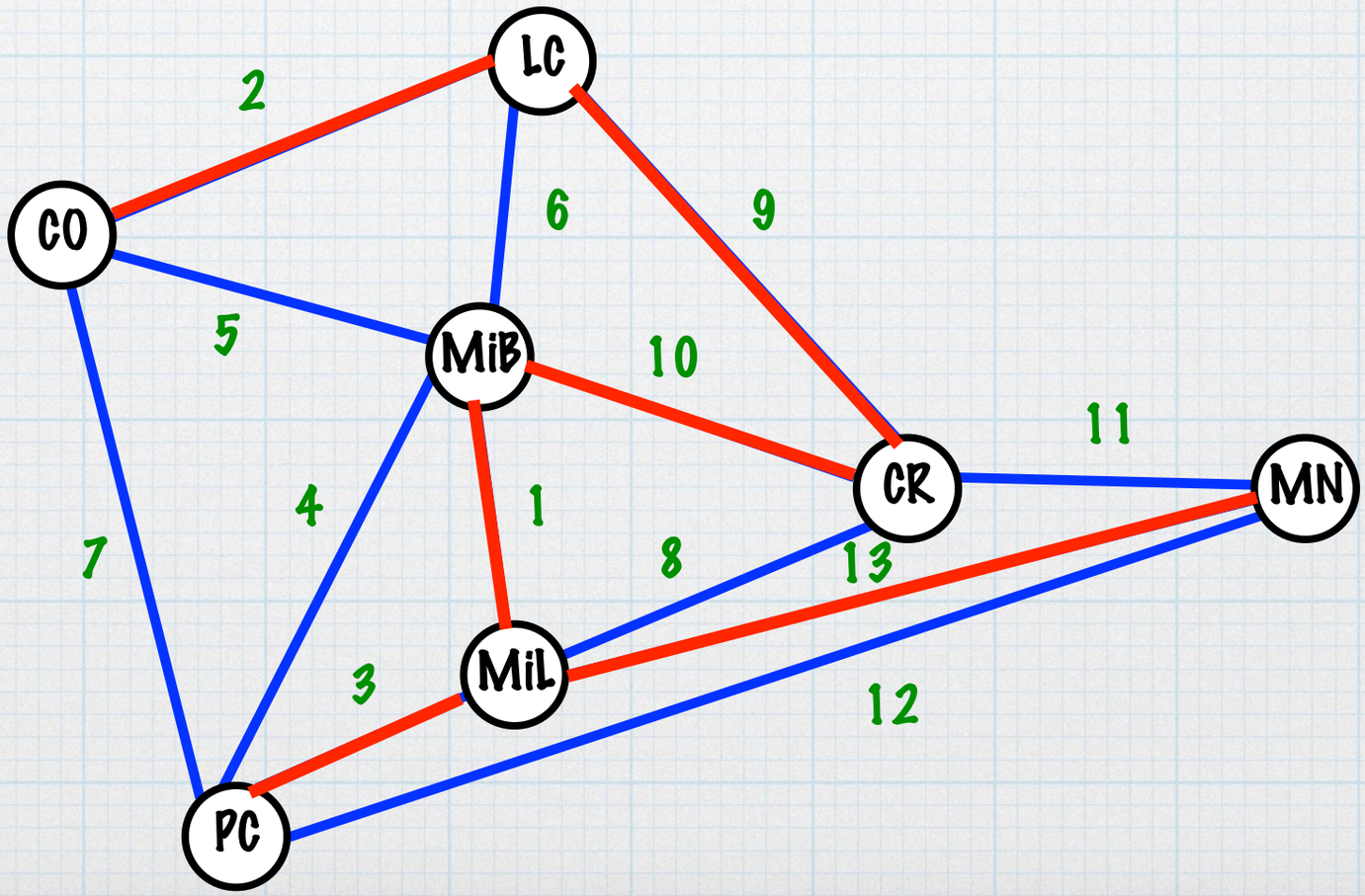


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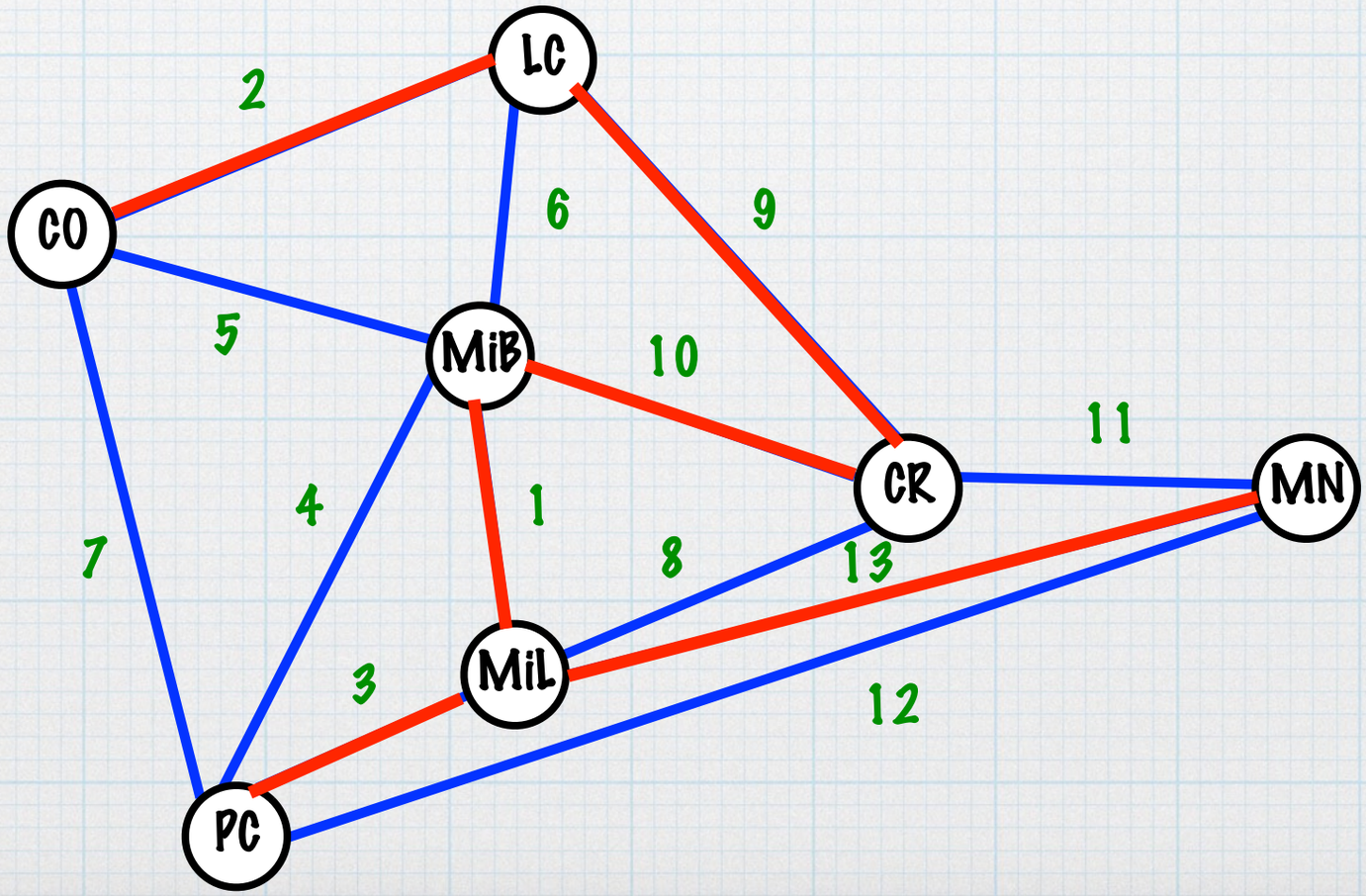
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find a connected subgraph $G' = (N, A'), A' \subseteq A$

minimizing $\sum_{(i,j) \in A'} w_{ij}$



Properties of minimal feasible solutions

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* No cycles: otherwise we could remove some arcs

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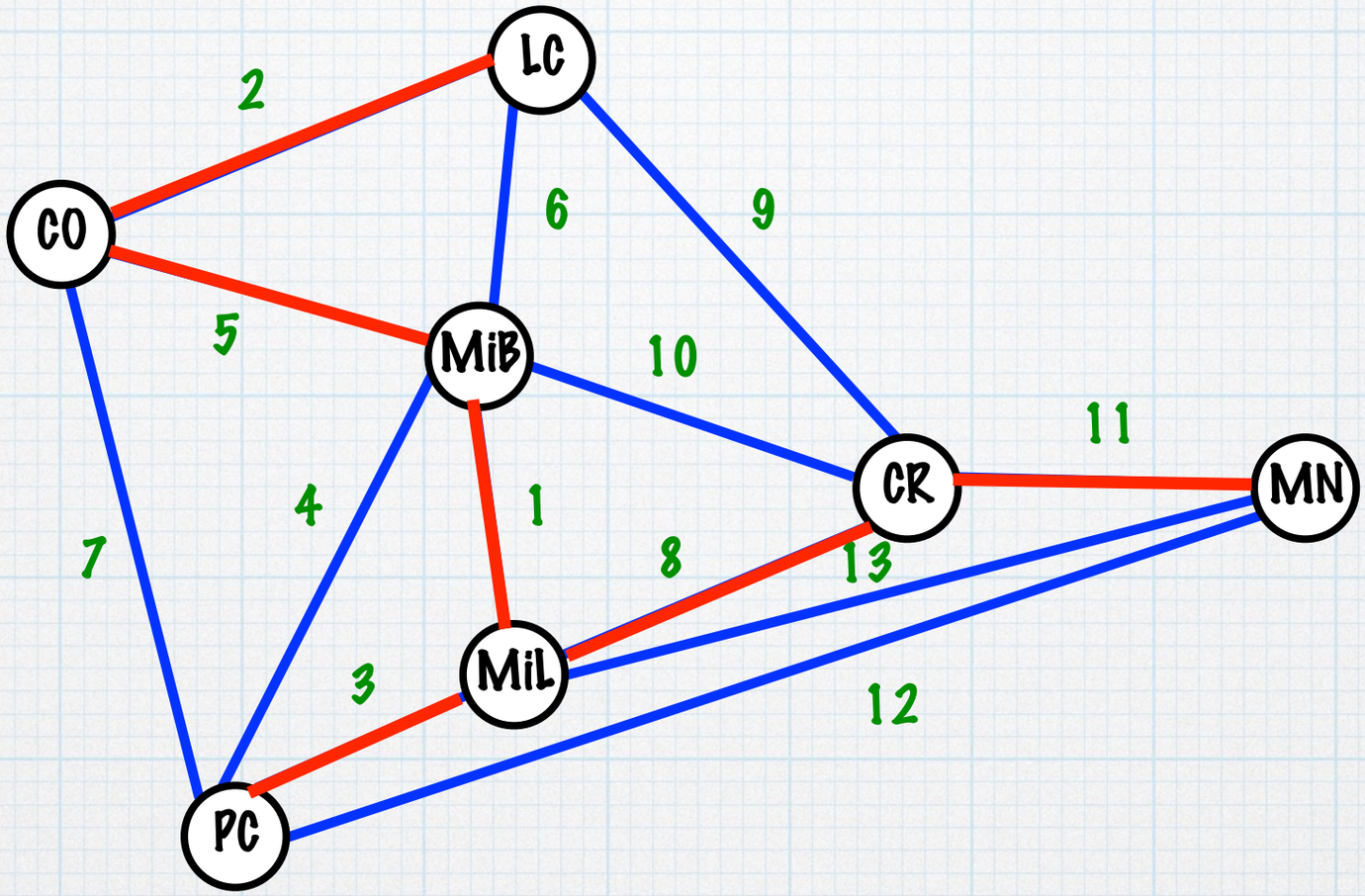
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Minimum Spanning Tree problem (MST)

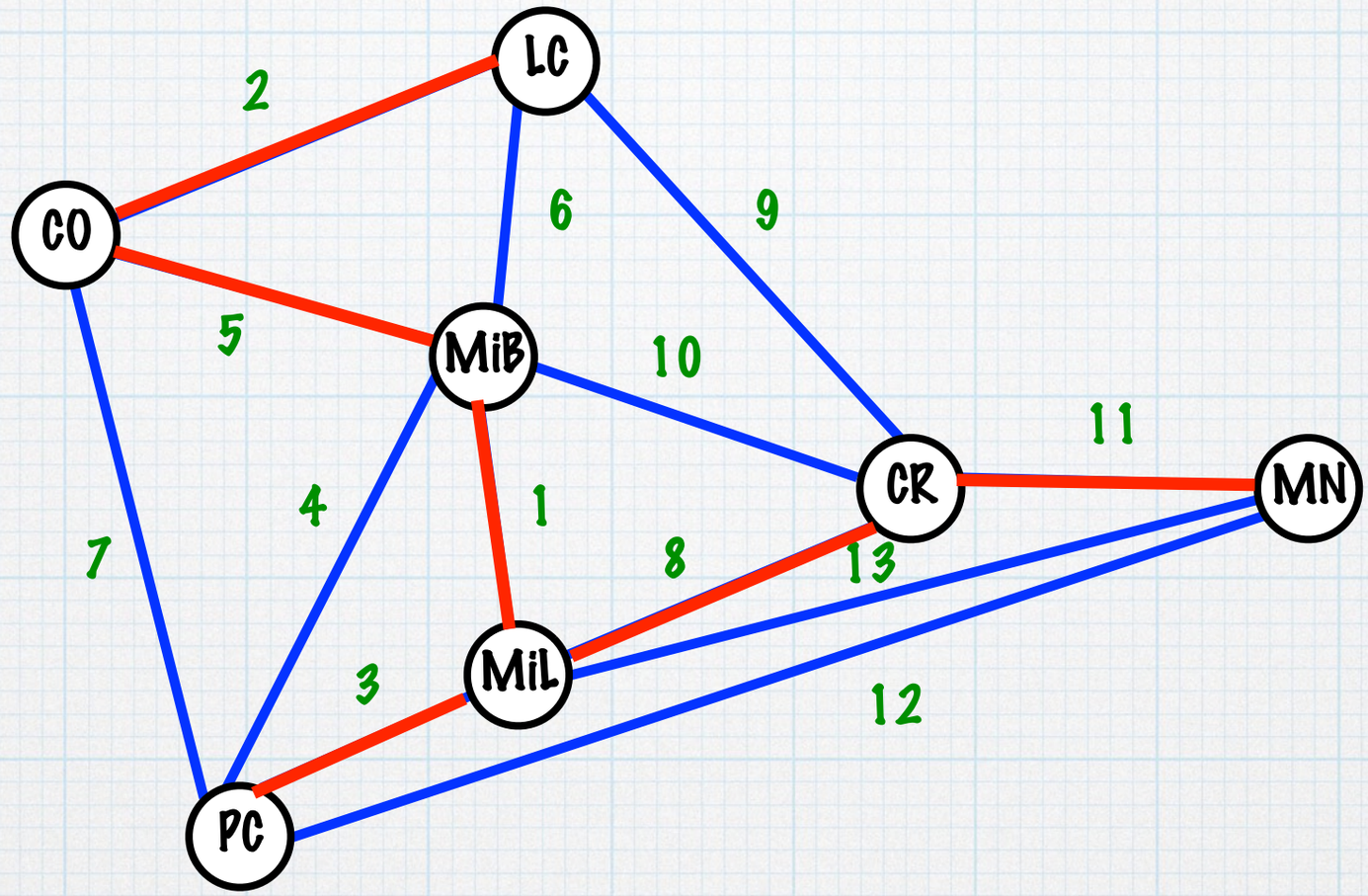
Property suggesting a solution algorithm

MST



Property suggesting a solution algorithm

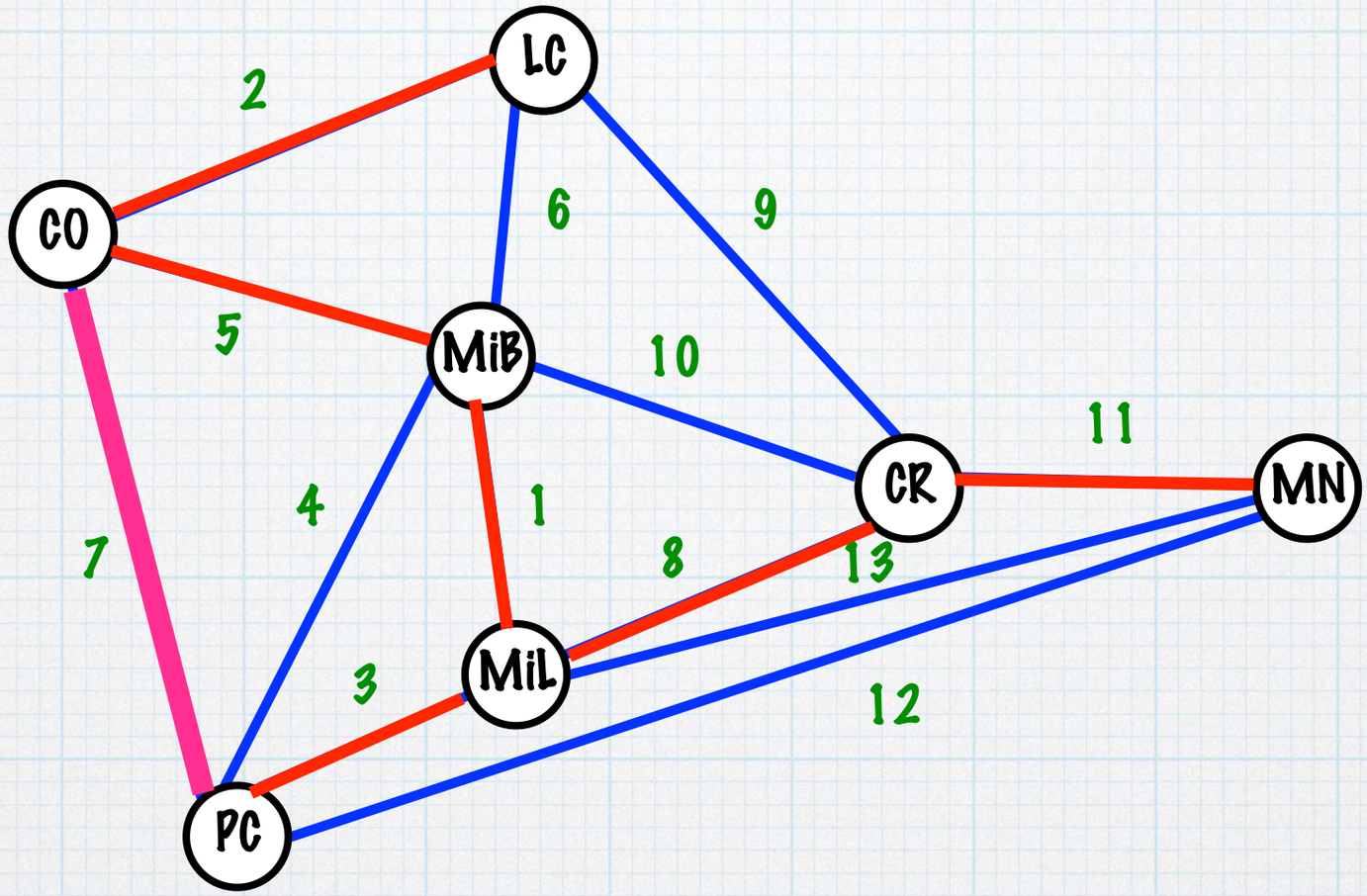
MST



Consider $A' \cup \{i, j\} : \{i, j\} \in A \setminus A'$

Property suggesting a solution algorithm

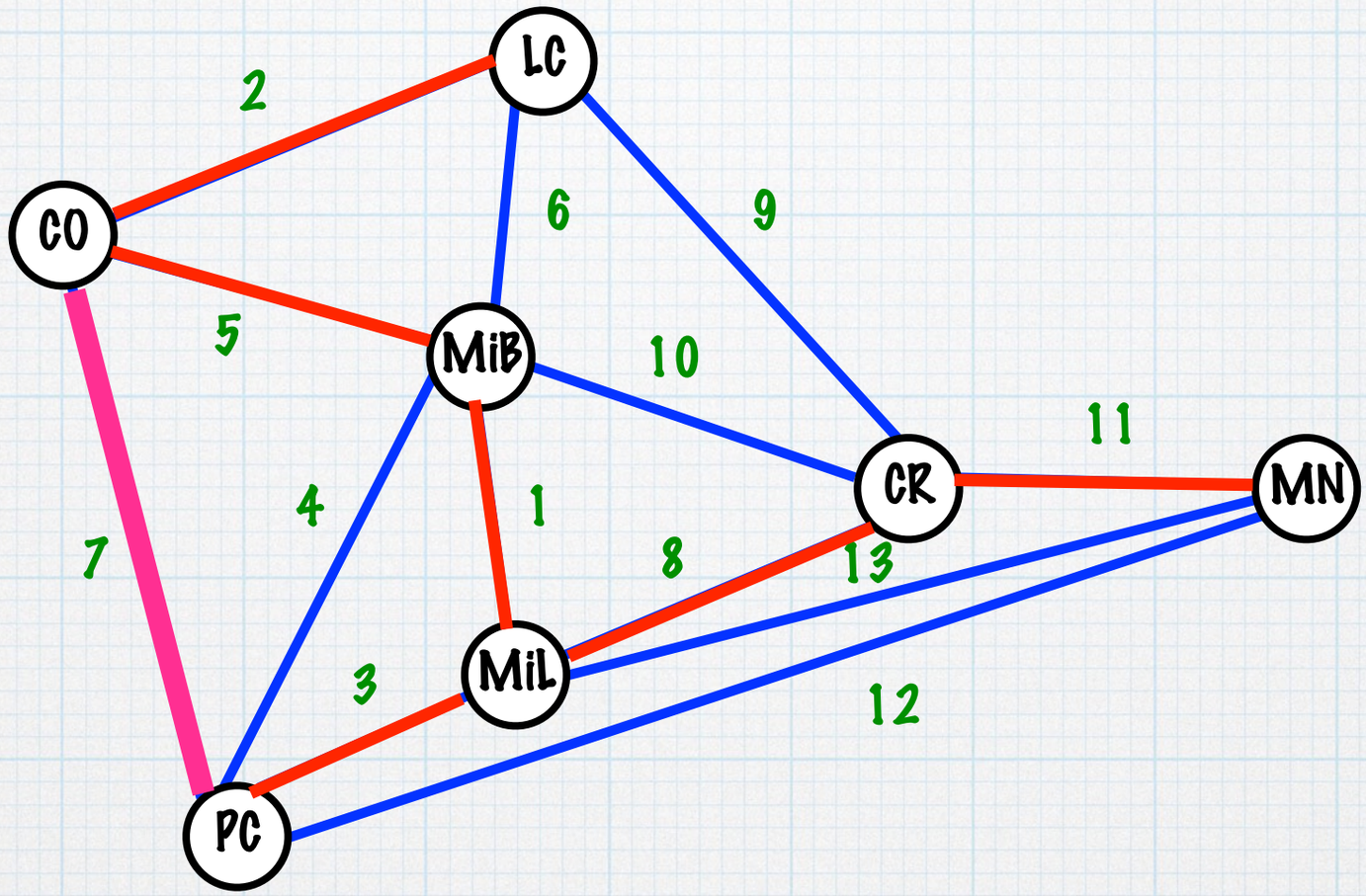
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MST

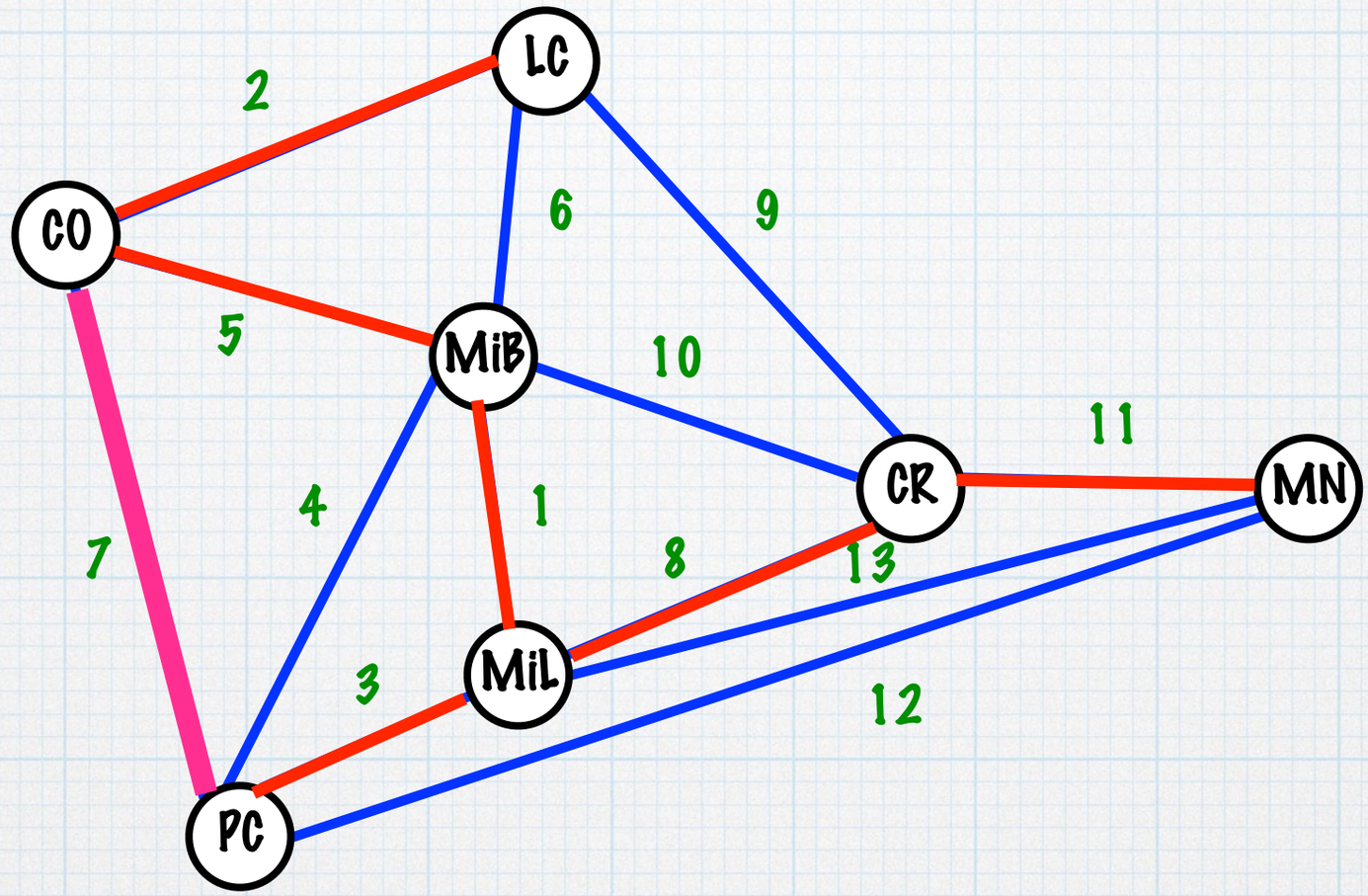


Consider $A' \cup \{i, j\} : \{i, j\} \in A \setminus A'$

the added arc induces a unique cycle C

Property suggesting a solution algorithm

MST



Consider $A' \cup \{i,j\} : \{i,j\} \in A \setminus A'$

the added arc induces a unique cycle C

then $w_{ij} \geq w_{hk} \qquad \forall \{h,k\} \in C$

Algorithm [Kruskal 1956]

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* Sort arcs in increasing order of w_{ij}

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Algorithm [Kruskal 1956]

* Sort arcs in increasing order of w_{ij}
* $A' = \emptyset$
* repeat
  * consider the next arc $\{i,j\}$ in the order
  * **if $A' \cup \{i,j\}$ does not induce a cycle**

    then $A' \leftarrow A' \cup \{i,j\}$ **find - union**
* until $|A'| = n-1$

Implementation

Critical point: discover the cycle

...

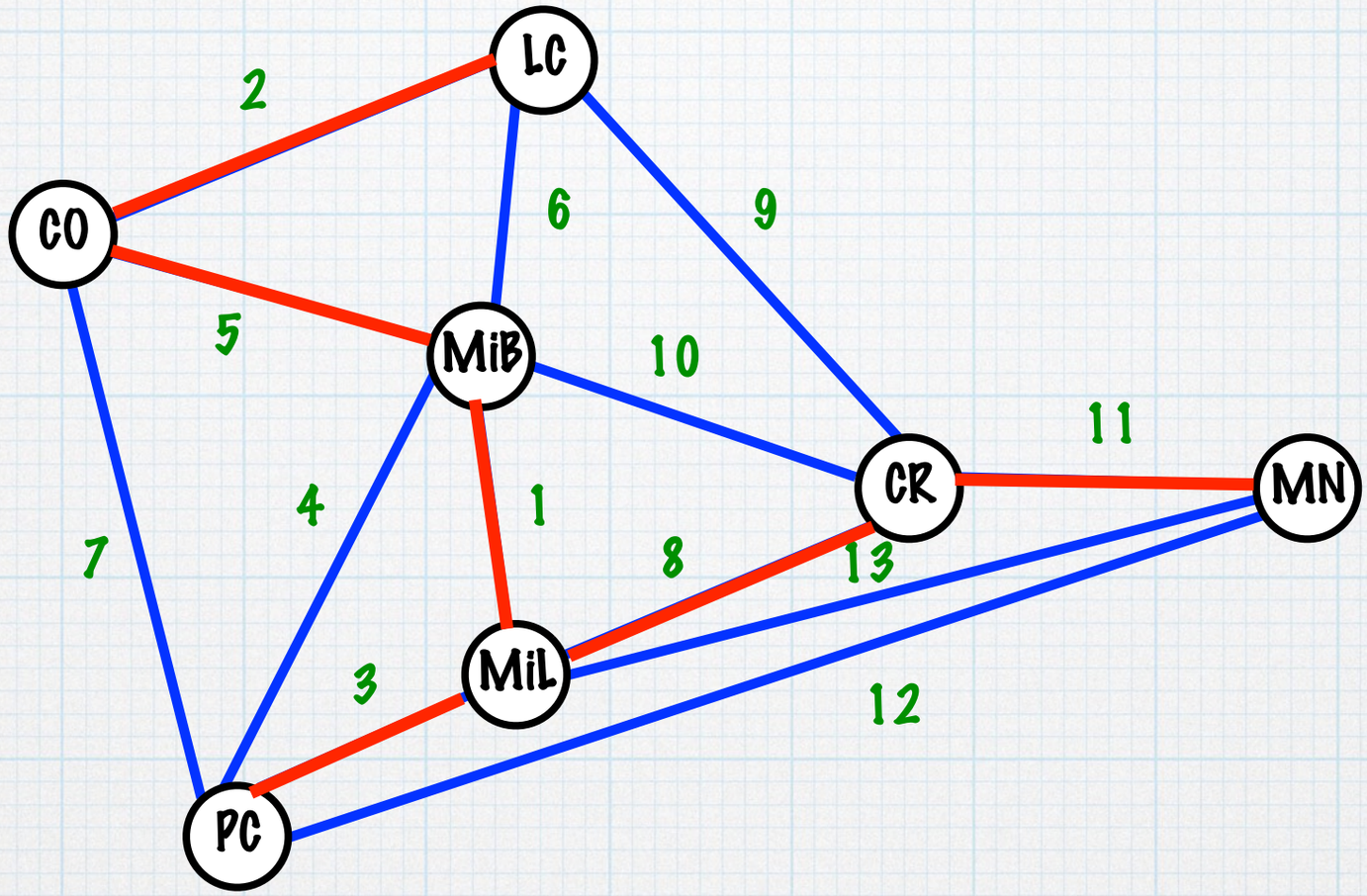
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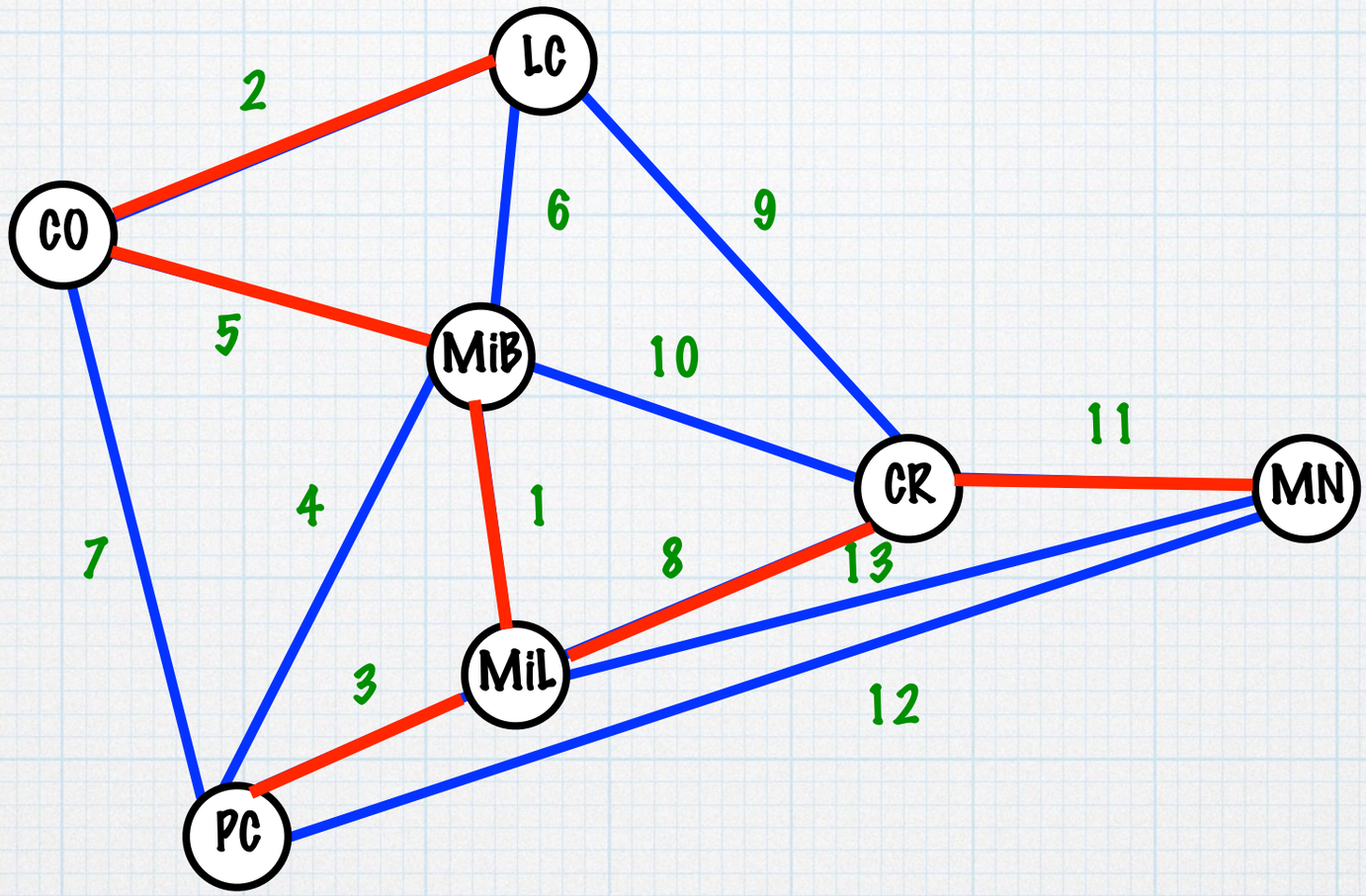
...

If you want to learn more on algorithmic aspects

Property suggesting a solution algorithm

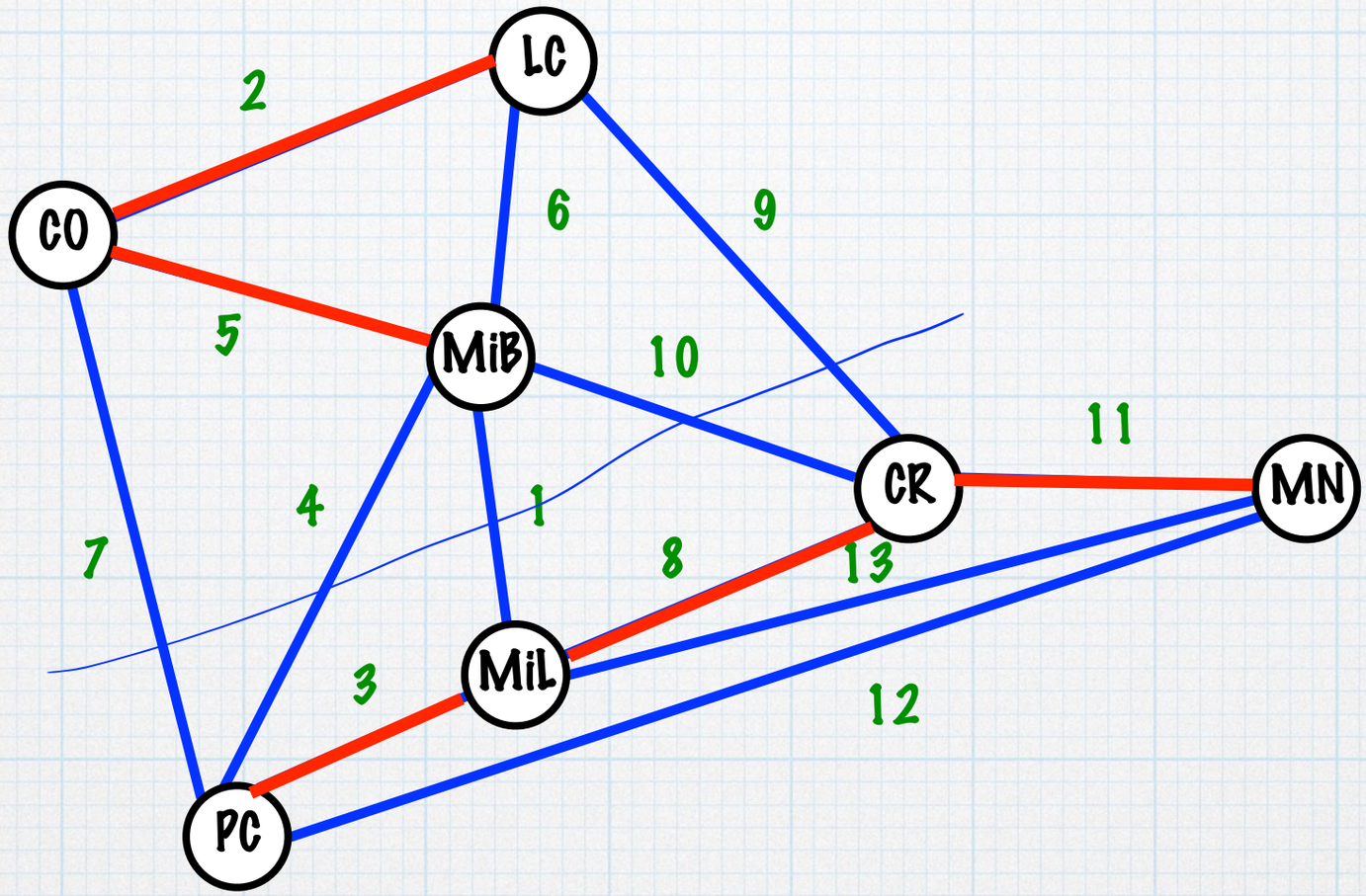


Property suggesting a solution algorithm



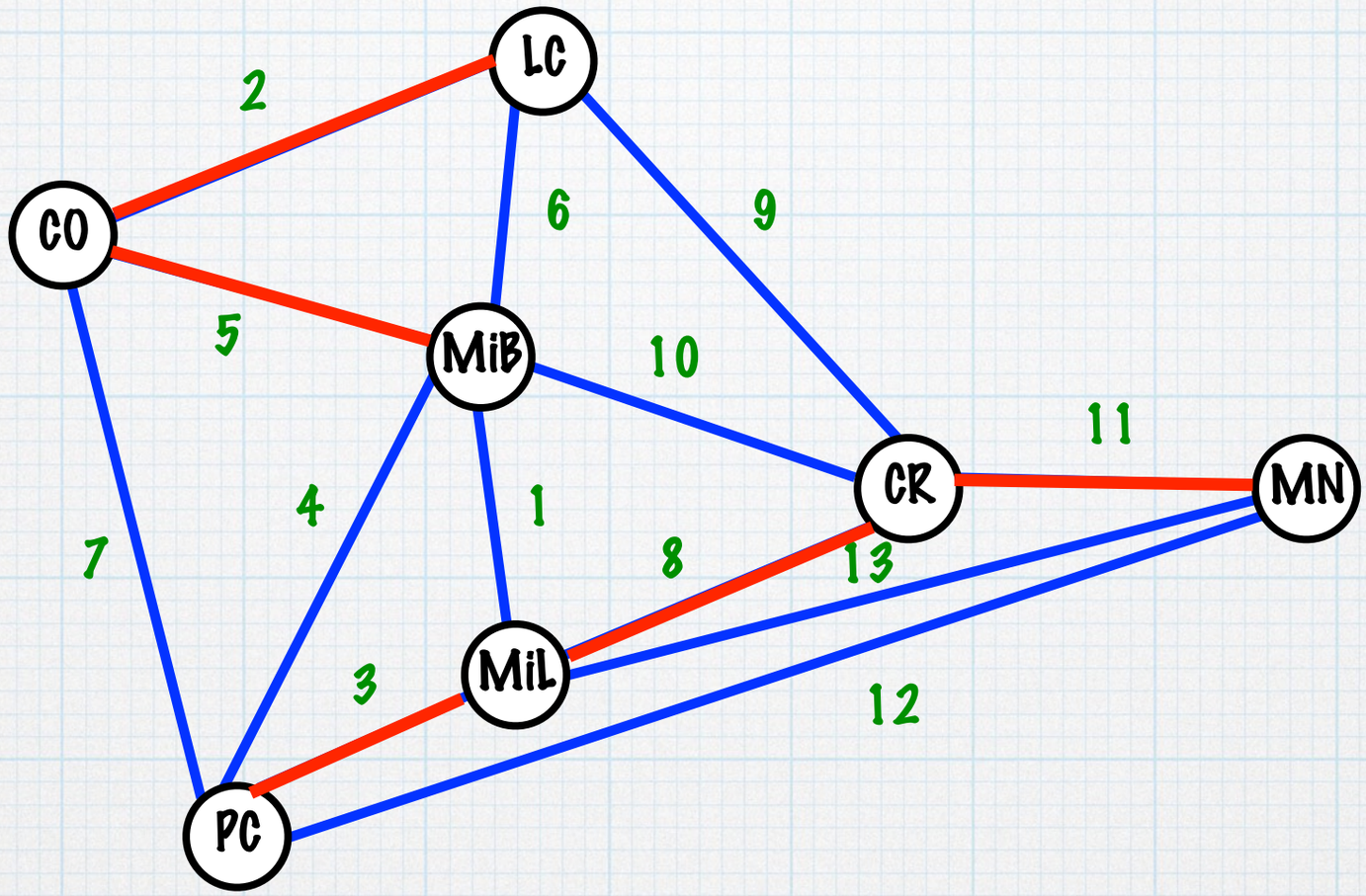
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Property suggesting a solution algorithm



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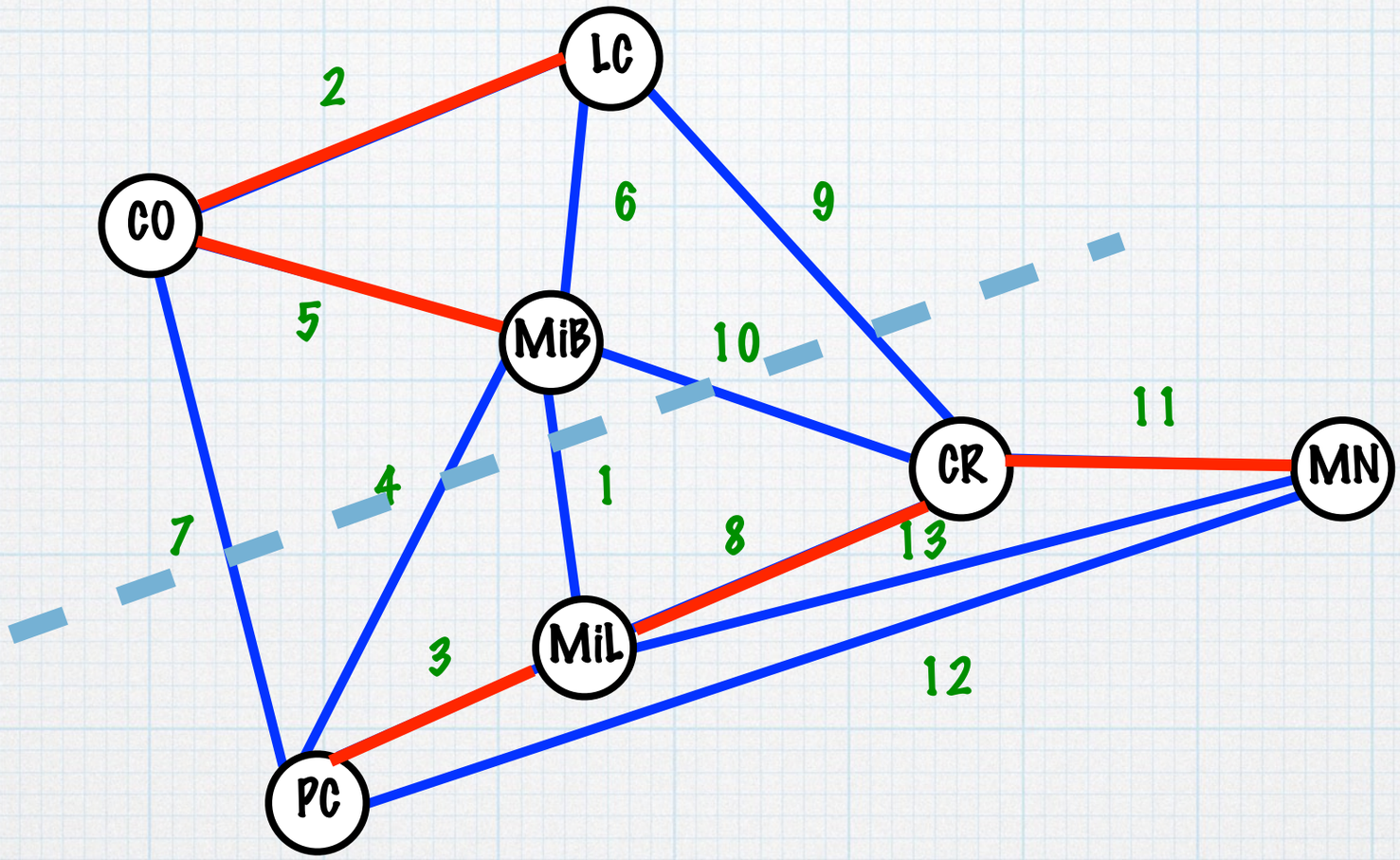


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the removed arc induces two connected components

(H, H')

Property suggesting a solution algorithm

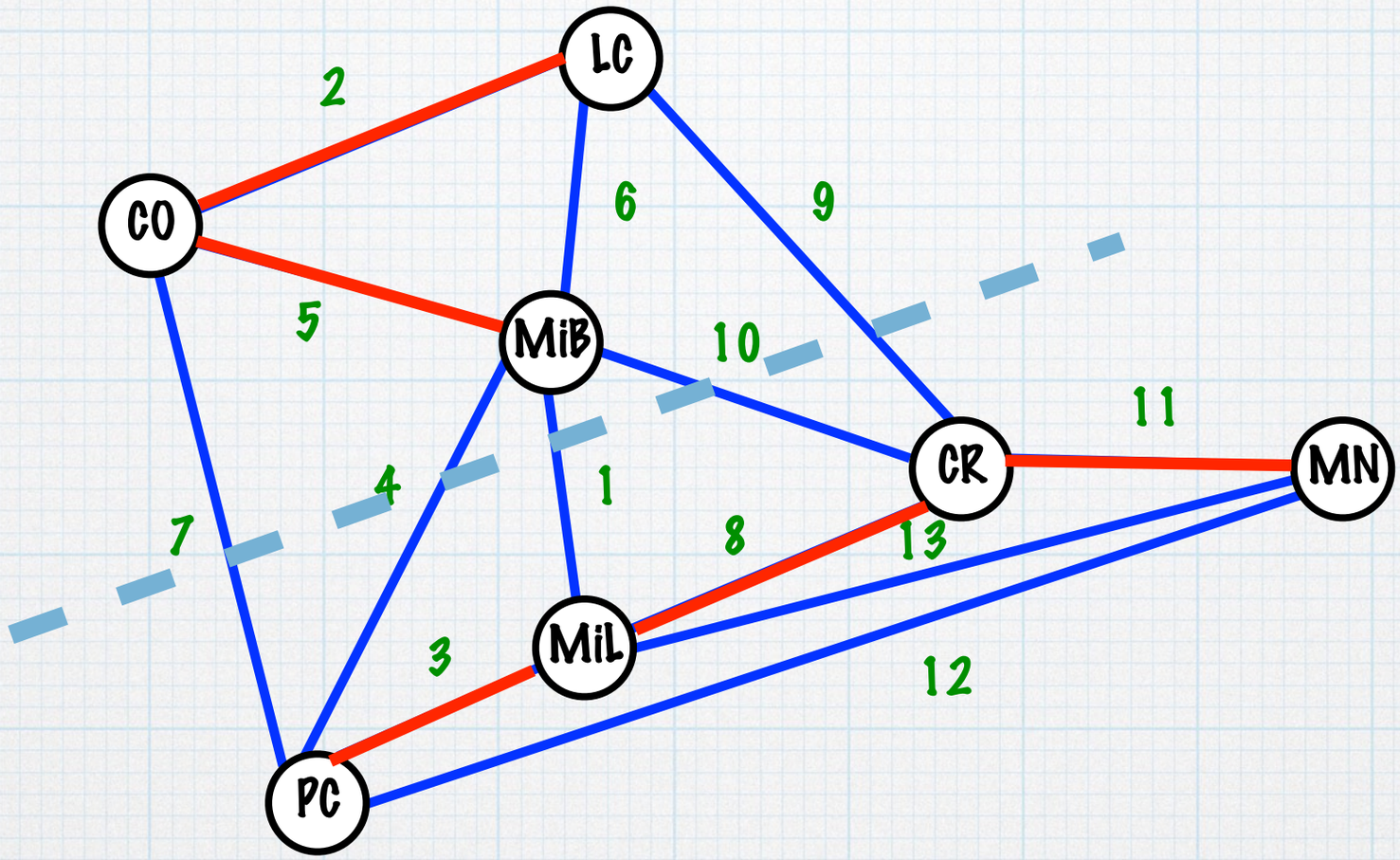


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Consider $A' \setminus \{i,j\} : \{i,j\} \in A'$

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(H, H')

then $w_{i,j} \leq w_{hk} \quad \forall \{h,k\} : h \in H, k \in H'$

Algorithms

Other algorithms can be derived from this property

...

Design a reliable network

* What happens to the Spanning Tree if a link breaks down?

Design a reliable network

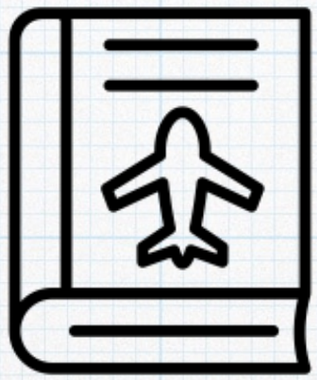
* What happens to the Spanning Tree if a link breaks down?
* How can we make the network reliable in case of a single failure?

Design a reliable network

* What happens to the Spanning Tree if a link breaks down?
* How can we make the network reliable in case of a single failure?

* The cheapest solution is the minimum cost Hamiltonian cycle

Self reflections

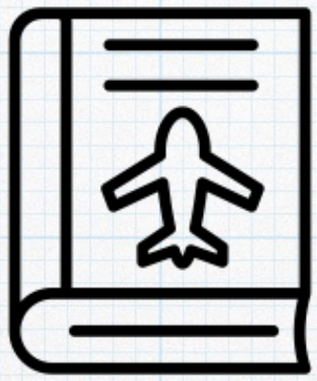


MST and TSP formulations are very similar

Why do you think it was so easy to provide an exact algorithm for MST and not for TSP?

Try to create an algorithm based on Kruskal to find a reasonable solution for the TSP, exploiting the similarities.

Self reflections



MST and TSP formulations are very similar

Why do you think it was so easy to provide an exact algorithm for MST and not for TSP?

Try to create an algorithm based on Kruskal to find a reasonable solution for the TSP, exploiting the similarities.



Call a colleague and compare your ideas

Curiosity: Hamiltonian cycle



In 1800s lord Hamilton studied the problem arising from the "Icosian game"

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This gave the start to the literature on the Traveling Salesperson Problem (TSP)

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www.tsp.gatech.edu

Curiosity: Hamiltonian cycle



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There are no necessary and sufficient conditions to say if a graph has a Hamiltonian Cycle

App Concorde TSP



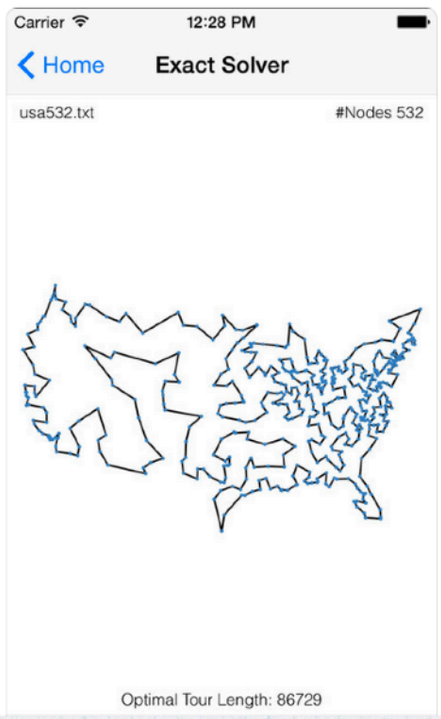
Concorde TSP 4+

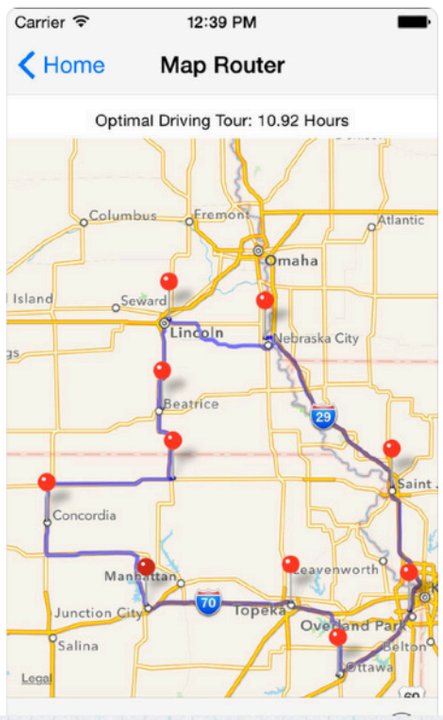
William Cook

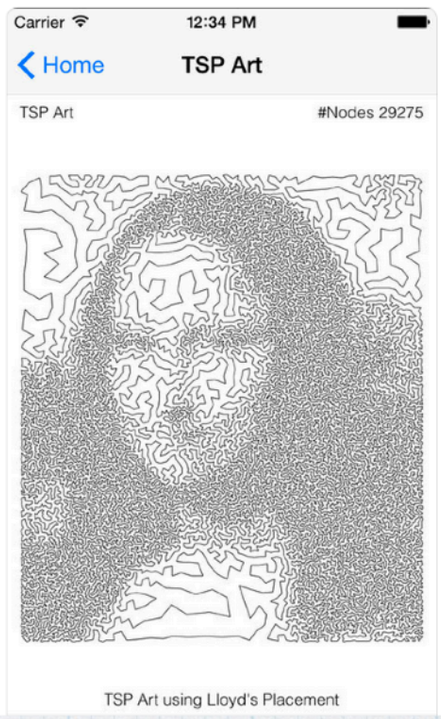
Gratis

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Pocket TSP solver for mobile devices



Popular TSP

Movie

https://www.imdb.com/title/tt1801123/

Fiction book

https://www.amazon.it/ricerca-della-breve-Unavventura-matematica/dp/8847010888

ILO of this module



* Graph notation
* Minimum Spanning Tree
  * Kruskal algorithm
  * Optimality conditions
  * Formulation
* Hamiltonian circuits and TSP



Chapter 2: Sections 1, 2 and 3